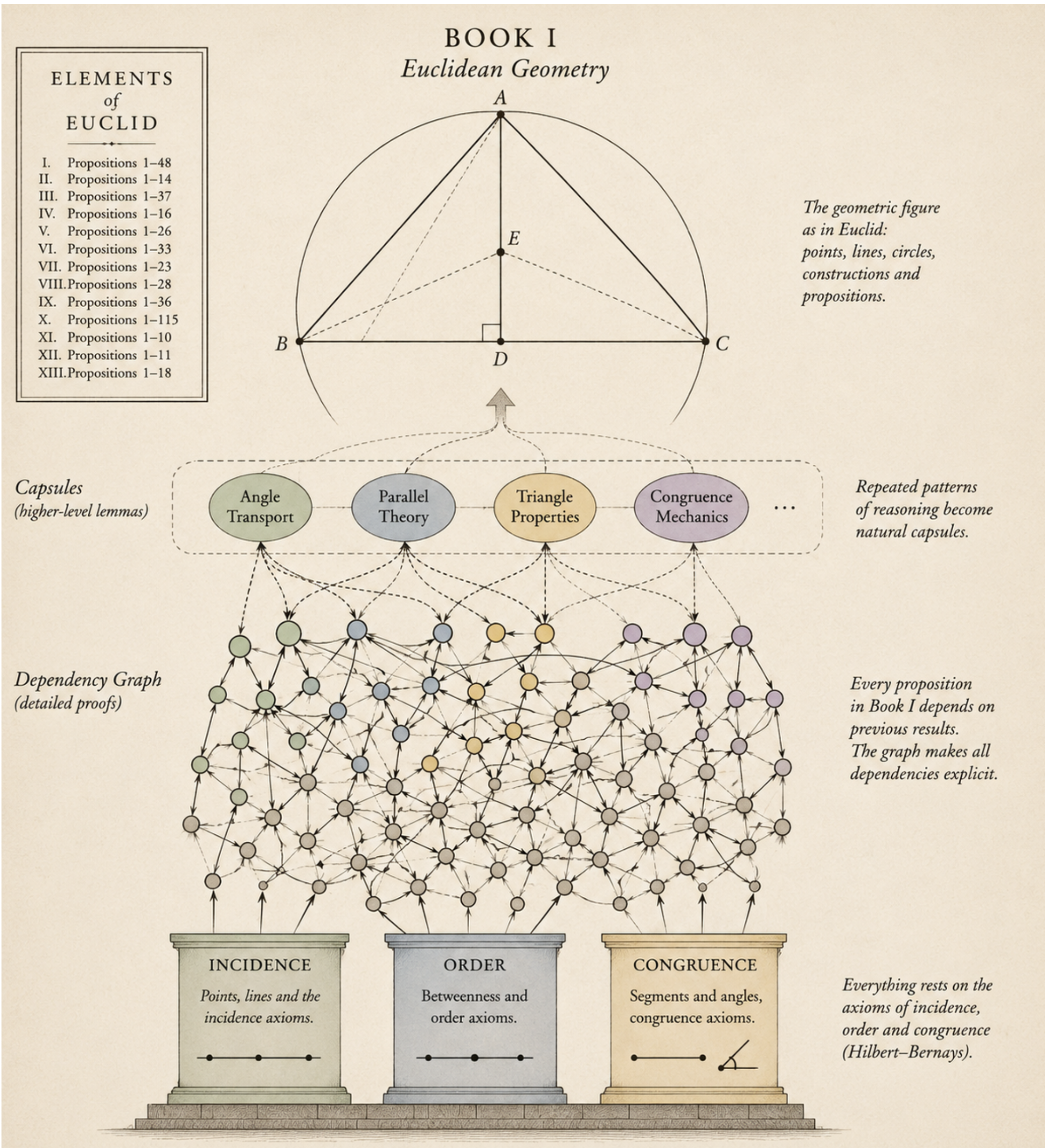
Book I of Euclid's Elements

Reconstruction



CGJteamLab

August 13, 2026

Contents

Chapter 1

Proposition I.1

Euclid's first proposition in Book I is usually presented as the straightforward construction obtained from the intersection of two circles. A formal reconstruction, however, reveals two logically independent problems: the existence of a common point of the corresponding circles and the proof that the resulting point does not lie on the line determined by the given base.

The proof of non-collinearity relies on the classical idea that a proper part of a segment cannot be congruent to the entire segment. This argument is well known from modern axiomatic reconstructions of geometry, including the work of Michael Beeson and his collaborators.

The purpose of this note is not to present a new geometric proof. Rather, it is to demonstrate that, once Book Zero has been developed, this argument becomes a direct application of the previously established theory. Proposition I.1 therefore serves as the first experimental test of Book Zero as an intermediate language for synthetic proofs.

1.1 The Classical Construction

Let $A \neq B$. Euclid constructs two circles of radius AB: the first centered at A, the second centered at B. One of their points of intersection is denoted by C.

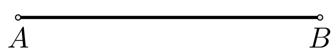
From the construction we obtain

$$AC \cong AB, \qquad BC \cong AB.$$

Formally, however, this is not yet a complete proof. One must also show that the points A, B, C are not collinear.

Step 1. Given Segment

The construction starts with a given segment AB.



Step 2. Draw the Circle Centered at A

Draw the circle centered at A through B.

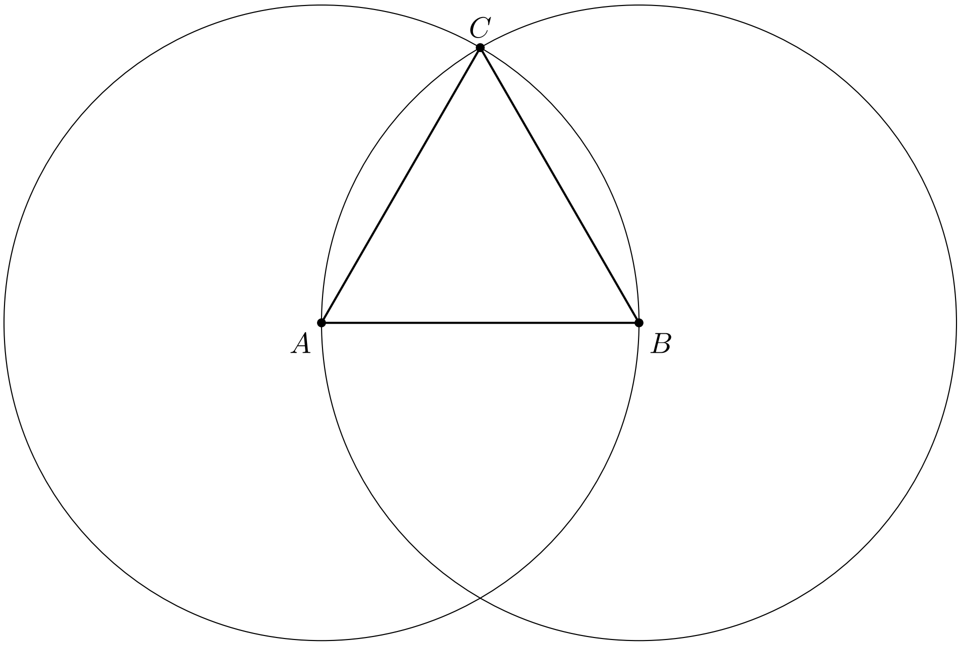
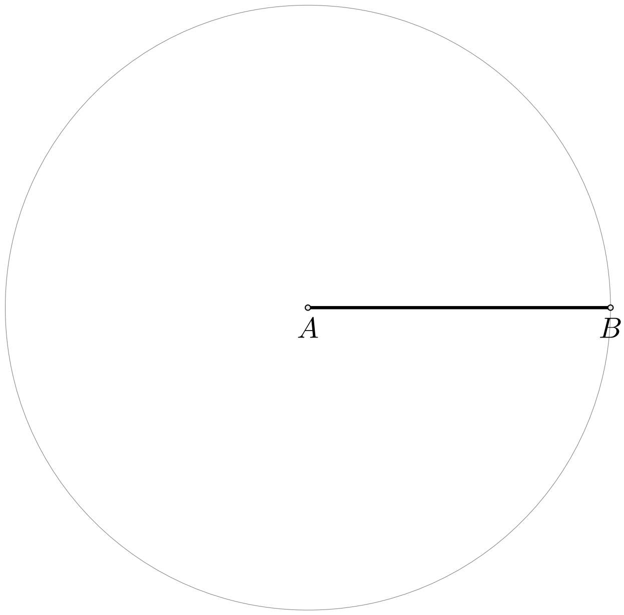
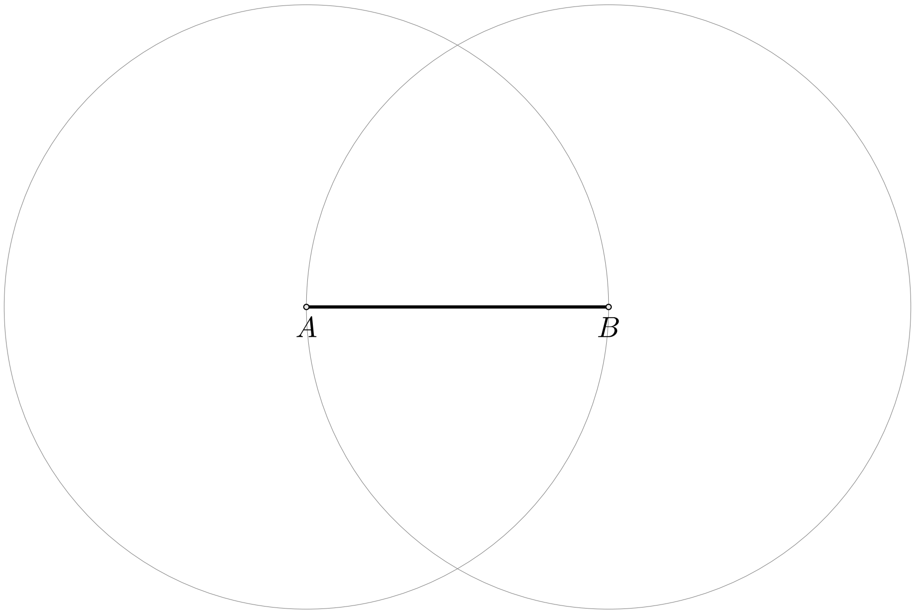


Figure 1.1: Euclid's classical construction. The point C is a common point of two circles of radius AB.



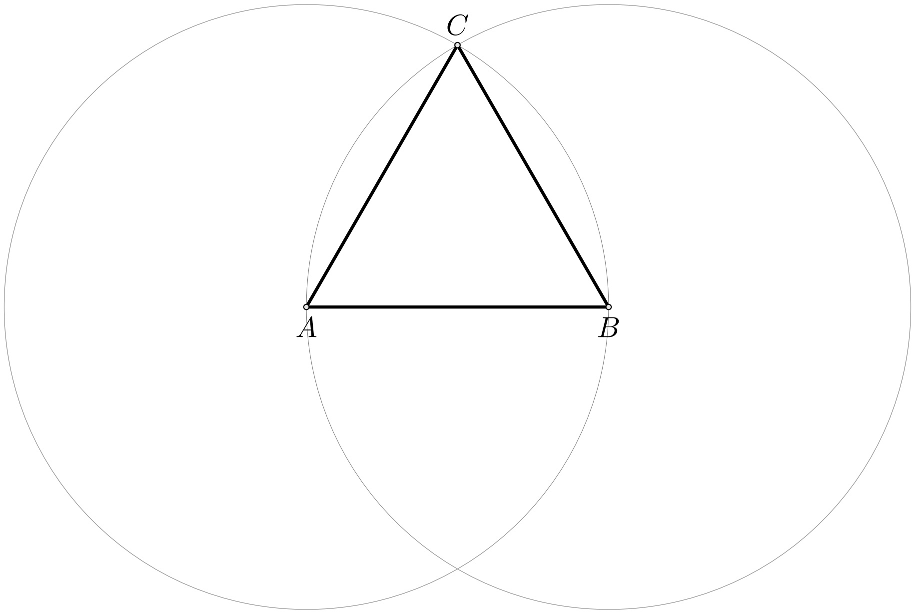
Step 3. Draw the Circle Centered at B

Draw the circle centered at B through A.



Step 4. Construct the Equilateral Triangle

Let C be one of the intersection points of the two circles. Join AC and BC to obtain the equilateral triangle.



1.2 Separating the Two Problems

The reconstruction reveals two logically independent stages.

1. **Existence stage.** One must obtain a point C such that

$$AC \cong AB, \qquad BC \cong AB.$$

In the current version of the project, this stage is provided by the temporary principle `hilbert_equidistant_point_exists`. Ultimately, it is intended to be derived from an appropriate theory of continuity in Hilbert geometry.

2. **Synthetic stage.** From the congruences above one must derive

$$\neg\,\mathrm{Collinear}(A, B, C).$$

This stage no longer requires any theory of circles.

1.3 Theorem

Theorem. Let $A \neq B$. Assume that there exists a point C such that

$$AC \cong AB, \qquad BC \cong AB.$$

Then the points A, B, C are not collinear.

1.4 Proof

First observe that $C \neq A$ and $C \neq B$. If $C = A$, then $AC \cong AB$ would imply that a null segment is congruent to the non-null segment AB. Similarly, $C \neq B$.

Now suppose, for contradiction, that the points A, B, C are collinear. Since they are pairwise distinct, the trichotomy of the order of points on a line gives exactly three possibilities:

$$A - B - C, \qquad B - A - C, \qquad A - C - B.$$

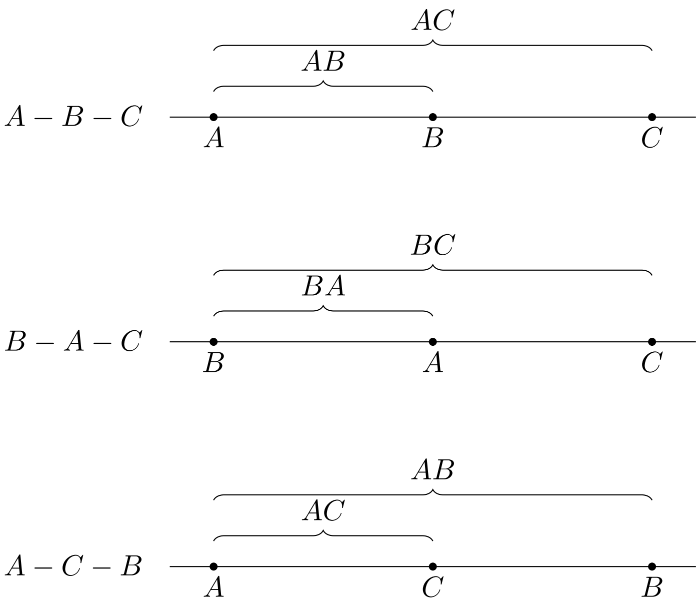


Figure 1.2: The three possible arrangements of pairwise distinct collinear points.

Case 1: $A - B - C$

The segment AB is a proper part of the segment AC. Book Zero contains the general theorem:

> A proper part of a segment cannot be congruent to the whole segment.

On the other hand, by assumption,

$$AC \cong AB.$$

Using symmetry of congruence, we obtain

$$AB \cong AC,$$

which is a contradiction.

Case 2: $B - A - C$

The segment BA is a proper part of the segment BC. Since

$$BC \cong AB,$$

and reversing the endpoints of a segment does not change its length, we have

$$BA \cong AB.$$

By transitivity of congruence, it follows that

$$BA \cong BC,$$

again contradicting the fact that a proper part cannot be congruent to the whole.

Case 3: $A - C - B$

The segment AC is a proper part of the segment AB. But the assumption directly gives

$$AC \cong AB.$$

This yields the third contradiction.

All possible collinear arrangements lead to a contradiction. Therefore,

$$\neg \, \mathrm{Collinear}(A, B, C).$$

□

1.5 Architecture of the Proof

1.6 Significance of the Reconstruction

The geometric result itself is not new. Constructing an equilateral triangle on a given segment is one of the classical theorems of elementary geometry.

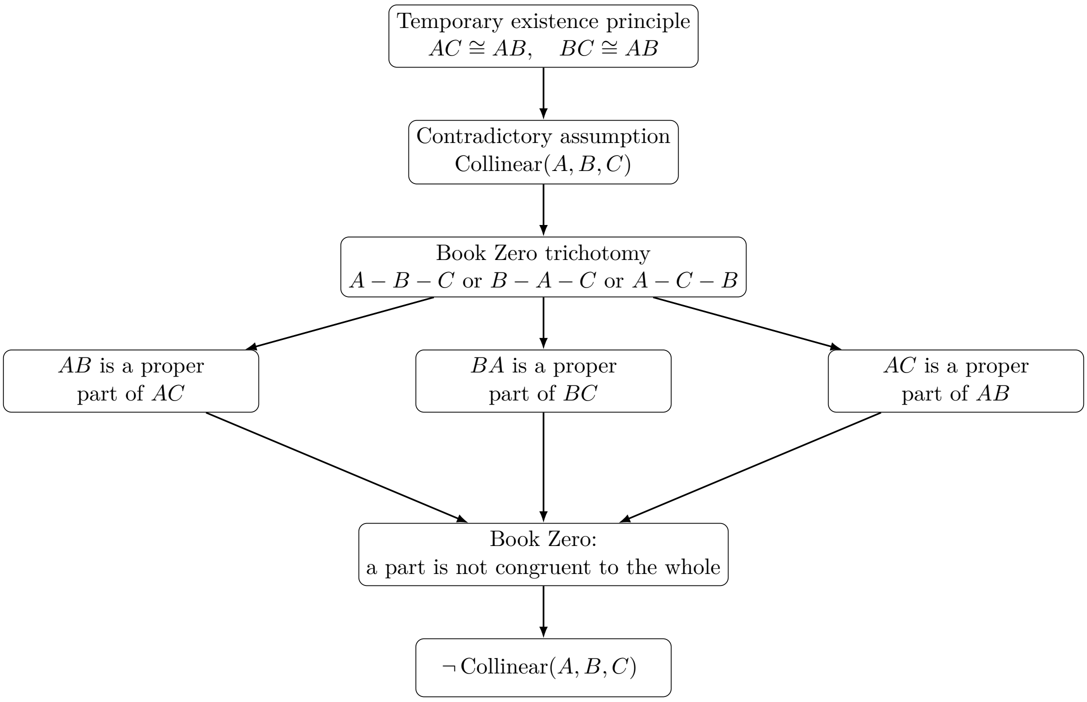


Figure 1.3: The structure of the proof of Proposition I.1 above the Book Zero layer.

Likewise, the idea underlying the proof of non-collinearity is not new. Modern axiomatic reconstructions, in particular the work of Michael Beeson and his collaborators, employ an analogous argument based on the theorem that a proper part of a segment cannot be congruent to the entire segment.

The novelty of the present reconstruction lies elsewhere.

The objective of the CGJteamLab project was not to discover a new proof of Proposition I.1, but to investigate whether the theory developed in Book Zero is sufficiently rich to serve as an intermediate language for synthetic proofs.

The experiment shows that, once the existence problem for an equidistant point has been isolated, the entire argument can be expressed almost exclusively in terms of the theorems previously established in Book Zero.

From the perspective of the architecture of the theory, this is more significant than Proposition I.1 itself. It demonstrates that Book Zero is not merely a collection of auxiliary lemmas, but rather forms a new layer of abstraction between Hilbert's axioms and classical synthetic reasoning.

Schematically, the situation may be represented as

$$\text{Hilbert's axioms} \longrightarrow \text{Book Zero} \longrightarrow \text{Proposition I.1.}$$

In this sense, Proposition I.1 constitutes the first experimental test confirming that Book Zero can serve as a language of synthetic proofs for the further reconstruction of Euclidean geometry.

Chapter 2

Proposition I.2

This note compares Euclid's original construction for Proposition I.2 with its reconstruction over the Hilbert foundation. In the CGJteamLab development the proposition follows immediately from Book Zero #49 (Layoff), illustrating the difference between the architectural roles of segment construction in Euclid's geometry and the Hilbert axiomatic framework.

2.1 Statement

Given a point A and a segment BC, construct a point D such that

$$AD \cong BC.$$

2.2 Euclid's Classical Construction

Euclid's construction proceeds as follows.

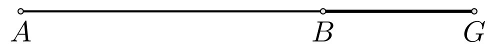
1. Construct the equilateral triangle DAB on the segment AB.
2. Produce the side DB beyond B.
3. Produce the side DA beyond A.
4. Draw the circle centered at B with radius BC.
5. Draw the circle centered at D with radius DG.
6. The constructed point L satisfies
$$AL \cong BC.$$

Step 1. Given data

The construction starts with a given point A and a given segment BG.

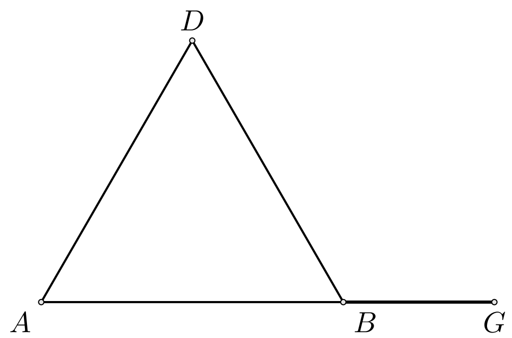
Step 2. Join AB

Join the given point A to the endpoint B of the given segment.



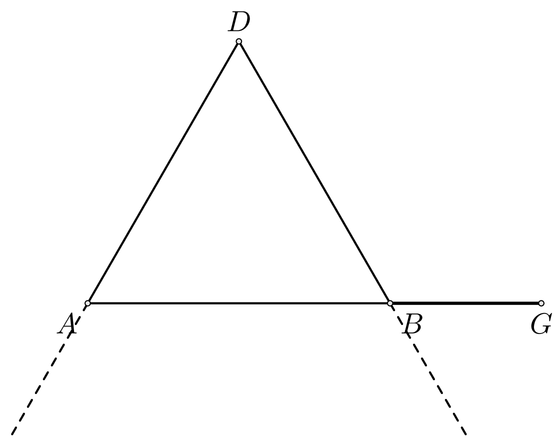
Step 3. Construct the equilateral triangle DAB

Construct the equilateral triangle DAB on the segment AB.



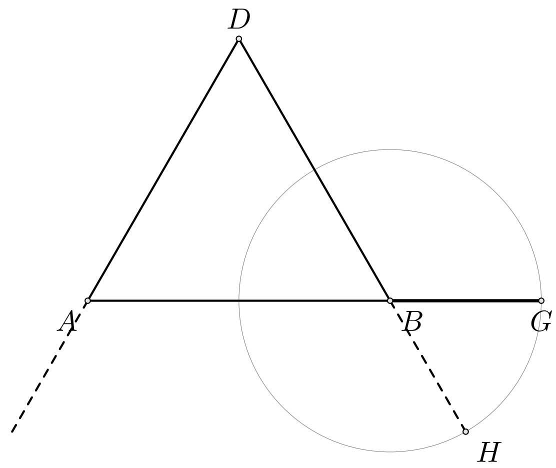
Step 4. Produce DA and DB

Produce DA beyond A and DB beyond B.



Step 5. Draw the circle centered at B

Draw the circle centered at B through G. Let H be the intersection of this circle with the line DB on the side beyond B.

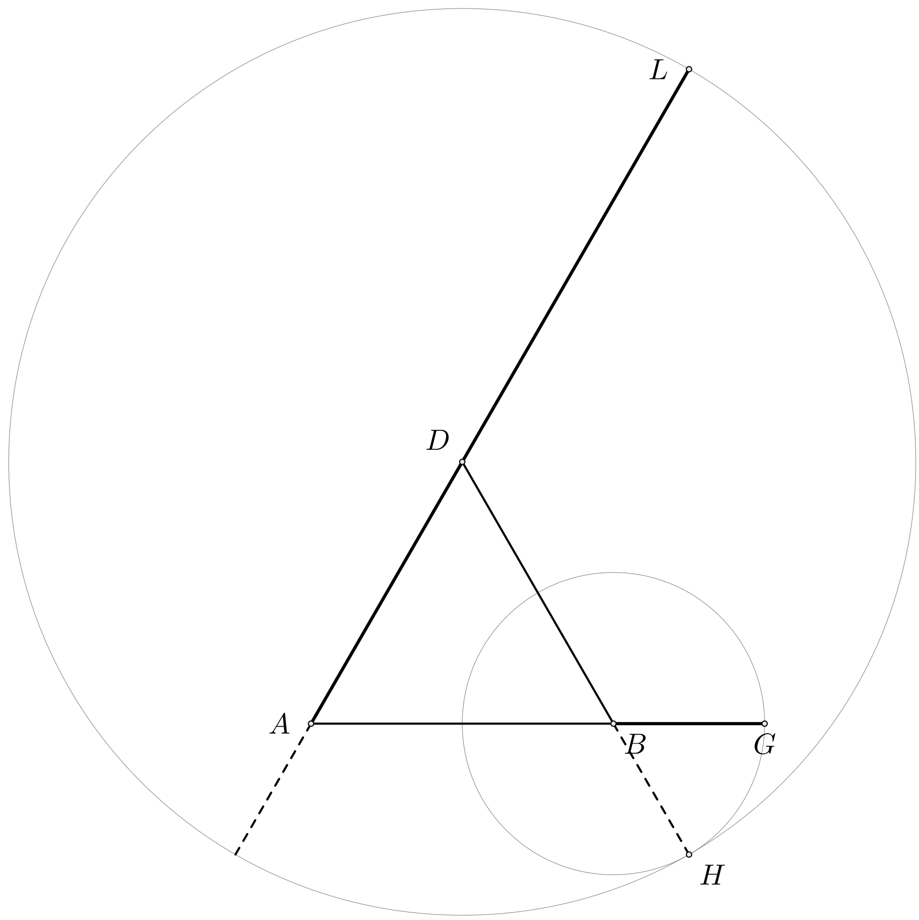


Since H lies on the circle centered at B through G,

$$BH \cong BG.$$

Step 6. Draw the circle centered at D

Draw the circle centered at D through H. Let L be the intersection of this circle with the line DA on the side beyond A.



Since L lies on the circle centered at D through H,

$$DL \cong DH.$$

The segment AL is therefore the required copy of the given segment.

2.3 Hilbert Reconstruction

The reconstruction over the Hilbert foundation is considerably shorter.

1. Choose any point $E \neq A$.
2. The ray AE is thereby determined.
3. Apply Book Zero #49 (Layoff).
4. Obtain a point D on the ray AE satisfying

$$AD \cong BC.$$

No auxiliary geometric construction is required.

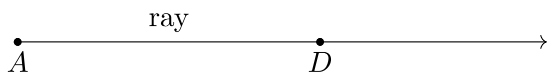


Figure 2.1: Lay off the segment on a chosen ray.

Book Zero #49 provides a point D on a chosen ray from A such that

$$AD \cong BC.$$

2.4 Formalization

The formal proof consists of a single application of Book Zero #49.

```
theorem euclid_proposition_2 ...
  := bookZero_49_layoff ...
```

2.5 Discussion

Thus Proposition I.2 illustrates a fundamental architectural difference between Euclid's geometry and the Hilbert approach.

In Euclid's *Elements*, the transport of a segment is obtained through a sequence of explicit geometric constructions beginning with Proposition I.1.

In contrast, Hilbert incorporates this operation into the congruence axioms. Book Zero encapsulates it in the Layoff Theorem, making Proposition I.2 an immediate consequence.

2.6 Book Zero #49: Layoff

The proof of Proposition I.2 relies on one of the elementary construction theorems established in Book Zero.

Book Zero #49 (Layoff). Let $A \neq B$ and $C \neq D$. Then there exists a point X on the ray AB such that

$$AX \cong CD.$$

In other words, every nondegenerate segment can be copied onto an arbitrarily chosen ray while preserving its length.

```
Input
-----

Ray AB

A------------------------------>

Segment CD

C-------------D


|
|  Book Zero #49
v


Output
------

A----------------X------------->

AX congruent CD
```

This theorem provides the fundamental operation of transporting segments. Once it is available, Euclid's Proposition I.2 becomes an immediate consequence.

Chapter 3

Proposition I.3

3.1 Introduction

Euclid's third proposition asks for the construction of a segment equal to a given shorter segment on a given greater segment.

Classically, this construction is obtained by combining Proposition I.2 with an additional circle construction. A segment equal to the shorter one is first laid off from an endpoint of the greater segment, and a circle is then used to determine the required point on the greater segment.

In the present reconstruction over Hilbert geometry, the logical structure is substantially different. The theory developed in Book Zero already contains an abstract notion of one segment being less than another. This notion is defined by the existence of a point between the endpoints of the greater segment such that the corresponding subsegment is congruent to the smaller one.

Consequently, the existential content of Euclid's Proposition I.3 is already embodied in the definition of segment inequality. The reconstruction therefore consists not in repeating Euclid's geometric construction, but in exposing the witness contained in the definition of `HilbertSegmentLess`.

This proposition provides the first clear example where a classical Euclidean construction becomes an immediate consequence of the abstract language developed in Book Zero.

3.2 The Classical Construction

Euclid's third proposition asks for the construction of a segment equal to a given shorter segment on a given greater segment.

Unlike Proposition I.2, the present construction does not introduce a new geometric technique. Instead, it combines Proposition I.2 with one additional circle construction.

The construction proceeds as follows.

3.2.1 Final Configuration

The objective is to determine a point E on the greater segment AB such that

$$AE \cong CD.$$

The final geometric configuration is shown in Figure 1.

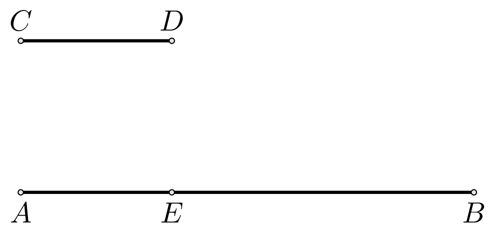


Figure 3.1: Final configuration of Euclid's construction for Proposition I.3.

From the construction we obtain

$$AE \cong CD.$$

The remaining task is to justify that the point E lies between the endpoints of the greater segment.

3.2.2 Construction Algorithm

The construction uses Proposition I.2 as an already established construction primitive.

Step 1. Given Segments

Let AB be the greater segment and CD the given shorter segment.

Step 2. Apply Proposition I.2

By Proposition I.2, construct a segment AF congruent to the given segment CD.

Thus

$$AF \cong CD.$$

Step 3. Draw the Circle Centered at A

Draw the circle centered at A passing through F.

Step 4. Determine the Required Point

Let E be the intersection of the circle with the greater segment.

Since

$$AE \cong AF$$

and

$$AF \cong CD,$$

it follows that

$$AE \cong CD.$$

The required segment has therefore been obtained.

3.3 Proof

The classical Euclidean proof constructs the required point by applying Proposition I.2 and then introducing an auxiliary circle.

The reconstruction over Hilbert geometry follows a different logical route. The relation that one segment is less than another is already represented in Book Zero by the predicate

$$\texttt{HilbertSegmentLess}\ CD\ AB.$$

By definition, this predicate means that there exists a point E between the endpoints of the greater segment satisfying

$$A - E - B$$

and

$$AE \cong CD.$$

Therefore, assuming

$$CD < AB,$$

we immediately obtain a point E such that

$$A - E - B \qquad \text{and} \qquad AE \cong CD.$$

This is precisely the conclusion required by Proposition I.3.

Consequently, over the present Hilbert/Book Zero layer, the proposition is an immediate consequence of the definition of `HilbertSegmentLess`.

□

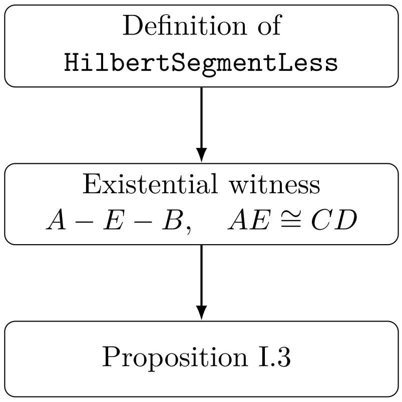


Figure 3.2: Logical structure of Proposition I.3 over the Book Zero layer.

3.4 Proof Architecture

3.5 Hilbert Reconstruction

Unlike Euclid's original proof, the present reconstruction does not repeat the geometric construction based on Proposition I.2.

The abstract notion of one segment being less than another has already been introduced in Book Zero by the predicate `HilbertSegmentLess`. By definition, this predicate contains exactly the existential information required by Proposition I.3.

Consequently, the reconstruction consists simply in extracting the witness from this definition.

This illustrates an important feature of the Book Zero layer. Classical geometric constructions are progressively replaced by abstract synthetic notions whose mathematical content has already been established.

3.6 Lean Formalization

The Lean formalization closely follows the mathematical argument.

The hypothesis

$$\texttt{HilbertSegmentLess}\; CD\; AB$$

already contains an existential witness providing the required point between the endpoints of the greater segment together with the corresponding congruence.

The formal proof therefore consists only of unpacking this witness and applying the symmetry of segment congruence.

Unlike the classical Euclidean proof, no additional geometric construction is required.

The resulting Lean theorem is therefore considerably shorter than the corresponding classical argument.

Chapter 4

Proposition I.4

4.1 Introduction

Euclid's fourth proposition is the first theorem on triangle congruence.

Unlike the preceding propositions, it does not describe a geometric construction. Instead, it establishes that two triangles are congruent whenever two corresponding sides and the included angle are equal.

Euclid's original proof relies on the method of superposition, one of the most discussed arguments in the *Elements*. Modern axiomatic systems generally avoid superposition as a primitive method of proof.

In the present reconstruction over Hilbert geometry, the proposition is obtained from the Side–Angle–Side (SAS) congruence theorem, which has already been established within the Hilbert theory.

Thus, while the mathematical content coincides with Euclid's Proposition I.4, the logical structure of the proof is fundamentally different.

4.2 The Classical Configuration

Unlike the first three propositions of Book I, Proposition I.4 is not a geometric construction. Instead, it establishes a criterion for the congruence of two triangles.

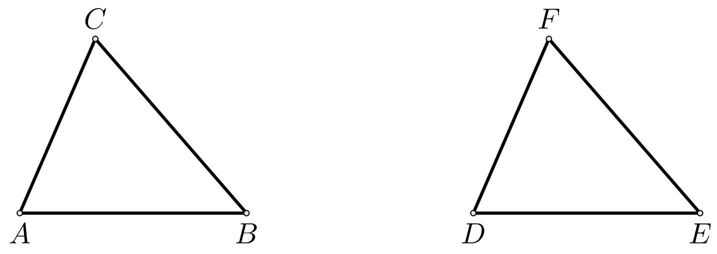
The proposition considers two triangles satisfying three geometric hypotheses:

- two pairs of corresponding sides are congruent;
- the included angles are congruent.

The objective is to prove that the triangles are congruent. As a consequence, the remaining corresponding side and the remaining corresponding angles are also congruent.

4.2.1 Final Configuration

The geometric configuration is shown in Figure 1.



Configuration of Proposition I.4.

The hypotheses are

$$AB \cong DE,$$

$$AC \cong DF,$$

and

$$\angle BAC \cong \angle EDF.$$

The desired conclusions are

$$BC \cong EF,$$

together with

$$\angle ABC \cong \angle DEF,$$

and

$$\angle ACB \cong \angle DFE.$$

4.3 Statement

Proposition I.4.

If two triangles have two corresponding sides congruent and the included angles congruent, then the triangles are congruent.

More precisely, let ABC and DEF be two non degenerate triangles:

$$AB \cong DE,$$

$$AC \cong DF,$$

and

$$\angle BAC \cong \angle EDF.$$

Then

$$BC \cong EF,$$

$$\angle ABC \cong \angle DEF,$$

and

$$\angle ACB \cong \angle DFE.$$

4.4 Proof

Euclid's original proof proceeds by superposition. One triangle is conceived as being placed upon the other in order to establish the coincidence of the corresponding sides and vertices.

The present reconstruction follows a different logical route. Superposition is not part of Hilbert's axiomatic system and therefore does not appear in the formal development.

Instead, the Side–Angle–Side congruence theorem has already been established within the Hilbert theory.

Applying the SAS theorem to the hypotheses

$$AB \cong DE,$$

$$AC \cong DF,$$

and

$$\angle BAC \cong \angle EDF$$

immediately yields the congruence of the two triangles.

Consequently,

$$BC \cong EF,$$

$$\angle ABC \cong \angle DEF,$$

and

$$\angle ACB \cong \angle DFE.$$

This is precisely the conclusion of Proposition I.4.

□

4.5 Proof Architecture

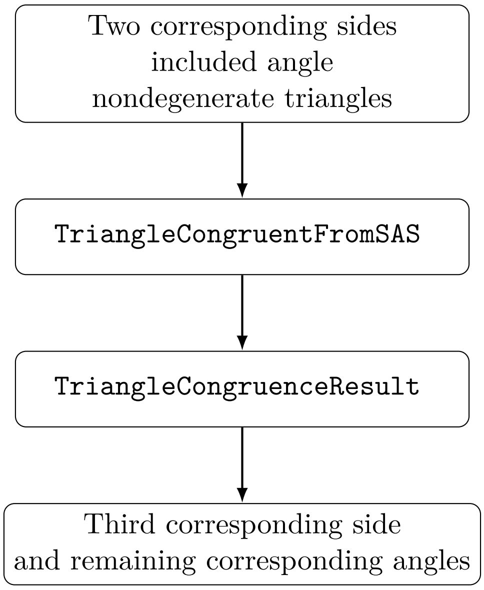


Figure 4.1: Logical structure of Proposition I.4 in the Hilbert reconstruction.

4.6 Hilbert Reconstruction

The principal difference between Euclid's original argument and the present reconstruction concerns the method of proof rather than the mathematical statement.

Euclid establishes Proposition I.4 by superposition, treating one triangle as if it could be placed upon the other. Although historically important, this method is not available within the Hilbert axiomatic framework.

Instead, the present development relies on the Side–Angle–Side congruence theorem, which has already been established as part of the Hilbert theory.

Once the hypotheses of the SAS theorem have been established, an application of `TriangleCongruentFromSAS` immediately yields the complete triangle congruence.

Thus the classical proposition is reconstructed without appealing to superposition while preserving exactly the same mathematical content.

4.7 Lean Formalization

The Lean theorem is a direct application of the previously established Hilbert SAS theorem, exposed through the compatibility theorem `TriangleCongruentFromSAS`.

In addition to the two side congruences and the congruence of the included angles, the formal statement explicitly records that both triples of vertices are noncollinear.

The theorem returns a value of type `TriangleCongruenceResult`. This structure contains the congruence of all three corresponding sides and all three corresponding angles.

Thus the formal proof of Proposition I.4 consists of a single application of `TriangleCongruentFromSAS`; no additional geometric argument is required.

From the viewpoint of formalization, Proposition I.4 is also the first result whose Lean proof is essentially a single application of a previously established interface theorem.

Chapter 5

Proposition I.5

5.1 Introduction

Proposition I.5 is the first theorem of Book I devoted specifically to the geometry of isosceles triangles. It establishes that the angles at the base of an isosceles triangle are congruent. Moreover, if the equal sides are extended beyond the base, the exterior angles formed by these extensions are also congruent.

In the Elements this proposition is one of the best known results of classical geometry and later became known under the traditional name *pons asinorum*. Although the theorem itself is elementary, it marks the beginning of a systematic study of congruence properties of triangles.

Euclid proves the proposition by combining the congruence theorem of Proposition I.4 with auxiliary constructions on the extensions of the equal sides.

In the present reconstruction the logical organization is different. The theorem is obtained by composing two previously established results of the Hilbert interface. The congruence of the base angles is provided by the theorem on isosceles triangles, while the congruence of the exterior angles follows from Hilbert's theorem on adjacent angles.

Consequently, Proposition I.5 is the first example in which a Euclidean proposition is reconstructed by combining several independent interface theorems rather than by introducing new geometric techniques.

5.2 The Classical Configuration

Euclid's proof of Proposition I.5 does not use only the original isosceles triangle and the extensions of its equal sides. It introduces additional auxiliary points and joins, producing a larger configuration to which Proposition I.4 can be applied.

Let ABC be an isosceles triangle satisfying

$$AB \cong AC.$$

Produce the equal sides AB and AC beyond the base vertices B and C. Euclid then chooses an auxiliary point on one extension, transfers a corresponding segment to the other extension, and joins the resulting points to the opposite vertices of the base.

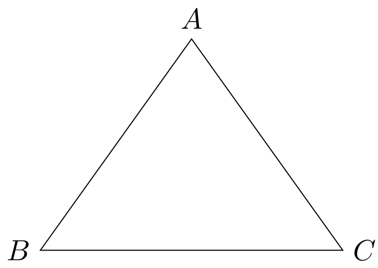
The following sequence shows the successive stages of Euclid's configuration.

5.2.1 Initial Isosceles Triangle

The construction begins with an isosceles triangle ABC satisfying

$$AB \cong AC.$$

At this stage no auxiliary points or segments have yet been introduced.



$AB \cong AC$

Figure 5.1: The initial isosceles triangle.

5.2.2 Extension of the Equal Sides

Euclid next produces the equal sides AB and AC beyond the base vertices. Let D lie beyond B on the line AB, and let E lie beyond C on the line AC.

Thus

$$A - B - D, \qquad A - C - E.$$

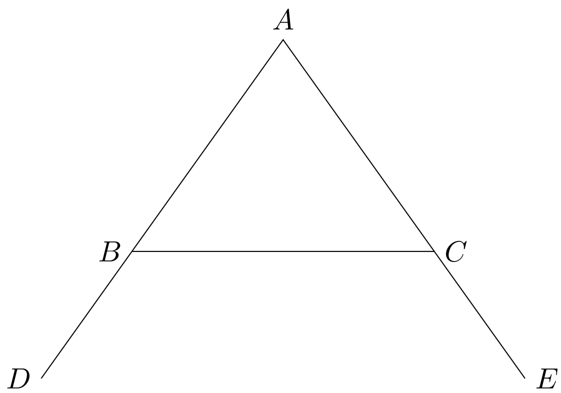


Figure 5.2: The equal sides AB and AC produced beyond B and C.

5.2.3 Choice of an Auxiliary Point

Euclid now chooses an arbitrary point F on the extension BD of the side AB.

Thus

$$A - B - F,$$

with F lying on the ray from A through B beyond the base vertex B.

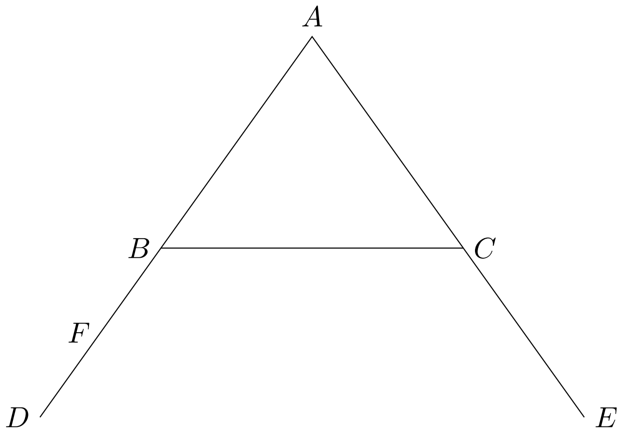


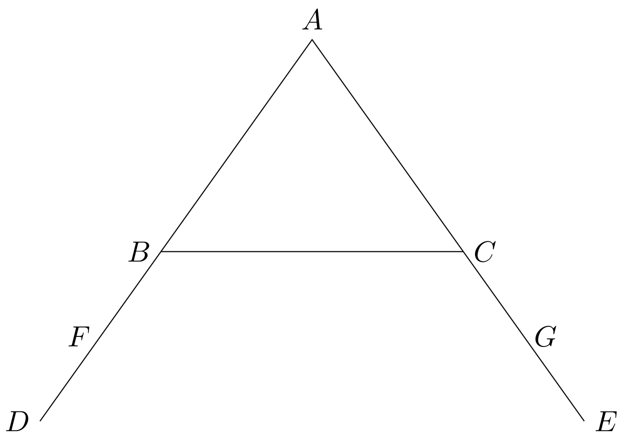
Figure 5.3: An arbitrary point F is chosen on the extension BD.

5.2.4 Construction of an Equal Segment

Using Proposition I.3, Euclid marks a point G on the extension of AC such that

$$AG \cong AF.$$

The two auxiliary points are now related by the equality of the segments measured from the common vertex A.



$$AF \cong AG$$

Figure 5.4: Construction of the equal segment $AG = AF$ using Proposition I.3.

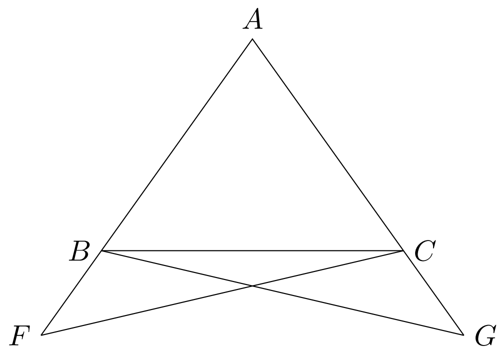
5.2.5 Auxiliary Joins

The final step of Euclid's construction is to join the newly introduced points to the opposite vertices of the base.

The segments

$$FC \quad \text{and} \quad GB$$

complete the auxiliary configuration required for the proof.



$$AF \cong AG$$

Figure 5.5: The complete auxiliary configuration used in Euclid's proof.

5.3 Statement

Let ABC be a nondegenerate isosceles triangle satisfying

$$AB \cong AC.$$

Let the equal sides be extended beyond the base vertices so that

$$A - B - D, \qquad A - C - E.$$

Then

$$\angle ABC \cong \angle ACB,$$

and

$$\angle CBD \cong \angle BCE.$$

Thus both the base angles of the isosceles triangle and the exterior angles formed after extending the equal sides are congruent.

5.4 Proof

Euclid's proof is considerably more elaborate than the modern reconstruction presented later in this chapter. Rather than reasoning directly from the properties of an isosceles triangle, he introduces an auxiliary construction that allows repeated applications of the Side– Angle–Side congruence theorem established in Proposition I.4.

After extending the equal sides of the triangle, an arbitrary point is chosen on one extension. Proposition I.3 is then used to construct an equal segment on the opposite extension. The newly obtained points are joined to the opposite vertices of the base, completing the auxiliary configuration.

The proof proceeds by applying Proposition I.4 twice. The first application establishes congruence relations between the auxiliary triangles and yields equal corresponding sides and angles. These newly obtained equalities provide the hypotheses required for a second application of Proposition I.4.

Finally, Euclid combines the resulting congruence relations with the elementary properties of equal segments and equal angles to conclude that the interior base angles of the original isosceles triangle are congruent. The same argument immediately establishes the congruence of the corresponding exterior angles formed after extending the equal sides.

The auxiliary construction is therefore not an end in itself. Its sole purpose is to transform Proposition I.5 into two successive applications of Proposition I.4, allowing the theorem to be derived using only previously established geometric results.

5.5 Implementation Architecture

The formal proof of Proposition I.5 is obtained by composing two previously established theorems of the Hilbert interface.

The first theorem establishes the congruence of the base angles of an isosceles triangle. The second theorem transfers this congruence to the adjacent exterior angles after the equal sides have been extended.

The resulting logical dependency is shown below.

Unlike the classical proof of Euclid, the formal development does not repeat the geometric argument establishing the base-angle theorem. Instead, the proposition is reconstructed by composing two independent results already available in the Hilbert interface. Consequently, Proposition I.5 illustrates the modular nature of the library: new Euclidean propositions are obtained by combining previously verified synthetic theorems.

5.6 Hilbert Reconstruction

The reconstruction of Proposition I.5 differs significantly from the classical proof presented in the *Elements*. Rather than repeating Euclid's geometric argument, the result is obtained by combining two previously established theorems of the Hilbert interface.

The first theorem,

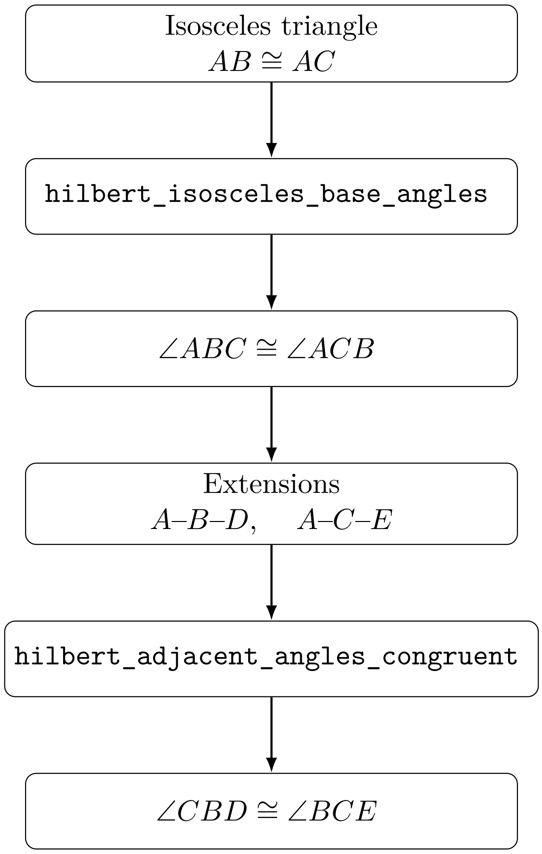


Figure 5.6: Implementation architecture of Proposition I.5.

`hilbert_isosceles_base_angles`,

states that the base angles of a nondegenerate isosceles triangle are congruent. Applied to the hypothesis

$$AB \cong AC,$$

it immediately yields

$$\angle ABC \cong \angle ACB.$$

The second theorem,

`hilbert_adjacent_angles_congruent`,

uses the betweenness relations describing the extensions of the equal sides to transfer this congruence to the corresponding exterior angles.

Consequently,

$$\angle CBD \cong \angle BCE.$$

The reconstruction therefore introduces no new geometric machinery. Proposition I.5 is obtained entirely by composing two fundamental results already available in the Hilbert interface.

5.7 Lean Formalization

The Lean implementation follows the mathematical reconstruction directly.

The theorem first applies `hilbert_isosceles_base_angles` to obtain the congruence of the base angles of the isosceles triangle.

The resulting angle congruence is then supplied to `hilbert_adjacent_angles_congruent`, together with the two betweenness hypotheses describing the extensions of the equal sides.

The theorem therefore returns both conclusions of Proposition I.5:

$$\angle ABC \cong \angle ACB,$$

and

$$\angle CBD \cong \angle BCE.$$

Unlike Proposition I.4, whose formal proof consists of a single application of the SAS theorem, Proposition I.5 is the first Euclidean proposition reconstructed as a composition of multiple interface theorems.

5.8 Discussion

Proposition I.5 marks an important transition in the reconstruction of Book I.

The earlier propositions are centred on elementary constructions and a single application of the SAS theorem. Beginning with Proposition I.5, most subsequent propositions of Book I are expected to be reconstructed by composing previously established interface theorems.

This modular organization differs substantially from the presentation of the *Elements*. Euclid develops each proposition as an independent geometric argument, whereas the present reconstruction emphasizes reusable formal components. The resulting proofs are shorter, their logical dependencies are explicit, and each proposition becomes a thin verification layer built upon the synthetic theory already developed in the Hilbert interface.

Chapter 6

Proposition I.6

6.1 Introduction

Proposition I.6 establishes the converse of the preceding proposition. Where Proposition I.5 proves that equal sides of a triangle imply equal base angles, Proposition I.6 shows that equal base angles imply the equality of the opposite sides.

This result completes the fundamental correspondence between the sides and angles of an isosceles triangle. Together, Propositions I.5 and I.6 establish the equivalence between the two defining properties of an isosceles triangle.

Euclid's original proof is considerably more involved than the modern reconstruction. It proceeds by contradiction, introducing an auxiliary construction obtained from Proposition I.3 and ultimately deriving an impossible configuration by repeated use of Proposition I.4.

In the Hilbert reconstruction developed in this project, the theorem admits a substantially shorter proof. The previously established Angle–Side–Angle theorem is sufficient to derive the equality of the opposite sides directly, eliminating the auxiliary construction and the reductio ad absurdum argument required by Euclid.

6.2 The Classical Construction

Euclid's proof begins with a triangle whose base angles are assumed to be equal. The desired conclusion is that the opposite sides are also equal.

Instead of arguing directly, Euclid assumes that the two sides are unequal and derives a contradiction. The proof therefore combines an auxiliary construction with a reductio ad absurdum argument.

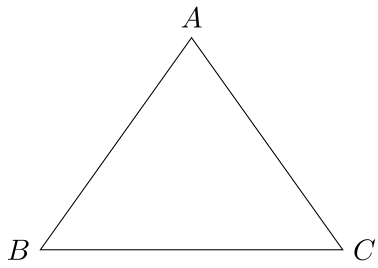
The following sequence illustrates the successive stages of Euclid's construction.

6.2.1 Initial Triangle

Let ABC be a triangle satisfying

$$\angle ABC \cong \angle ACB.$$

At this stage no auxiliary construction has yet been introduced.



$$\angle ABC \cong \angle ACB$$

Figure 6.1: The initial triangle with equal base angles.

6.2.2 Assuming Unequal Sides

To obtain a contradiction, Euclid assumes that the two sides opposite the equal angles are not equal. Without loss of generality, suppose

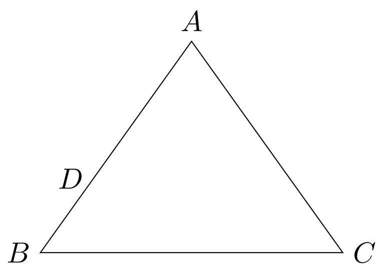
$$AB > AC.$$

Using Proposition I.3, a point D is chosen on the segment AB so that

$$DB \cong AC.$$

Consequently,

$$A - D - B.$$



$$A - D - B, \qquad DB \cong AC$$

Figure 6.2: Assuming $AB > AC$, Proposition I.3 constructs D on AB with $DB = AC$.

6.2.3 The Auxiliary Triangle

The auxiliary point is joined to the remaining vertex of the original triangle.

Since

$$DB \cong AC,$$

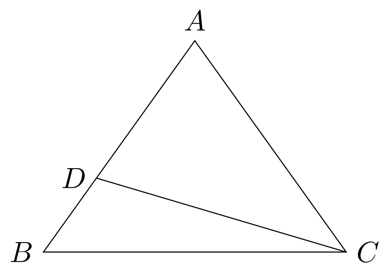
and the angle

$$\angle DBC$$

coincides with the original angle

$$\angle ABC,$$

the auxiliary triangle is prepared for an application of Proposition I.4.



$$DB \cong AC$$

Figure 6.3: The auxiliary triangle DBC obtained after joining D to C.

6.2.4 The Contradiction

Applying Proposition I.4 to the triangles DBC and ACB establishes their congruence. Consequently, the corresponding angles and sides are equal.

Since the point D lies between A and B, one of the resulting equalities contradicts the assumption that the whole segment AB is strictly greater than its proper part DB.

The contradiction shows that the assumption

$$AB > AC$$

is impossible. By symmetry, neither side can be greater than the other, and therefore the two sides are equal.

6.3 Statement

Theorem 1 (Euclid I.6)**.** *Let ABC be a nondegenerate triangle.*

If

$$\angle ABC \cong \angle ACB,$$

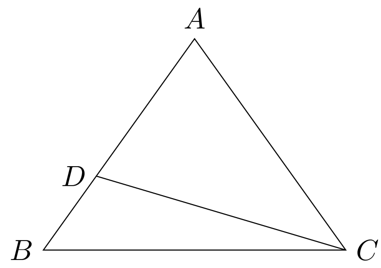


Figure 6.4: The auxiliary configuration from which Proposition I.4 forces a contradiction with the assumption $AB > AC$.

then

$$AB \cong AC.$$

That is, equal angles in a triangle subtend equal opposite sides.

6.4 Proof

Euclid proves the theorem by contradiction.

Assume that the sides opposite the equal angles are unequal. Without loss of generality, suppose that AB is greater than AC. By Proposition I.3, a point is constructed on the longer side so that the remaining segment is equal to the shorter side.

Joining the auxiliary point to the remaining vertex produces a second triangle. Proposition I.4 is then applied to show that the auxiliary triangle is congruent to the original one.

The resulting congruence forces equalities that are incompatible with the assumption that one side is strictly longer than the other. Since the assumption leads to a contradiction, the two sides must be equal.

Thus equal angles in a triangle necessarily subtend equal opposite sides.

6.5 Implementation Architecture

The formal proof follows a substantially different route from Euclid's classical argument. Rather than introducing an auxiliary point, constructing additional triangles, and reasoning by contradiction, the implementation exploits a higher-level theorem already available in the Hilbert interface.

The proof compares the given triangle with itself after exchanging the roles of the two equal angles. Since the common side is trivially congruent to itself, the derived Angle–Side–Angle theorem immediately implies the equality of the remaining sides.

Consequently, the entire classical construction disappears from the formal proof and is replaced by a single application of a previously established congruence theorem.

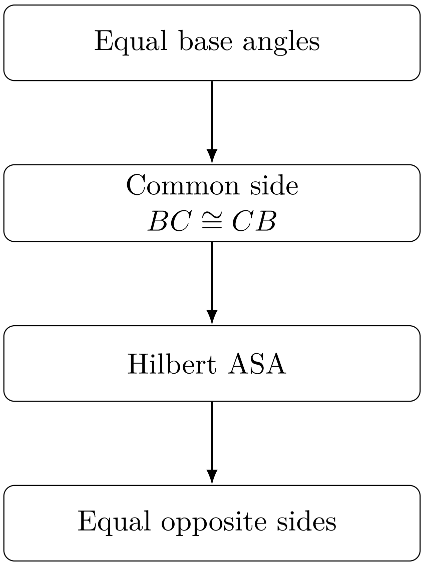


Figure 6.5: Architecture of the formal reconstruction of Proposition I.6.

6.6 Hilbert Reconstruction

The formal reconstruction replaces Euclid's indirect argument by a direct consequence of the developed Hilbert theory.

The key observation is that the given triangle can be compared with itself after exchanging the two equal base angles. The side BC is common to both descriptions of the triangle and is therefore congruent to itself.

The hypotheses of the derived Angle–Side–Angle theorem are thus immediately satisfied:

$$\triangle BCA \quad \longleftrightarrow \quad \triangle CBA.$$

Applying the Hilbert ASA theorem yields the congruence of the remaining corresponding sides,

$$BA \cong CA,$$

which is equivalent to

$$AB \cong AC.$$

Unlike Euclid's original proof, no auxiliary construction, no appeal to Proposition I.3, and no proof by contradiction are required. The entire argument is reduced to a single application of a previously established triangle congruence theorem.

6.7 Lean Formalization

The Lean implementation closely follows the Hilbert reconstruction.

The proof first rewrites the given triangle by exchanging the roles of its equal base angles. The common side BC is identified with itself using reflexivity of segment congruence, while standard symmetry properties convert the supplied angle congruence into the orientation required by the Angle–Side–Angle theorem.

A single application of the derived theorem `hilbert_asa_sides` then establishes the congruence of the remaining sides. Finally, elementary symmetry of segment congruence rewrites the result into the statement of Proposition I.6.

Compared with Euclid's original argument, the formal proof is substantially shorter because it relies on previously developed interface theorems rather than reconstructing the auxiliary geometric configuration.

6.8 Discussion

Proposition I.6 provides one of the clearest examples of the difference between a classical geometric proof and its modern formal reconstruction.

Euclid establishes the theorem by introducing an auxiliary point, constructing an additional triangle, and arguing by contradiction. The proof occupies a central place in Book I because it demonstrates how previous propositions can be combined to derive a converse statement.

Within the Hilbert framework, the same mathematical result becomes an almost immediate consequence of the previously established Angle–Side–Angle theorem. Once ASA has been derived from Hilbert's axioms, the converse property of isosceles triangles follows directly, without auxiliary constructions or reductio ad absurdum.

This proposition therefore illustrates an important principle of the CGJteamLab reconstruction: as the interface theory grows, increasingly complex classical arguments collapse into short applications of high-level synthetic theorems.

Chapter 7

Proposition I.7

7.1 Introduction

Proposition I.7 occupies a special position in Book I of Euclid's *Elements*. Unlike the preceding propositions, its principal purpose is not to establish a new geometric construction or a direct property of triangles, but to prove a uniqueness principle.

Euclid shows that, on the same side of a given base, there cannot exist two distinct vertices determining triangles whose corresponding sides are equal. This uniqueness result immediately precedes Proposition I.8, where it is used as a crucial ingredient in the classical proof of the Side–Side–Side congruence theorem.

The Hilbert reconstruction follows a different logical order. The general Side–Side–Side theorem has already been established within the Hilbert interface. Consequently, Proposition I.7 becomes a concise uniqueness argument derived from the developed congruence theory, without reproducing Euclid's original chain of auxiliary arguments.

This proposition therefore provides one of the clearest examples of the difference between the historical organization of Euclid's *Elements* and the logical organization of a modern axiomatic development.

7.1.1 Initial Configuration

Let AB be a given finite straight line and let C and D be two points lying on the same side of the line AB.

Assume that

$$AC \cong AD$$

and

$$BC \cong BD.$$

The proposition asks whether the two vertices C and D can be distinct under these assumptions.

Euclid's objective is to prove that such a configuration is impossible.

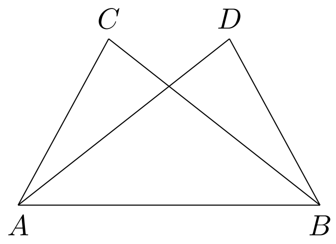


Figure 7.1: The assumed existence of two distinct vertices on the same side of the base AB.

7.1.2 The Assumed Second Vertex

Suppose that the two vertices are distinct.

Both triangles share the same base AB, and their corresponding sides are assumed to satisfy

$$AC \cong AD, \qquad BC \cong BD.$$

Thus two different triangles would have the same prescribed side lengths while lying on the same side of the base.

Euclid proceeds by showing that this assumption inevitably leads to a contradiction.

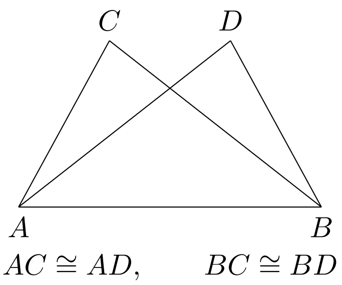


Figure 7.2: The two assumed triangles have the same base and equal corresponding sides.

7.1.3 The Contradictory Configuration

Euclid joins the two assumed vertices by the segment CD.

Since

$$AC \cong AD,$$

the triangle ACD is isosceles. By Proposition I.5,

$$\angle ACD \cong \angle CDA.$$

From the geometric position of the points, one of these angles is strictly greater than the corresponding angle formed with the base vertex B.

On the other hand, since

$$BC \cong BD,$$

the triangle BCD is also isosceles. Proposition I.5 therefore gives

$$\angle BCD \cong \angle CDB.$$

The same pair of angles is thus forced to be both equal and unequal, which is impossible. Hence the two assumed vertices cannot be distinct.

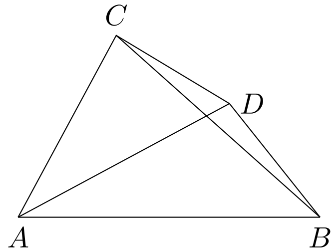


Figure 7.3: Euclid joins the two assumed vertices C and D and applies Proposition I.5 to the isosceles triangles ACD and BCD.

7.2 Statement

Theorem 2 (Euclid I.7)**.** *Let AB be a given base, and let C and D lie on the same side of the line AB.*

Assume

$$AC \cong AD$$

and

$$BC \cong BD.$$

Then

$$C = D.$$

Equivalently, on a fixed side of a given base there is at most one point having prescribed distances from the two endpoints of that base.

7.3 Proof

Assume, for contradiction, that C and D are distinct.

Join C to D. Since

$$AC \cong AD,$$

the triangle ACD is isosceles. By Proposition I.5, its base angles are congruent:

$$\angle ACD \cong \angle CDA.$$

Similarly, since

$$BC \cong BD,$$

the triangle BCD is isosceles, and Proposition I.5 gives

$$\angle BCD \cong \angle CDB.$$

The position of the points on the same side of the base forces one of these angles to contain the other as a proper part. Hence an angle would have to be congruent to one of its proper parts, which is impossible.

Therefore the assumption that C and D are distinct leads to a contradiction. Hence

$$C = D.$$

Thus, on a fixed side of a given base, a point is uniquely determined by its distances from the two endpoints of the base.

7.4 Implementation Architecture

The formal proof bears little resemblance to Euclid's original argument.

Instead of constructing auxiliary triangles and reasoning by contradiction, the implementation first applies the already established Side–Side–Side theorem of the Hilbert interface. This immediately shows that the two candidate triangles are congruent.

From the resulting congruence, the corresponding angle at the common base vertex is obtained. The uniqueness of angle construction then forces both candidate vertices to lie on the same ray issued from the base vertex.

Finally, since the two points lie on the same ray and are at the same distance from the common endpoint, the uniqueness of segment construction implies that the two points coincide.

7.5 Hilbert Reconstruction

The Hilbert reconstruction follows a completely different line of reasoning from Euclid's original proof.

The three assumed side congruences are first combined with the nondegeneracy of one of the triangles. The previously established Hilbert Side–Side–Side theorem immediately yields

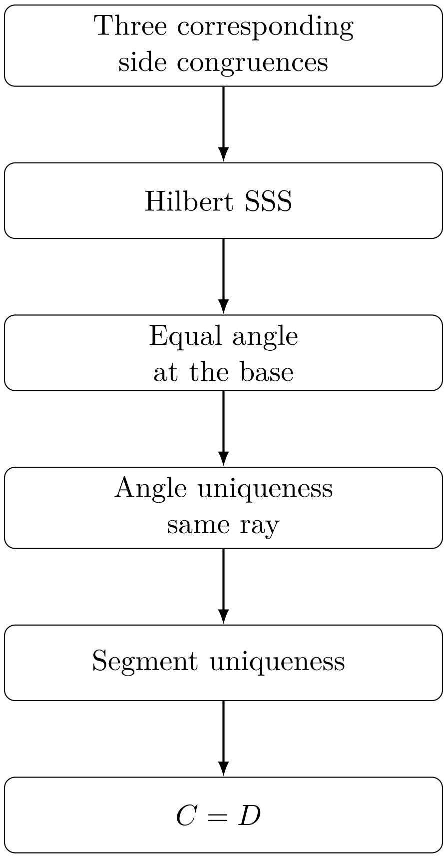


Figure 7.4: Architecture of the Hilbert reconstruction of Proposition I.7.

congruence of the two triangles and, in particular, equality of the corresponding angles at the common base vertex.

Since both candidate vertices lie on the same side of the base, the uniqueness theorem for angle construction implies that they determine the same ray issuing from the base endpoint.

Finally, the two points lie on the same ray and are at the same distance from the common endpoint. The uniqueness of segment construction therefore forces the two points to coincide.

Thus the uniqueness statement of Proposition I.7 is obtained directly from the developed Hilbert theory, without reproducing Euclid's auxiliary geometric argument.

7.6 Lean Formalization

The Lean proof mirrors the Hilbert reconstruction almost step by step.

The proof first derives nondegeneracy of the two triangles from the assumption that the candidate vertices lie on the same side of the given base.

A single application of the theorem `HilbertSSS` establishes congruence of the two triangles and yields equality of the corresponding base angles.

The uniqueness theorem for angle construction then shows that both candidate vertices determine the same ray from the common base vertex. Finally, the uniqueness of segment construction implies that two points lying on the same ray at the same distance from the origin must coincide.

The complete formal proof is therefore a short composition of high-level interface theorems and contains no explicit geometric construction.

7.7 Discussion

Proposition I.7 illustrates a fundamental difference between the historical development of Euclid's geometry and its modern axiomatic reconstruction.

In the *Elements*, Proposition I.7 serves as an essential lemma for the proof of the Side–Side–Side congruence theorem in Proposition I.8. Euclid therefore establishes the uniqueness result before proving the general congruence theorem.

The Hilbert reconstruction follows the opposite logical direction. Once the general Side–Side–Side theorem has been derived from the Hilbert axioms, Proposition I.7 becomes an immediate uniqueness corollary obtained by combining triangle congruence with the uniqueness of angle and segment constructions.

This proposition therefore demonstrates that the historical order of the *Elements* need not coincide with the logical organization of a modern formal theory. As the interface develops, results that were once foundational become concise consequences of more general synthetic theorems.

Chapter 8

Proposition I.8

8.1 Introduction

Proposition I.8 is one of the central results of Book I of Euclid's *Elements*. It establishes the Side–Side–Side criterion for triangle congruence, proving that triangles having three corresponding sides equal must also have their corresponding angles equal.

Within Euclid's development, this proposition concludes a carefully prepared sequence of results. Proposition I.7 provides the essential uniqueness argument that allows the final congruence theorem to be established.

The Hilbert reconstruction follows a different logical organization. The general Side–Side–Side theorem has already been derived from the Hilbert axioms before the Euclidean propositions are reconstructed. Consequently, Proposition I.8 becomes a direct application of this general theorem rather than the culmination of a long geometric argument.

This proposition therefore marks the point where the historical proof of Euclid and the modern Hilbert reconstruction differ most significantly while establishing exactly the same mathematical result.

8.1.1 Initial Configuration

Consider two triangles,

$$\triangle ABC \quad \text{and} \quad \triangle DEF,$$

whose corresponding sides satisfy

$$AB \cong DE,$$

$$BC \cong EF,$$

and

$$AC \cong DF.$$

The objective is to prove that the corresponding angles of the two triangles are also congruent.

Figure 8.1: Two triangles with three corresponding sides congruent.

8.1.2 Superposition

Euclid conceptually places one triangle upon the other so that the vertex D coincides with A and the side DE lies along the side AB.

Since

$$AB \cong DE,$$

the endpoint E coincides with B.

The remaining vertex F must then lie at the intersection of two circles centered at A and B with radii equal to the corresponding sides of the first triangle.

Proposition I.7 shows that, on a fixed side of the base, such a point is unique. Consequently, the vertex F must coincide with C.

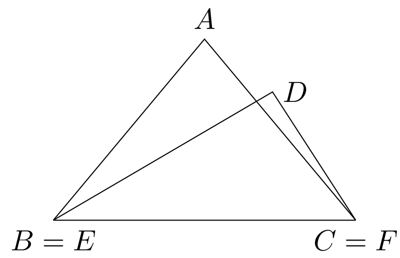


Figure 8.2: After superposition, the corresponding side $BC = EF$ coincides. If the remaining vertices A and D were distinct, Proposition I.7 would exclude this configuration.

8.1.3 Application of Proposition I.7

Once the triangles have been superposed, the only remaining possibility is that the third vertices do not coincide.

In that case there would exist two distinct points on the same side of the common base satisfying the same two distance conditions.

Proposition I.7 excludes precisely this situation.

Therefore the remaining vertices must also coincide, and the two triangles agree completely. Consequently, all corresponding angles are congruent.

8.2 Statement

Theorem 3 (Euclid I.8). *Let*

$$\triangle ABC \quad \text{and} \quad \triangle DEF$$

be nondegenerate triangles.

Assume

$$AB \cong DE,$$

$$BC \cong EF,$$

and

$$AC \cong DF.$$

Then

$$\angle BAC \cong \angle EDF,$$

$$\angle ABC \cong \angle DEF,$$

and

$$\angle ACB \cong \angle DFE.$$

That is, triangles having three corresponding sides congruent also have their corresponding angles congruent.

8.3 Proof

Superpose the triangle ABC onto the triangle DEF so that the vertex B coincides with E and the side BC lies along the side EF.

Since

$$BC \cong EF,$$

the point C coincides with F.

Suppose that the remaining vertex A does not coincide with D. Then there would exist two distinct points on the same side of the base EF satisfying

$$EA \cong ED$$

and

$$FA \cong FD.$$

This contradicts Proposition I.7, which asserts the uniqueness of such a point.

Therefore the vertex A must coincide with D, and the two triangles coincide completely.

Hence all corresponding angles are congruent:

$$\angle BAC \cong \angle EDF,$$

$$\angle ABC \cong \angle DEF,$$

and

$$\angle ACB \cong \angle DFE.$$

This establishes the Side–Side–Side congruence theorem.

8.4 Implementation Architecture

The formal reconstruction is dramatically simpler than Euclid's classical proof.

No superposition argument is required. Once the three corresponding side congruences are available, the previously established Side–Side–Side theorem of the Hilbert interface immediately provides the congruence of the two triangles together with the congruence of all corresponding angles.

The entire classical argument is therefore replaced by a single application of a high-level interface theorem.

8.5 Hilbert Reconstruction

The Hilbert reconstruction consists of a direct application of the general Side–Side–Side theorem.

The three corresponding side congruences, together with the nondegeneracy of one triangle, satisfy exactly the hypotheses of the derived Hilbert theorem.

Consequently, the theorem immediately yields congruence of the two triangles and, in particular, congruence of all corresponding angles.

Unlike Euclid's original proof, no superposition argument and no uniqueness reasoning are required. These ideas have already been absorbed into the previously established Hilbert theory.

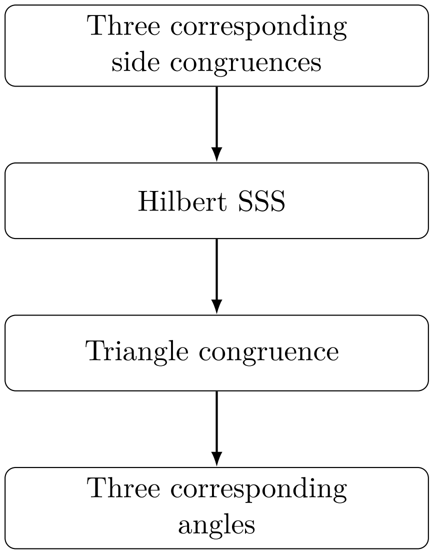


Figure 8.3: Architecture of the Hilbert reconstruction of Proposition I.8.

8.6 Lean Formalization

The Lean implementation follows the Hilbert reconstruction directly.

After assuming nondegeneracy of the first triangle and congruence of the three corresponding sides, the proof consists of a single application of the theorem `HilbertSSS`.

This theorem returns a complete triangle congruence result, from which the congruence of the three corresponding angles is obtained by projecting the appropriate fields of the resulting structure.

The formal proof therefore contains no auxiliary constructions, no superposition argument, and no separate uniqueness reasoning. These steps have already been incorporated into the previously established Hilbert interface.

8.7 Discussion

Proposition I.8 concludes the first major block of triangle congruence theory developed in Book I.

Historically, Euclid establishes the Side–Side–Side criterion only after proving the uniqueness result of Proposition I.7 and employing the method of superposition. The classical proof therefore depends on a carefully ordered sequence of earlier propositions.

The Hilbert reconstruction follows a different logical organization. The general Side–Side–Side theorem has already been established as a theorem of the Hilbert interface. Consequently, Proposition I.8 becomes a direct application of this previously developed result, requiring no additional geometric construction.

Together, Propositions I.4 through I.8 illustrate the progressive development of triangle congruence theory. The classical arguments of Euclid are systematically replaced by applications of increasingly powerful synthetic theorems derived from the Hilbert axioms.

This transition marks the completion of the first coherent layer of the CGJteamLab Hilbert interface and demonstrates how a modern axiomatic framework can substantially simplify the formal verification of classical geometry.

Chapter 9

Proposition I.9

9.1 Introduction

Proposition I.9 presents Euclid's construction of the bisector of a given rectilinear angle.

The classical proof combines several earlier propositions. A segment of one side of the angle is first copied onto the other side, after which an auxiliary equilateral triangle is constructed. The Side–Side–Side criterion established in Proposition I.8 is then used to prove that the joining segment bisects the angle.

The Hilbert reconstruction follows a different route. Rather than constructing an auxiliary equilateral triangle, it first lays off a segment congruent to one side of the angle, constructs the midpoint of the resulting segment, and finally applies the general Side–Side–Side theorem.

Although the geometric constructions differ substantially, both approaches establish the existence of an angle bisector within neutral geometry.

9.1.1 Initial Configuration

Let $\angle BAC$ be the given rectilinear angle.

The objective is to construct a ray issuing from the vertex A that divides the angle into two congruent angles.

No assumption is made on the lengths of the sides forming the angle; only the angle itself is given.

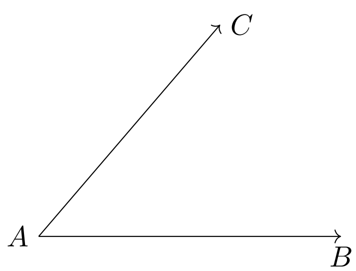


Figure 9.1: The given angle $\angle BAC$.

9.1.2 Construction

Choose an arbitrary point D on the side AB.

On the side AC, construct a point E such that

$$AE \cong AD.$$

Join the points D and E, and construct the equilateral triangle DEF on the segment DE.

Finally, join the vertex A to the vertex F.

The claim is that the segment AF bisects the given angle.

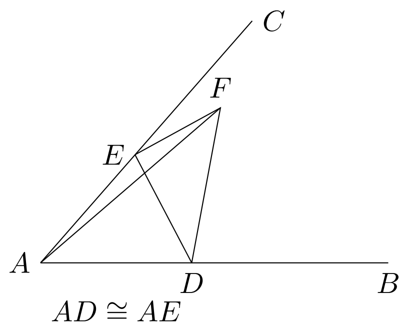


Figure 9.2: Euclid's auxiliary construction: $AD \cong AE$ and DEF is equilateral.

9.1.3 Application of Proposition I.8

Since the triangle DEF is equilateral,

$$DF \cong EF.$$

Moreover,

$$AD \cong AE$$

by construction, while

$$AF \cong AF$$

is common to the two triangles.

Therefore the triangles

$$\triangle ADF \quad \text{and} \quad \triangle AEF$$

have three corresponding sides congruent.

By Proposition I.8,

$$\angle DAF \cong \angle FAE.$$

Since D lies on the ray AB and E lies on the ray AC, these are precisely the two parts of the original angle $\angle BAC$.

Hence AF bisects the given angle.

9.2 Statement

Theorem 4 (Euclid I.9)**.** *Let $\angle BAC$ be a nondegenerate angle.*

Then there exists a point M such that

$$\angle BAM \cong \angle MAC.$$

Equivalently, the ray AM bisects the angle $\angle BAC$.

9.3 Proof

Choose a point D on the side AB and construct a point E on the side AC such that

$$AD \cong AE.$$

Join the points D and E, and construct the equilateral triangle DEF.

Finally, join the vertex A to the vertex F.

Since

$$AD \cong AE,$$

$$DF \cong EF,$$

and

$$AF \cong AF,$$

the triangles

$$\triangle ADF \quad \text{and} \quad \triangle AEF$$

have three corresponding sides congruent.

By Proposition I.8,

$$\angle DAF \cong \angle FAE.$$

Because D lies on the ray AB and E lies on the ray AC, these angles are exactly

$$\angle BAF \quad \text{and} \quad \angle FAC.$$

Hence the ray AF bisects the given angle.

This completes the construction.

9.4 Implementation Architecture

The Hilbert reconstruction replaces Euclid's auxiliary equilateral triangle by two fundamental constructions already available in the Hilbert interface.

First, a segment congruent to one side of the angle is laid off on the other side of the angle. The midpoint of the resulting segment is then constructed, and the Side–Side–Side theorem is applied to two auxiliary triangles.

Thus the existence of the angle bisector follows entirely from previously established neutral-geometric results.

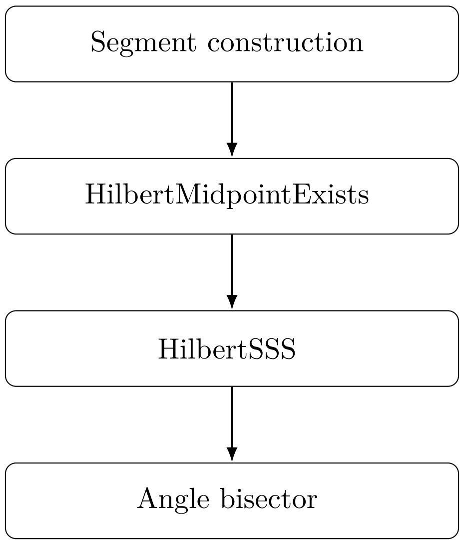


Figure 9.3: Architecture of the Hilbert reconstruction of Proposition I.9.

9.5 Hilbert Reconstruction

Starting from the given angle, a point is first constructed on one ray so that the new segment is congruent to the side on the opposite ray.

The midpoint of the segment joining these two constructed endpoints is then obtained using the neutral midpoint theorem.

Finally, the triangles determined by the midpoint satisfy the hypotheses of the general Side–Side–Side theorem. The resulting angle congruence shows that the segment joining the vertex of the angle to the midpoint is an angle bisector.

The proof therefore avoids both the construction of an equilateral triangle and the explicit application of Euclid's Proposition I.8, replacing them by previously established interface theorems.

9.6 Lean Formalization

The Lean implementation follows the Hilbert reconstruction directly.

Starting from a nondegenerate angle, the proof first constructs a point on one ray whose distance from the vertex equals the length of the opposite side of the angle.

The midpoint of the segment joining the two constructed endpoints is then obtained using the neutral midpoint existence theorem.

Finally, a single application of the general Side–Side–Side theorem establishes the congruence of the two auxiliary triangles. The desired angle congruence follows immediately after identifying the constructed segment with the original ray.

Unlike Euclid's original argument, the formal proof contains no equilateral triangle construction and makes no direct use of Proposition I.8. Instead, it relies entirely on previously established results of the Hilbert interface.

9.7 Discussion

Proposition I.9 is the first construction in Book I whose Hilbert reconstruction differs substantially from Euclid's original proof.

Euclid constructs an auxiliary equilateral triangle and applies the Side–Side–Side criterion established in Proposition I.8. The proof is therefore closely tied to the historical sequence of propositions in the *Elements*.

The Hilbert reconstruction adopts a different strategy. A congruent segment is first laid off on one side of the angle, the midpoint of the resulting segment is constructed, and the general Side–Side–Side theorem is applied directly to two auxiliary triangles.

This approach illustrates a general principle of the CGJteamLab library: classical geometric constructions are not reproduced verbatim when a simpler synthetic argument is available within the existing Hilbert theory.

Proposition I.9 therefore demonstrates how a modern axiomatic development can replace a classical construction by a shorter proof while preserving the same mathematical content.

Chapter 10

Proposition I.10

10.1 Introduction

Proposition I.10 constructs the midpoint of a given finite straight line.

In Euclid's development this construction depends on several earlier results. An equilateral triangle is first erected on the given segment using Proposition I.1. The vertex angle of this triangle is then bisected by Proposition I.9, and the Side–Angle–Side congruence criterion of Proposition I.4 is used to prove that the intersection point is the midpoint of the original segment.

The Hilbert reconstruction follows a fundamentally different approach. The existence of a midpoint has already been established as an independent theorem of the Hilbert interface. Consequently, Proposition I.10 becomes a direct application of this previously derived result.

This proposition therefore provides one of the clearest examples of the difference between Euclid's historical development and a modern axiomatic reconstruction.

10.1.1 Initial Configuration

Let AB be the given finite straight line.

The objective is to construct a point lying on the segment AB that divides it into two congruent parts.

Such a point is called the midpoint of the segment.

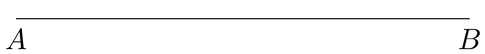


Figure 10.1: The given finite straight line AB.

10.1.2 Construction

Construct the equilateral triangle ABC on the segment AB by Proposition I.1.

Next, bisect the angle $\angle ACB$ by Proposition I.9.

Let the angle bisector meet the segment AB at the point D.

The claim is that D is the midpoint of the segment AB.

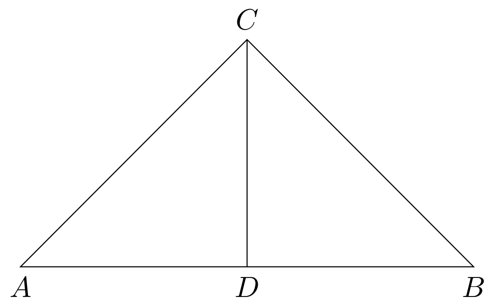


Figure 10.2: Euclid's construction of the midpoint of a segment.

10.1.3 Application of Proposition I.4

Consider the triangles

$$\triangle ACD \quad \text{and} \quad \triangle BCD.$$

Since the triangle ABC is equilateral,

$$AC \cong BC.$$

The segment

$$CD$$

is common to both triangles, and by construction,

$$\angle ACD \cong \angle DCB.$$

Therefore the hypotheses of Proposition I.4 are satisfied.

It follows that

$$AD \cong DB.$$

Since the point D lies on the segment AB, it is the midpoint of the given finite straight line.

10.2 Statement

Theorem 5 (Euclid I.10)**.** *Let AB be a finite straight line with*

$$A \neq B.$$

Then there exists a midpoint of the segment AB. That is, there exists a point M such that

$$A, M, B$$

are collinear,

$$M$$

lies between A and B, and

$$AM \cong MB.$$

Equivalently, every nondegenerate segment admits a midpoint.

10.3 Proof

Construct the equilateral triangle ABC on the segment AB by Proposition I.1.

Bisect the angle $\angle ACB$ by Proposition I.9, and let the bisecting ray meet the segment AB at the point D.

Since

$$AC \cong BC,$$

$$CD \cong CD,$$

and

$$\angle ACD \cong \angle DCB,$$

the triangles

$$\triangle ACD \quad \text{and} \quad \triangle BCD$$

satisfy the hypotheses of Proposition I.4.

Hence

$$AD \cong DB.$$

Because the point D lies on the segment AB, it is the midpoint of the given finite straight line.

This completes the construction.

10.4 Implementation Architecture

The Hilbert reconstruction replaces the entire classical construction by a previously established theorem of the Hilbert interface.

Rather than constructing an auxiliary equilateral triangle, bisecting an angle, and applying the Side–Angle–Side criterion, the existence of a midpoint is obtained directly from the general midpoint existence theorem.

Consequently, Proposition I.10 becomes a simple wrapper around an existing interface theorem.

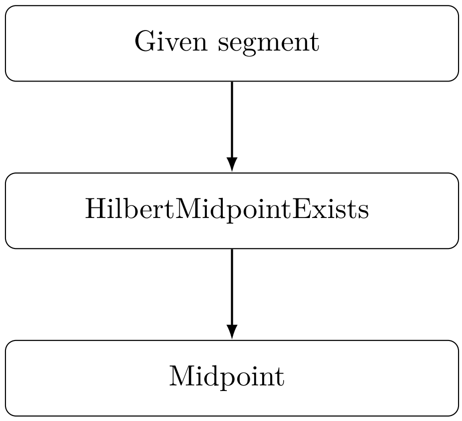


Figure 10.3: Architecture of the Hilbert reconstruction of Proposition I.10.

10.5 Hilbert Reconstruction

The reconstruction consists of a direct application of the general midpoint existence theorem.

Given a nondegenerate segment, the theorem `HilbertMidpointExists` immediately produces a point lying between the endpoints and equidistant from them.

No auxiliary geometric construction is required, since the existence of midpoints has already been established independently within the Hilbert development.

This represents one of the largest simplifications obtained by the layered architecture of the Hilbert interface.

10.6 Lean Formalization

The Lean implementation mirrors the Hilbert reconstruction exactly.

After assuming that the endpoints of the given segment are distinct, the proof consists of a single application of the theorem

`HilbertMidpointExists`.

This theorem immediately returns a point lying between the endpoints and equidistant from them, precisely satisfying the definition of a midpoint.

Consequently, the formal proof contains no auxiliary constructions and makes no explicit use of Propositions I.1, I.4, or I.9. All of the classical geometric reasoning has already been incorporated into the previously established Hilbert interface.

10.7 Discussion

Proposition I.10 provides one of the clearest illustrations of the difference between Euclid's historical development and a layered axiomatic reconstruction.

In the *Elements*, the midpoint construction depends on three earlier propositions. Euclid first constructs an equilateral triangle, then bisects one of its angles, and finally applies the Side–Angle–Side criterion to prove that the resulting point divides the original segment into two equal parts.

The Hilbert reconstruction is fundamentally different. The existence of midpoints has already been established as an independent theorem of the Hilbert interface. Consequently, Proposition I.10 becomes a direct application of this theorem, requiring no additional geometric construction.

This proposition demonstrates the principal advantage of the layered architecture developed in CGJteamLab. Once sufficiently rich interface theorems have been established, many classical propositions cease to be independent proofs and instead become concise applications of the existing theory.

Proposition I.10 is perhaps the simplest example of this philosophy, reducing an entire classical construction to a single theorem call.

Chapter 11

Proposition I.11

11.1 Euclid's Proposition I.11

Proposition I.11 states:

> To draw a straight line at right angles to a given straight line from a given point on it.

Let AB be the given straight line and let C be the given point on AB.

Euclid's construction proceeds as follows. Choose a point D on AC. By Proposition I.3, construct a point E on the opposite side of C such that

$$CD \cong CE.$$

On the segment DE construct the equilateral triangle DFE by Proposition I.1, and join C to F.

Since

$$CD \cong CE, \qquad DF \cong EF,$$

and CF is common to the two triangles DCF and ECF, Proposition I.8 gives

$$\angle DCF \cong \angle ECF.$$

The points D, C, and E lie on one straight line. Hence the two congruent angles DCF and FCE are adjacent angles formed when the straight line CF stands on the straight line DE.

By Definition 10, each of these angles is therefore a right angle. Consequently, CF is drawn at right angles to the given straight line AB at the given point C.

Thus the required perpendicular has been constructed.

11.2 Statement

The synthetic content of Proposition I.11 is the existence of a perpendicular through a prescribed point of a given straight line.

Theorem 6 (Euclid I.11)**.** *Let A, C, B be three points such that*

$$A * C * B.$$

Then there exists a point X such that the angle ACX is a right angle.

In the Hilbert development, a right angle is represented directly by the Euclidean definition: an angle is right when it is congruent to its adjacent angle obtained by extending one of its sides through the vertex.

More precisely, we define

$$\text{RightAngle}(A, O, B)$$

to mean that there exists a point C such that

$$A * O * C$$

and

$$\angle AOB \cong \angle BOC.$$

Thus Proposition I.11 is represented formally by

$$A * C * B \quad \Longrightarrow \quad \exists X \ \text{RightAngle}(A, C, X).$$

The corresponding Lean definition is

```
def HilbertRightAngle
(Geo : Geometry.Geo)
(A O B : Geo.Point) : Prop :=
exists C : Geo.Point,
Geo.Between A O C /\
Geo.AngleCongruent A O B B O C
```

and the proposition itself has the form

```
theorem euclid_proposition_11
[HilbertCongruence Geo]
(A C B : Geo.Point)
(hACB : Geo.Between A C B) :
exists X : Geo.Point,
HilbertRightAngle Geo A C X
```

Notice that the formal statement does not attempt to encode Euclid's particular construction of the perpendicular. It expresses the geometric result of that construction: through the prescribed point C on the given line there exists a ray CX forming a right angle with the line.

11.3 Proof

Let AB be the given straight line and let C be the prescribed point on it.

Choose a point D on AC. By Proposition I.3, cut off from the ray starting at C in the direction of B a segment CE congruent to CD. Thus

$$CD \cong CE,$$

and the points D, C, E lie on one straight line, with C between D and E.

Construct the equilateral triangle DFE on the segment DE by Proposition I.1, and join C to F. Since the triangle is equilateral,

$$DF \cong EF.$$

Moreover,

$$CF \cong CF$$

by reflexivity.

Consider the triangles

$$\triangle DCF \qquad \text{and} \qquad \triangle ECF.$$

Their three corresponding sides satisfy

$$CD \cong CE, \qquad DF \cong EF, \qquad CF \cong CF.$$

Therefore Proposition I.8 yields

$$\angle DCF \cong \angle ECF.$$

Since D, C, E are collinear and C lies between D and E, the rays CD and CE are opposite rays. Hence

$$\angle DCF \qquad \text{and} \qquad \angle FCE$$

are adjacent angles formed by the straight line CF standing on the straight line DE.

The two adjacent angles are congruent. By Euclid's Definition 10, each of them is therefore a right angle.

Consequently,

$$\angle DCF$$

is a right angle, and the straight line through C and F is perpendicular to the given straight line AB at the prescribed point C.

This proves Proposition I.11.

11.4 Implementation Architecture

The formal reconstruction separates the notion of a right angle from the particular statement of Proposition I.11.

At the foundational level, the predicate `HilbertRightAngle` expresses Euclid's Definition 10. An angle AOB is right when there exists a point C on the extension of AO through O such that

$$\angle AOB \cong \angle BOC.$$

The substantial geometric result is then established independently as

```
hilbert_right_angle_exists
```

which proves that, whenever

$$A * C * B,$$

there exists a point X such that

$$\text{HilbertRightAngle}(A, C, X).$$

The proof of this existence theorem is considerably more elaborate than the final formalization of Proposition I.11. It uses the developed Hilbert theory of angle construction, segment construction, midpoints, triangle congruence, ray order, plane separation, and angle addition.

Once this general result has been established, Proposition I.11 itself becomes a direct interface theorem:

```
hilbert_right_angle_exists
|
v
euclid_proposition_11
```

More explicitly, the architecture is

$$\boxed{\texttt{HilbertRightAngle}}$$

$$\downarrow$$

$$\boxed{\text{Hilbert synthetic geometry}}$$

angle construction
segment construction
midpoint existence
Hilbert SSS
SameSide/OppositeSide theory
ray-order theorem
angle addition

$$\downarrow$$

$$\boxed{\texttt{hilbert\_right\_angle\_exists}}$$

$$\downarrow$$

$$\boxed{\texttt{euclid\_proposition\_11}}$$

Thus the formal dependency structure differs sharply from Euclid's historical construction. Euclid obtains the perpendicular by constructing an equilateral triangle and applying Proposition I.8. The Hilbert development instead proves a general right-angle existence theorem from the already developed synthetic theory and then obtains Proposition I.11 as an immediate consequence.

This distinction is intentional. The proposition file records the mathematical content of Euclid I.11, while the underlying Hilbert layer is free to establish that content by a more general synthetic argument.

11.5 Hilbert Reconstruction

The Hilbert proof of the existence theorem does not reproduce Euclid's equilateral-triangle construction. Instead, it derives a right angle from the general machinery already available in the Hilbert layer.

Assume

$$A * C * B.$$

Thus A, C, B lie on a common line, which we denote by b.

Choose a point D outside b. The angle-construction theorem is then applied at C to construct a point E, on the same side of b as D, such that

$$\angle ACD \cong \angle BCE.$$

The uniqueness part of angle construction determines the ray CE.

Next choose a point F on the ray CD such that

$$CF \cong CE.$$

Let X be the midpoint of FE. Hence

$$F * X * E$$

and

$$FX \cong XE.$$

Consider the triangles

$$\triangle CFX \qquad \text{and} \qquad \triangle CEX.$$

They satisfy

$$CF \cong CE, \qquad FX \cong XE, \qquad CX \cong CX.$$

Hilbert SSS therefore gives

$$\angle FCX \cong \angle ECX.$$

Since F and D lie on the same ray from C, replacing the ray CF by the ray CD does not change the corresponding angle. Consequently,

$$\angle DCX \cong \angle ECX.$$

We have therefore obtained the two component congruences

$$\angle ACD \cong \angle BCE$$

and

$$\angle DCX \cong \angle ECX.$$

The remaining problem is to justify that these component angles may be added in the required configuration.

This is the technically most substantial part of the reconstruction. The ray-order theorem applied to the noncollinear configuration determines two possible cases:

$$\text{RayMeetsSegment}(CD, EA)$$

or

$$\text{RayMeetsSegment}(CE, DA).$$

In the first case, the ray CD meets the open segment EA at a point Y. The SameSide/OppositeSide calculus shows that

$$A \text{ and } X$$

lie on opposite sides of the line CD, while

$$B \text{ and } X$$

lie on opposite sides of the line CE.

In the second case, the ray CE meets the open segment DA at a point Y. Plane-separation arguments show instead that

$$A \text{ and } X$$

lie on the same side of CD, while

$$B \text{ and } X$$

lie on the same side of CE.

Thus in both cases the two configurations have the same side pattern:

$$\text{SameSide}(A, X; CD) \quad \Longleftrightarrow \quad \text{SameSide}(B, X; CE).$$

The proof also establishes that X does not lie on the original line AB. Hence both

$$\angle ACX \qquad \text{and} \qquad \angle BCX$$

are nondegenerate angles.

The angle-addition theorem can now be applied to

$$\angle ACD \cong \angle BCE$$

and

$$\angle DCX \cong \angle ECX.$$

It yields

$$\angle ACX \cong \angle BCX.$$

Since $A * C * B$, the rays CA and CB are opposite. Reversing the order of the sides in the second angle gives

$$\angle ACX \cong \angle XCB.$$

Therefore the two adjacent angles formed by CX with the straight line AB are congruent.

By the definition of `HilbertRightAngle`,

$$\text{HilbertRightAngle}(A, C, X).$$

Hence

$$\exists X \;\; \text{HilbertRightAngle}(A, C, X),$$

which is the required Hilbert form of Proposition I.11.

11.6 Lean Formalization

The formalization is divided into three levels: the definition of a right angle, the general Hilbert existence theorem, and the final Euclid I.11 interface theorem.

The notion of a right angle is represented by the following definition:

```
def HilbertRightAngle
(Geo : Geometry.Geo)
(A O B : Geo.Point) : Prop :=
exists C : Geo.Point,
Geo.Between A O C /\
Geo.AngleCongruent A O B B O C
```

This is a direct formalization of the adjacent-angle characterization of a right angle. The witness C extends the ray OA through O, and the congruence

$$\angle AOB \cong \angle BOC$$

states that the two adjacent angles are equal.

The main work is contained in the general existence theorem:

```
theorem hilbert_right_angle_exists
[HilbertCongruence Geo]
(A C B : Geo.Point)
(hACB : Geo.Between A C B) :
exists X : Geo.Point,
HilbertRightAngle Geo A C X := by
...
```

The proof first extracts the incidence and nondegeneracy information contained in `hACB`. It then chooses an auxiliary point off the line AB and constructs a congruent angle on the opposite ray at C.

A point F is subsequently constructed on the auxiliary ray with

$$CF \cong CE,$$

and the midpoint X of FE is introduced. The central triangle congruence step is encoded by an application of Hilbert SSS:

```
have hSSS :=
HilbertSSS ...
```

This produces the congruence of the two angles at C required for the second component of the final angle-addition argument.

The principal technical issue is not the congruence calculation but the relative position of the rays. The proof therefore invokes the ray-order theorem and splits into its two possible cases:

```
rcases hRayOrder with hCaseD | hCaseE
```

Each branch establishes the same logical condition required by the angle-addition theorem:

```
have hSideConfiguration :
HilbertSameSide Geo A X lineCD <->
HilbertSameSide Geo B X lineCE := by
...
```

In one branch both propositions are false; in the other both are true. This case distinction isolates the plane-separation reasoning from the final angle calculation.

After proving the necessary noncollinearity conditions, the two component angle congruences are combined by

```
have hACX_BCX :
Geo.AngleCongruent A C X B C X :=
hilbert_angle_addition
...
hSideConfiguration
...
hAngleE
hDCX_ECX
```

The second angle is then reversed:

```
have hACX_XCB :
Geo.AngleCongruent A C X X C B :=
(Geo.angle_congruent_reverse_second
A C X B C X).mp hACX_BCX
```

Since the original hypothesis already supplies

$$A * C * B,$$

the point B itself is the witness required by the definition of `HilbertRightAngle`. The existence theorem is therefore completed by

```
refine <X, ?_>
refine <B, hACB, ?_>
exact hACX_XCB
```

Once this theorem is available, the formal statement of Proposition I.11 contains no additional geometric argument:

```
theorem euclid_proposition_11
[HilbertCongruence Geo]
(A C B : Geo.Point)
(hACB : Geo.Between A C B) :
exists X : Geo.Point,
HilbertRightAngle Geo A C X := by
exact
hilbert_right_angle_exists
Geo A C B hACB
```

Thus the Lean theorem corresponding to Euclid I.11 is deliberately small. The substantial synthetic argument has been factored into `hilbert_right_angle_exists`, making right-angle existence available independently of Proposition I.11 and reusable in the subsequent development.

11.7 Discussion

Proposition I.11 provides an instructive example of the distinction between the organization of Euclid's *Elements* and the organization of the Hilbert formalization.

In Euclid's development, Proposition I.11 is itself a construction. Its proof depends explicitly on earlier propositions:

$$\text{I.1}, \qquad \text{I.3}, \qquad \text{I.8},$$

together with the definition of a right angle. The perpendicular is obtained by constructing an equilateral triangle and comparing two triangles by SSS.

The formal Hilbert development reaches the same result through a different dependency structure. Proposition I.11 does not carry the construction itself. Instead, the substantial argument is isolated as the general theorem

```
hilbert_right_angle_exists
```

and Proposition I.11 merely exposes this theorem at the Euclidean interface.

This is similar in form to Proposition I.10. In both cases the final Euclid theorem is short because the required existence result has already been established in the underlying Hilbert theory. The amount of work hidden behind the interface, however, is very different.

For Proposition I.10, midpoint existence was already a natural primitive result of the developed theory. Proposition I.11 required the introduction of a new general notion,

```
HilbertRightAngle
```

and a substantial synthetic existence proof.

The proof also reveals where the formal difficulty of the result lies. The elementary congruence argument is comparatively short: after constructing F and the midpoint X of FE, SSS gives

$$\angle FCX \cong \angle ECX.$$

The difficult part is proving that the component angles occur in the correct relative configuration so that angle addition is legitimate.

This requires the developed theory of

$$\text{SameSide}, \qquad \text{OppositeSide}, \qquad \text{rays}, \qquad \text{betweenness},$$

together with the ray-order theorem. The resulting two-case analysis is substantially longer than the final congruence calculation.

Thus Proposition I.11 illustrates an important feature of the formalization. Statements that are elementary in an informal diagrammatic proof may require a significant amount of explicit order and separation theory once all geometric configurations must be justified formally.

At the same time, this complexity is not local overhead that must be repeated in later propositions. The result has been packaged as `hilbert_right_angle_exists`. Subsequent arguments may use right-angle existence directly without reconstructing the SameSide/OppositeSide analysis.

The resulting architecture is therefore

$$\text{foundational Hilbert theory} \longrightarrow \text{general synthetic lemmas} \longrightarrow \text{Euclid propositions.}$$

Proposition I.11 is a particularly clear instance of this pattern: its Euclidean interface is almost trivial precisely because the geometric content has been absorbed into the reusable Hilbert layer.

Finally, the comparison with Euclid should not be interpreted as a requirement that the formal proof reproduce Euclid's historical proof step by step. The purpose of the proposition layer is to recover the mathematical statements of Book I over the developed Hilbert theory. When that theory already supplies a stronger or more reusable result, the Euclid proposition can appropriately appear as a thin interface theorem.

Chapter 12

Proposition I.12

12.1 Introduction

Euclid's Proposition I.12 asks for the construction of a perpendicular to a given straight line from a given point lying outside that line.

Classically, Euclid obtains the perpendicular by choosing a point on the opposite side of the given line, drawing a circle centered at the external point, and using the two intersections of the circle with the given line. The segment joining these intersections is bisected by Proposition I.10, and Proposition I.8 is then applied to prove that the line joining the external point to the midpoint is perpendicular to the given line.

The reconstruction over the Hilbert foundation follows a different synthetic route. No circle-line intersection theorem is used. Instead, a second point is constructed on the opposite side of the given line so that two distinct points of the line are equidistant from the original external point and the constructed point.

This leads naturally to the perpendicular-bisector configuration. The required midpoint property is isolated in the general theorem

```
hilbert_equidistant_line_bisects_segment
```

which states that if two distinct points of a line are equidistant from two points lying on opposite sides of that line, then the intersection of the joining segment with the line bisects that segment.

A second auxiliary result,

```
hilbert_collinear_angle_transport
```

encapsulates the angle-order argument needed in the proof of this perpendicular-bisector theorem.

Consequently, Proposition I.12 provides another example of the role of the Book Zero layer: a comparatively elaborate geometric configuration is separated into reusable synthetic lemmas before the final Euclidean proposition is formulated.

12.2 The Classical Construction

Euclid's twelfth proposition asks for a perpendicular to a given straight line from a point outside it.

Let AB be the given straight line and let C be the given point outside it.

12.2.1 Final Configuration

The objective is to determine a point H on the straight line AB such that

$$CH \perp AB.$$

Euclid constructs two points E and G on AB satisfying

$$CE \cong CG,$$

and then bisects EG at H.

The final configuration therefore satisfies

$$E - H - G, \qquad EH \cong HG, \qquad CE \cong CG.$$

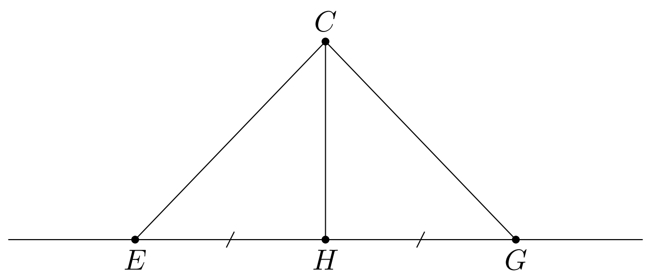


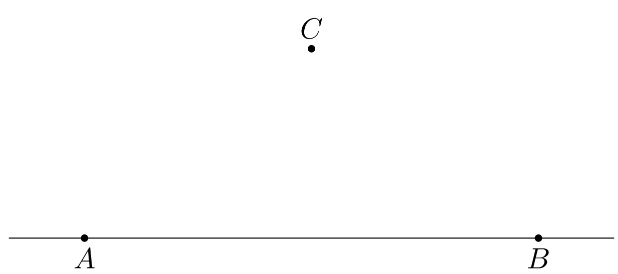
Figure 1: Final configuration of Euclid's construction for Proposition I.12.

12.2.2 Construction Algorithm

The construction uses Proposition I.10 as an already established construction primitive.

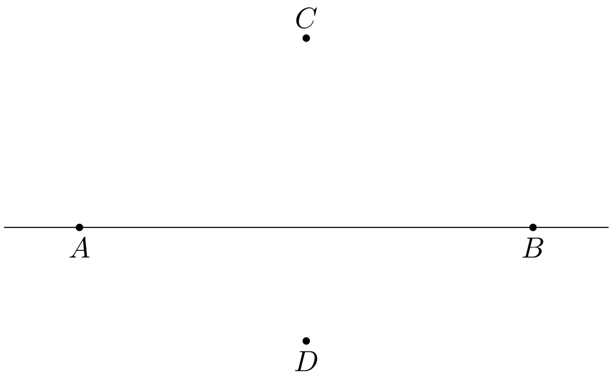
Step 1. Given Line and External Point

Let AB be the given straight line and let C be a point not lying on AB.



Step 2. Choose a Point on the Opposite Side

Choose a point D on the opposite side of the straight line AB from C.



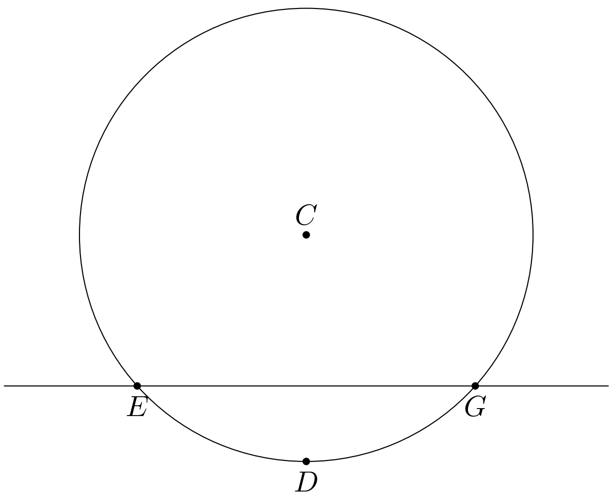
Step 3. Draw the Circle Centered at C

Draw the circle centered at C and passing through D.

Let the circle meet the given straight line at E and G.

Since E and G lie on the same circle centered at C,

$$CE \cong CG.$$



Step 4. Bisect EG

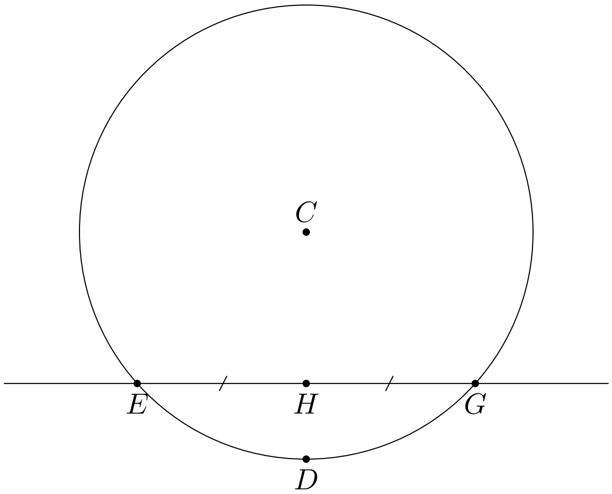
Apply Proposition I.10 to bisect the segment EG at H.

Thus

$$E - H - G$$

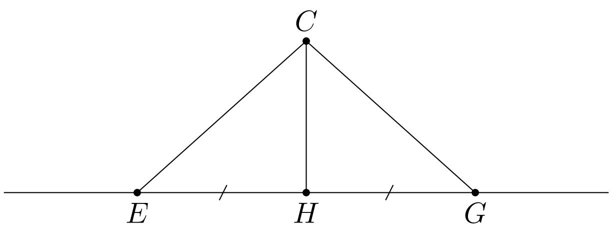
and

$$EH \cong HG.$$



Step 5. Join CH

Join the external point C to the midpoint H.



The two triangles EHC and GHC now satisfy

$$EH \cong HG, \qquad HC \cong HC, \qquad CE \cong CG.$$

By Proposition I.8,

$$\angle EHC \cong \angle GHC.$$

Since E, H, G are collinear, these are adjacent angles. By Definition 10 they are right angles.

Therefore

$$CH \perp AB.$$

The required perpendicular has been constructed.

12.3 Proof

The Hilbert reconstruction does not reproduce Euclid's circle construction.

Let A and B be two distinct points on the given straight line b, and let P be the prescribed point outside b.

The first part of the construction produces a point Q on the opposite side of b from P such that

$$AP \cong AQ$$

and

$$BP \cong BQ.$$

Thus the two distinct points A and B of the line b are both equidistant from P and Q.

Since P and Q lie on opposite sides of b, the segment PQ meets b. Let H be the intersection point. Then

$$P - H - Q.$$

The central step is supplied by the Book Zero theorem

```
hilbert_equidistant_line_bisects_segment
```

Applied to the present configuration, it yields

$$PH \cong HQ.$$

Hence H is the midpoint of the segment PQ.

It remains to prove that the line through P and H is perpendicular to b.

Since $A \neq B$, at least one of the points A and B is distinct from H. Choose such a point and call it R.

Because R is either A or B, the construction already provides

$$RP \cong RQ.$$

Consider the triangles

$$\triangle RHP \qquad \text{and} \qquad \triangle RHQ.$$

They satisfy

$$RH \cong RH, \qquad PH \cong HQ, \qquad RP \cong RQ.$$

Hilbert SSS therefore gives

$$\angle RHP \cong \angle RHQ.$$

Now extend RH through H to a point C, so that

$$R - H - C.$$

Since

$$P - H - Q,$$

the two straight lines RC and PQ intersect at H. The vertical-angle theorem gives

$$\angle RHQ \cong \angle PHC.$$

Consequently,

$$\angle RHP \cong \angle PHC.$$

The angles RHP and PHC are adjacent angles, because $R - H - C$. By the definition of a right angle,

$$\text{HilbertRightAngle}(R, H, P).$$

Therefore the straight line through P and H is perpendicular to the given straight line b.

This proves Proposition I.12.

12.4 Proof Architecture

The formal proof separates Proposition I.12 into two reusable Book Zero lemmas and one final construction theorem.

The first auxiliary result is

```
hilbert_collinear_angle_transport
```

It transports an angle congruence when one side of the angle is replaced by another point on the same supporting line. The proof handles the two possible ray configurations: direct transport along the same ray, and transport through supplementary angles when the new ray is opposite.

The second auxiliary result is

```
hilbert_equidistant_line_bisects_segment
```

Suppose two distinct points A and B lie on a line b, while P and Q lie on opposite sides of b, and

$$AP \cong AQ, \qquad BP \cong BQ.$$

If the segment PQ meets b at H, then

$$PH \cong HQ.$$

The logical structure is therefore

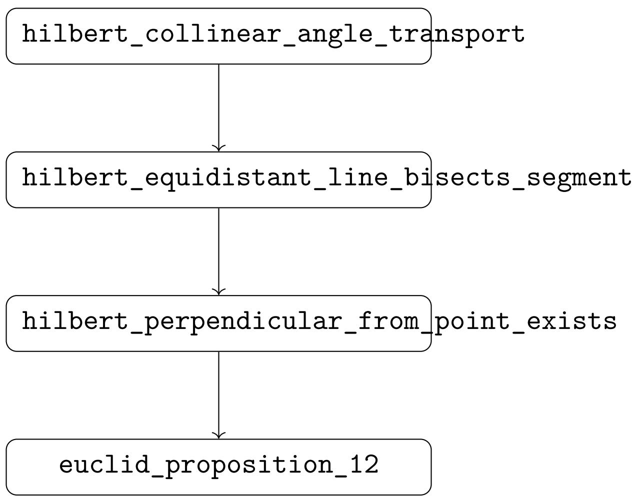


Figure 2: Logical structure of Proposition I.12 over the Book Zero layer.

The main construction theorem

```
hilbert_perpendicular_from_point_exists
```

uses the perpendicular-bisector result to obtain the midpoint H of PQ. It then chooses a point R on the given line distinct from H, applies Hilbert SSS to the triangles RHP and RHQ, and uses vertical angles together with the definition of a right angle.

Finally,

```
euclid_proposition_12
```

is a thin interface theorem exposing this general Hilbert result as Euclid's Proposition I.12.

12.5 Hilbert Reconstruction

The Hilbert reconstruction preserves the geometric objective of Proposition I.12 but does not reproduce Euclid's circle construction.

Let A and B be distinct points of the given line b, and let P be the prescribed point outside b.

The first task is to construct a point Q on the opposite side of b from P such that

$$AP \cong AQ, \qquad BP \cong BQ.$$

To obtain Q, extend PA through A to a point S. Thus P and S lie on opposite sides of b.

Using Hilbert angle construction at A, construct a ray on the side of b containing S whose angle with AB is congruent to the angle formed by AP and AB. On this ray lay off a segment congruent to AP. Denote its endpoint by Q.

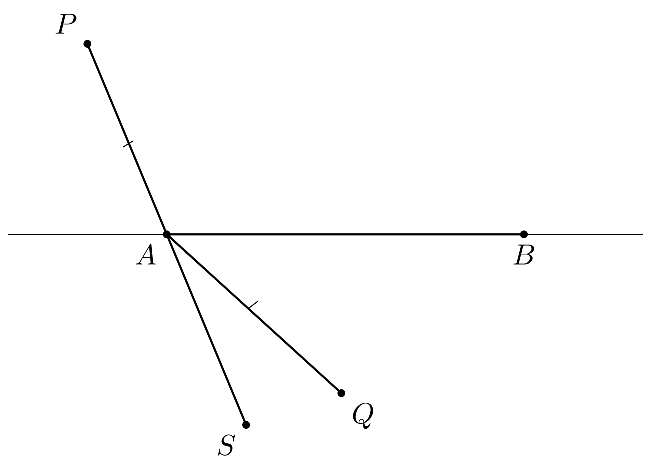


Figure 3: Construction of the point Q. The point S is obtained by extending PA through A. A ray is constructed on the side of the base containing S, and AP is laid off on this ray so that $AQ \cong AP$.

The construction gives

$$AP \cong AQ.$$

The two triangles

$$\triangle APB \qquad \text{and} \qquad \triangle AQB$$

now have

$$AP \cong AQ, \qquad AB \cong AB,$$

and the included angles at A are congruent. By SAS,

$$BP \cong BQ.$$

Moreover, the construction places Q on the opposite side of b from P. Hence the segment PQ intersects b at a point H.

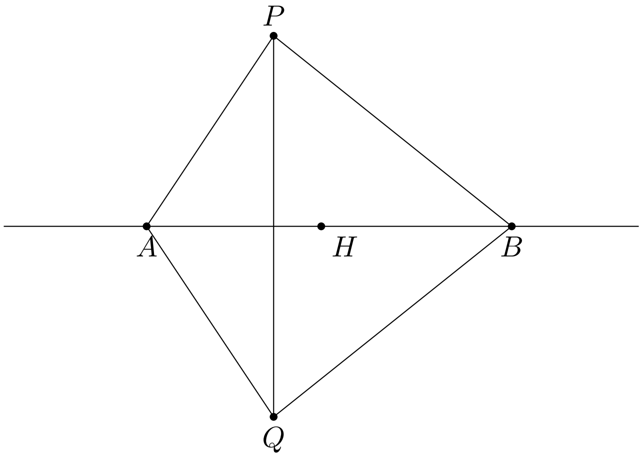


Figure 4: The perpendicular-bisector configuration used in the Hilbert reconstruction.

At this point the general Book Zero theorem

```
hilbert_equidistant_line_bisects_segment
```

applies. Since

$$AP \cong AQ, \qquad BP \cong BQ,$$

and A and B are distinct points of b, the intersection point H bisects PQ:

$$PH \cong HQ.$$

Thus the difficult incidence and angle-order argument has been encapsulated before Proposition I.12 itself is completed.

To obtain the right angle, choose one of A and B distinct from H and call it R. Since R is one of the two equidistant points,

$$RP \cong RQ.$$

Together with

$$PH \cong HQ \qquad \text{and} \qquad RH \cong RH,$$

Hilbert SSS applied to

$$\triangle RHP \qquad \text{and} \qquad \triangle RHQ$$

gives congruent angles at H.

Since $P - H - Q$, these congruent angles determine a perpendicular to b at H. In the formal development this last step is expressed using vertical angles and the predicate

```
HilbertRightAngle
```

The reconstruction therefore replaces Euclid's use of a circle and Propositions I.10 and I.8 by a Hilbert construction based on angle construction, segment layoff, SAS, the perpendicular-bisector theorem, and SSS.

The resulting geometric object is nevertheless the same: a point H on the given line such that the line through P and H is perpendicular to the given line.

12.6 Lean Formalization

The Lean formalization separates the reusable geometric content from the final statement of Proposition I.12.

The main construction theorem is

```
theorem hilbert_perpendicular_from_point_exists
[HilbertIncidence Geo]
[HilbertCongruence Geo]
(A B P : Geo.Point)
(base : Geo.Line)
(hAB : A != B)
(hAbase : HilbertIncidence.OnLine A base)
(hBbase : HilbertIncidence.OnLine B base)
(hPbase : not HilbertIncidence.OnLine P base) :
exists H R : Geo.Point,
HilbertIncidence.OnLine H base /\
HilbertIncidence.OnLine R base /\
HilbertRightAngle Geo R H P
```

Here H is the foot of the perpendicular and R is a second point of the given line used to determine its direction at H.

The presence of R in the conclusion is important. The intersection point H may coincide with either A or B, so neither of the original points can be fixed in advance as the first point in the right-angle predicate. Since

$$A \neq B,$$

at least one of them is distinct from H and can therefore be chosen as R.

The formal proof first constructs a point Q on the opposite side of the given line from P. The construction establishes

$$AP \cong AQ$$

and, by SAS,

$$BP \cong BQ.$$

The opposite-side relation supplies an intersection point H with

$$P - H - Q.$$

The reusable Book Zero theorem

```
hilbert_equidistant_line_bisects_segment
```

then gives

$$PH \cong HQ.$$

The remaining argument selects R, applies Hilbert SSS to the triangles RHP and RHQ, and converts the resulting angle congruence into a right angle by means of vertical angles.

The Euclidean proposition itself is exposed by the wrapper

```
theorem euclid_proposition_12
[HilbertIncidence Geo]
[HilbertCongruence Geo]
(A B P : Geo.Point)
(base : Geo.Line)
(hAB : A != B)
(hAbase : HilbertIncidence.OnLine A base)
(hBbase : HilbertIncidence.OnLine B base)
(hPbase : not HilbertIncidence.OnLine P base) :
exists H R : Geo.Point,
HilbertIncidence.OnLine H base /\
HilbertIncidence.OnLine R base /\
HilbertRightAngle Geo R H P := by

exact
hilbert_perpendicular_from_point_exists
Geo
A B P
base
hAB
hAbase
hBbase
hPbase
```

Thus the theorem carrying Euclid's name is deliberately only a thin wrapper. The geometric work is performed by the Hilbert construction theorem and by the reusable results already established in the Book Zero layer.

12.7 Discussion

Proposition I.12 reveals a substantial difference between Euclid's constructional language and the Hilbert/Book Zero architecture.

Euclid obtains the perpendicular through a concrete circle construction. The circle centered at the external point produces two points of the given line at equal distance from that point. Proposition I.10 supplies their midpoint, and Proposition I.8 converts the resulting three side equalities into equality of the adjacent angles at the midpoint.

The Hilbert reconstruction does not attempt to reproduce this sequence literally. In particular, no circle is introduced. Instead, the essential geometric configuration behind Euclid's construction is isolated:

$$AP \cong AQ, \qquad BP \cong BQ.$$

Thus two distinct points A and B of the given line are equidistant from P and Q. The line through A and B therefore plays the role of the perpendicular-bisector axis of PQ.

This observation leads to the theorem

```
hilbert_equidistant_line_bisects_segment
```

which extracts the reusable geometric content of the configuration. Once the segment PQ intersects the line at H, the theorem gives

$$PH \cong HQ.$$

The proposition is therefore no longer organized around a particular circle construction. It is organized around an abstract property of two equidistant points and the line determined by them.

The proof of the perpendicular-bisector theorem itself exposed another reusable operation. When an angle is known relative to one point of a line, replacing that point by another collinear point requires attention to the order of the points: the relevant ray may remain unchanged or become the opposite ray. This argument is encapsulated in

```
hilbert_collinear_angle_transport
```

rather than being repeated inside Proposition I.12.

This is characteristic of the role of Book Zero. Elementary incidence, order, congruence, ray, and angle manipulations are packaged as synthetic lemmas. Later Euclidean propositions can then operate with larger geometric units instead of repeatedly descending to the Hilbert axioms.

There is also an important difference from the preceding elementary propositions. Proposition I.12 does not collapse to a single previously available Book Zero operation. Its reconstruction required the identification of genuinely reusable intermediate geometry. In this sense, the formalization of I.12 contributed new infrastructure to Book Zero rather than merely consuming existing infrastructure.

The final theorem

```
euclid_proposition_12
```

is consequently a wrapper around

```
hilbert_perpendicular_from_point_exists
```

The wrapper records the Euclidean proposition in the formal library, while the underlying theorem records the more general Hilbert construction on which it depends.

Proposition I.12 therefore illustrates the intended division of labor in the development:

$$\begin{array}{ccl} \text{Hilbert axioms} & \longrightarrow & \text{elementary incidence, order, and congruence theory} \\ \text{Book Zero} & \longrightarrow & \text{reusable synthetic geometric mechanisms} \\ \text{Hilbert construction theorem} & \longrightarrow & \text{existence of the required perpendicular} \\ \text{Euclid wrapper} & \longrightarrow & \text{Proposition I.12} \end{array}$$

The result is not a literal transcription of Euclid's proof. It is a reconstruction of the same geometric theorem over the Hilbert foundation, with the intermediate reasoning reorganized into reusable components.

Chapter 13

Proposition I.13

13.1 Introduction

Euclid's Proposition I.13 concerns the two adjacent angles formed when one straight line stands on another straight line.

Let C, B, D lie on one straight line, with B between C and D, and let A be a point outside that line. The two adjacent angles are

$$\angle CBA \qquad \text{and} \qquad \angle ABD.$$

Euclid states that these angles are either two right angles or are together equal to two right angles.

The classical proof distinguishes two cases. If the adjacent angles are equal, then by Definition 10 they are both right angles. If they are not equal, Euclid erects a perpendicular at B using Proposition I.11 and compares the resulting angle decompositions.

The reconstruction over the Hilbert/Book Zero layer is much shorter. Book Zero already contains an explicit relation expressing that one angle is supplementary to another:

```
BookZeroSupplement
```

For the configuration of Proposition I.13, this relation contains exactly the required geometric data. Consequently, the formal theorem does not reconstruct Euclid's case distinction or introduce an arithmetic operation on angles. It identifies the two adjacent angles directly as a supplementary pair.

Thus Proposition I.13 provides a particularly clear example of the compression obtained from the Book Zero language: a classical geometric argument becomes essentially an instance of an already available synthetic relation.

13.2 The Classical Configuration

Proposition I.13 is not a construction theorem. The geometric configuration is already given.

Let C, B, D lie on one straight line with

$$C - B - D,$$

and let A lie outside the straight line CD.

The straight line BA stands on the straight line CD and forms the adjacent angles

$$\angle CBA \qquad \text{and} \qquad \angle ABD.$$

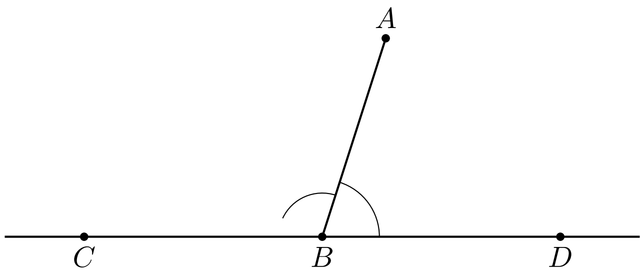


Figure 13.1: Configuration of Proposition I.13. The point B lies between C and D, and the ray BA forms the adjacent angles $\angle CBA$ and $\angle ABD$ with the straight line CD.

13.3 Statement

Assume

$$C - B - D$$

and let A be a point distinct from B.

Then the adjacent angles

$$\angle CBA \qquad \text{and} \qquad \angle ABD$$

form a supplementary pair.

In Euclid's formulation, they are therefore either two right angles or together equal to two right angles.

In the Book Zero language this conclusion is represented by

$$\text{BookZeroSupplement}(C, B, A, A, D).$$

By definition,

$$\text{BookZeroSupplement}(C, B, A, A, D)$$

means

$$\text{SameRay}(B, A, A)$$

together with

$$C - B - D.$$

The first condition expresses the trivial fact that the ray BA coincides with itself, while the second is precisely the given straight-line configuration.

13.4 Classical Proof

Euclid proves Proposition I.13 by distinguishing two cases.

13.4.1 Case 1. The Adjacent Angles Are Equal

Suppose first that
$$\angle CBA \cong \angle ABD.$$

Since C, B, D lie on one straight line, the two angles are adjacent angles formed by the straight line BA standing on CD.

By Euclid's Definition 10, when a straight line standing on another straight line makes the adjacent angles equal, each of those angles is a right angle.

Therefore
$$\angle CBA \qquad \text{and} \qquad \angle ABD$$
are two right angles.

This is the first alternative in Proposition I.13.

13.4.2 Case 2. The Adjacent Angles Are Not Equal

Suppose now that
$$\angle CBA \ncong \angle ABD.$$

By Proposition I.11, construct through B a perpendicular BE to the straight line CD.

Thus
$$\angle CBE \qquad \text{and} \qquad \angle EBD$$
are right angles.

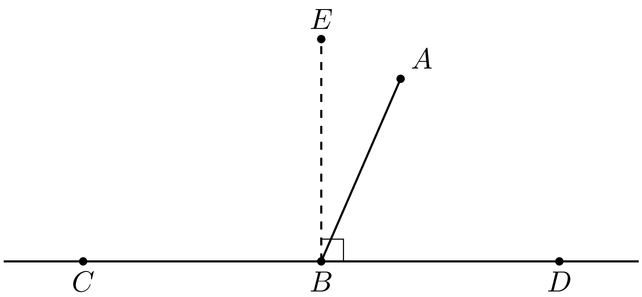


Figure 13.2: Euclid's auxiliary construction in the second case. By Proposition I.11 the perpendicular BE is erected at B on the straight line CD.

Since BE is perpendicular to CD,
$$\angle CBE + \angle EBD$$
is equal to two right angles.

The original ray BA divides one of these right-angle configurations. In the arrangement shown in Figure 13.2,
$$\angle CBE = \angle CBA + \angle ABE.$$

Adding $\angle EBD$ gives
$$\angle CBE + \angle EBD = \angle CBA + \angle ABE + \angle EBD.$$

But

$$\angle ABE + \angle EBD = \angle ABD.$$

Hence

$$\angle CBE + \angle EBD = \angle CBA + \angle ABD.$$

The left-hand side consists of two right angles. Therefore

$$\angle CBA + \angle ABD$$

is equal to two right angles.

This is the second alternative in Proposition I.13.

Thus, whether the two original adjacent angles are equal or unequal, they are either themselves two right angles or are together equal to two right angles.

13.5 Proof Architecture

Euclid's proof proceeds through the geometry of right angles and angle addition:

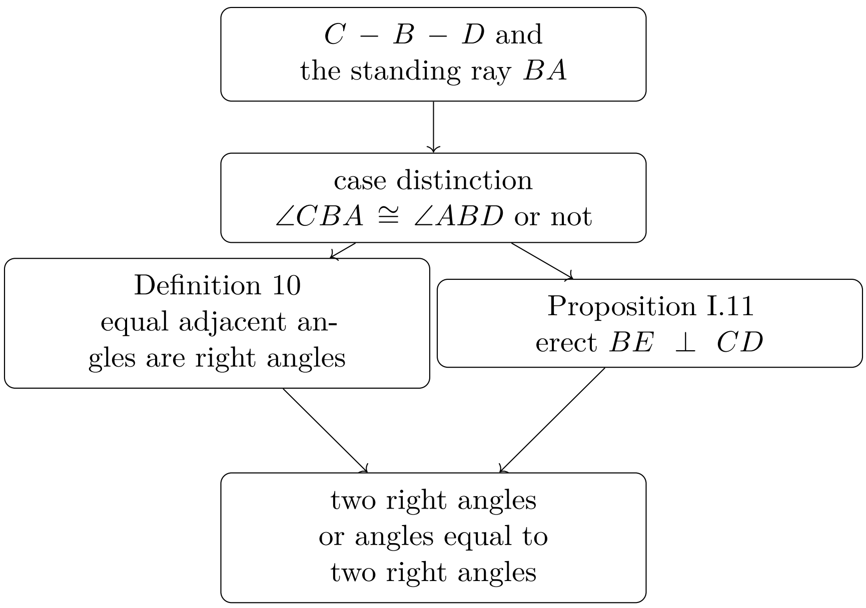


Figure 3: Logical structure of Euclid's proof of Proposition I.13.

The Book Zero reconstruction has a very different architecture:

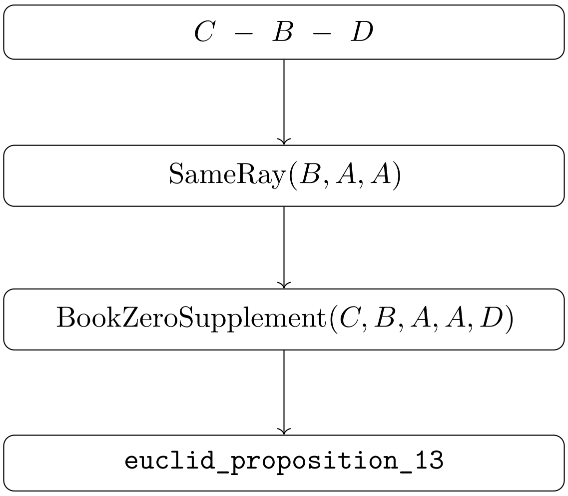


Figure 4: Book Zero architecture of Proposition I.13.

The contrast is substantial. Euclid proves the proposition by introducing a perpendicular and reasoning about decompositions and combinations of angles. At the Book Zero level, the same geometric configuration has already been abstracted as the supplementary-angle relation.

No new geometric construction is therefore required, and Proposition I.11 disappears from the formal dependency path.

13.6 Hilbert Reconstruction

The Hilbert reconstruction of Proposition I.13 does not reproduce Euclid's proof.

The essential observation is that the conclusion of the proposition has already been isolated in Book Zero as the relation

```
BookZeroSupplement
```

For five points A, B, C, D, F, this relation is defined by

$$\mathrm{BookZeroSupplement}(A, B, C, D, F)$$

if and only if

$$\mathrm{SameRay}(B, C, D)$$

and

$$A - B - F.$$

Thus the relation records synthetically the configuration in which one side of an angle is continued through its vertex while the other side is preserved along the same ray.

For Proposition I.13 we instantiate this relation as

$$\text{BookZeroSupplement}(C, B, A, A, D).$$

Expanding the definition gives exactly two conditions:

$$\text{SameRay}(B, A, A)$$

and

$$C - B - D.$$

The second condition is one of the hypotheses of Proposition I.13.

The first condition says only that the ray BA is the same ray as itself. Since $A \neq B$, it follows immediately from reflexivity of the same-ray relation.

Consequently,

$$\text{BookZeroSupplement}(C, B, A, A, D)$$

follows directly.

No perpendicular is constructed. No case distinction between equal and unequal adjacent angles is required. No decomposition or addition of angles occurs in the proof.

This does not mean that Euclid's geometric argument has been discarded. Rather, the geometric notion expressed by that argument has already been incorporated into the language of Book Zero. Proposition I.13 therefore becomes a recognition theorem: the given configuration is an instance of an existing synthetic relation.

13.7 Lean Formalization

The Lean theorem is correspondingly short:

```
theorem euclid_proposition_13
[HilbertIncidence Geo]
[HilbertOrder Geo]
(C B D A : Geo.Point)
(hCBD : Geo.Between C B D)
(hBA : B != A) :
BookZeroSupplement Geo C B A A D := by

constructor

· exact
hilbert_sameRay_refl
Geo B A hBA.symm

· exact hCBD
```

The proof follows the definition of

```
BookZeroSupplement
```

literally.

The first component of the conclusion is

$$\mathrm{SameRay}(B, A, A).$$

It is obtained from

```
hilbert_sameRay_refl
```

using $A \neq B$, which is supplied by the symmetric form of the hypothesis $B \neq A$.

The second component is

$$C - B - D,$$

which is exactly the hypothesis

```
hCBD
```

Thus the formal proof contains no intermediate geometric construction and requires no theorem corresponding to Euclid's use of Proposition I.11.

13.8 Discussion

Proposition I.13 is a particularly clear test of the role assigned to Book Zero in the reconstruction.

In Euclid's original development, the statement requires an actual proof. The argument distinguishes whether the two adjacent angles are equal. In the nontrivial case, Proposition I.11 is invoked to erect a perpendicular, and the conclusion is obtained by decomposing and recombining angles.

At the Hilbert level, one could reconstruct this argument explicitly. The existing theory already contains the necessary machinery for right angles, supplementary angles, and angle addition.

That reconstruction is unnecessary, however, once the Book Zero language is used.

The relation

```
BookZeroSupplement
```

was introduced independently as a reusable description of a standard synthetic configuration. Proposition I.13 is precisely an instance of that configuration.

The resulting compression is therefore structural rather than tactical. Lean is not discovering a shorter proof of Euclid's argument. The formal language has been organized at a level at which the conclusion of I.13 can already be stated directly.

This produces an unusually short dependency path:

$$\text{betweenness} \quad + \quad \text{same-ray reflexivity} \quad \longrightarrow \quad \text{BookZeroSupplement} \quad \longrightarrow \quad \text{Euclid I.13.}$$

In particular, Proposition I.11, although essential to Euclid's classical proof, is not a dependency of the reconstructed theorem.

This distinction is important. The objective of the reconstruction is not to force every Euclidean proposition to retain the dependency graph of the original text. The objective is to express Euclidean geometry over a stable synthetic interface derived from the Hilbert foundation.

Earlier propositions exhibited several possible relationships with this interface. Some consumed already available Book Zero results, while others exposed missing reusable geometry that had to be added to the interface.

Proposition I.13 exhibits a third and especially simple situation: the proposition is already expressed by the vocabulary of Book Zero.

Schematically,

$$\begin{array}{ccl} \text{Euclid} & : & \text{case distinction} + \text{I.11} + \text{angle composition} \\ \text{Hilbert/Book Zero} & : & \text{recognition of a supplementary configuration} \\ \text{Lean} & : & \text{same-ray reflexivity} + \text{given betweenness} \end{array}$$

The four-line proof is therefore not merely an accidental shortening of Proposition I.13. It is evidence that the Book Zero layer is functioning as intended: elementary synthetic arguments can be encapsulated once and subsequently used as part of the language in which later geometry is formulated.

13.9 Book Zero Components Used

The reconstruction of Proposition I.13 relies on the Book Zero relation

```
BookZeroSupplement
```

defined by

$$\text{BookZeroSupplement}(A, B, C, D, F)$$

if and only if

$$\text{SameRay}(B, C, D)$$

and

$$A - B - F.$$

Geometrically, this represents a pair of supplementary angles. The angle determined by the rays BA and BC has as its supplement the angle determined by the rays BD and BF, where BD lies on the same ray as BC and BF is the opposite extension of BA.

For Proposition I.13 the relation is instantiated as

$$\text{BookZeroSupplement}(C, B, A, A, D).$$

Thus the required conditions become

$$\text{SameRay}(B, A, A)$$

and

$$C - B - D.$$

The first is immediate from same-ray reflexivity, while the second is the given betweenness hypothesis.

The proof also uses

```
hilbert_sameRay_refl
```

which expresses reflexivity of a nondegenerate ray:

$$A \neq B \quad \Longrightarrow \quad \text{SameRay}(A, B, B).$$

Chapter 14

Proposition I.14

14.1 Introduction

Euclid's Proposition I.14 is the converse in spirit of the preceding Proposition I.13.

Proposition I.13 starts from a straight-line configuration and proves that the adjacent angles are together equal to two right angles. Proposition I.14 reverses this direction: if two adjacent angles are together equal to two right angles, and their outer rays lie on opposite sides of the common straight line, then those outer rays form one straight line.

In Euclid's original proof the argument is indirect. The ray opposite to one of the given rays is constructed, Proposition I.13 is applied, and equality of the resulting angles forces the constructed ray to coincide with the remaining given ray.

The Hilbert reconstruction follows the same geometric idea at a more structural level. The condition "equal to two right angles" is encoded without numerical angle measure, using an opposite ray and angle congruence. The decisive step is then supplied by the uniqueness clause of Hilbert's angle-construction axiom.

Proposition I.14 is therefore an instructive example of the interaction between Euclid's angle language, Hilbert plane separation, angle construction, and the ray calculus developed in Book Zero.

14.2 The Classical Configuration

Let AB be a straight line, and let two further straight lines BC and BD meet it at the common point B, with C and D lying on opposite sides of AB.

The two adjacent angles are therefore

$$\angle ABC \qquad \text{and} \qquad \angle ABD.$$

The hypothesis of Proposition I.14 is that these two angles are together equal to two right angles. The conclusion is that the two outer rays BC and BD are opposite rays and hence form one straight line.

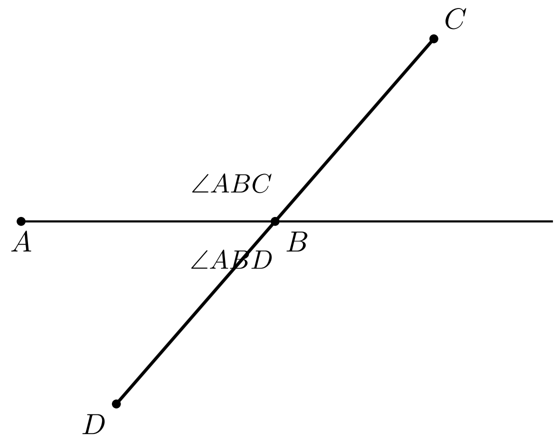


Figure 14.1: The classical configuration of Proposition I.14. The rays BC and BD lie on opposite sides of the straight line through A and B.

14.3 Statement

Theorem 7 (Euclid I.14)**.** *Let AB be a straight line, and let the straight lines BC and BD meet it at the point B on opposite sides of AB.*

If the adjacent angles

$$\angle ABC \qquad \textit{and} \qquad \angle ABD$$

are together equal to two right angles, then BC and BD form one straight line.

Equivalently,

$$C - B - D.$$

The proposition may therefore be viewed as a converse to the straight-line angle relation established in Proposition I.13.

The geometric content of the conclusion is stronger than mere collinearity. The point B lies between C and D, so the rays BC and BD are opposite rays with common endpoint B.

14.4 Classical Proof

Euclid argues by extending one of the two outer rays through the common vertex.

Extend CB through B to a point E. Then

$$C - B - E,$$

so BC and BE form one straight line.

By Proposition I.13, the adjacent angles

$$\angle ABC \qquad \text{and} \qquad \angle ABE$$

are together equal to two right angles.

But by hypothesis,

$$\angle ABC \qquad \text{and} \qquad \angle ABD$$

are also together equal to two right angles.

Therefore the remaining angles must be equal:

$$\angle ABE = \angle ABD.$$

Therefore the remaining angles must be equal:

$$\angle ABE = \angle ABD.$$

But if BE and BD were distinct, one of these two angles would be a proper part of the other. A proper part cannot be equal to the whole. Hence the rays BE and BD cannot be distinct.

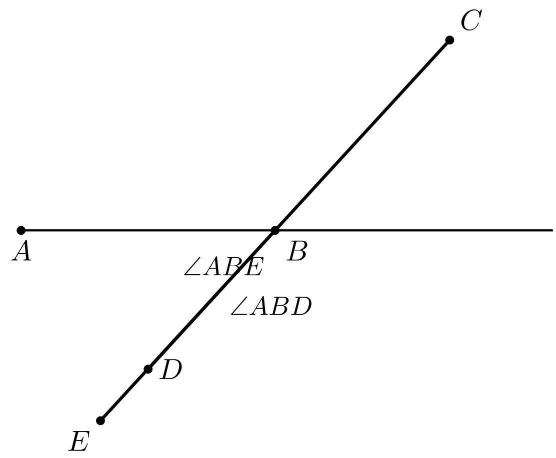
Therefore BE coincides with BD. Since

$$C - B - E,$$

it follows that

$$C - B - D.$$

Thus BC and BD form one straight line.



The essential structure of Euclid's argument is therefore

$$\text{extension} \longrightarrow \text{I.13} \longrightarrow \text{equality of the remaining angles} \longrightarrow \text{coincidence of the rays.}$$

14.5 Proof Architecture

The Hilbert reconstruction preserves the central geometric idea of Euclid's proof but reorganizes its logical dependencies.

Euclid extends CB through B to a point E and uses Proposition I.13 to obtain the supplementary angle relation. He then compares this relation with the hypothesis of Proposition I.14 and concludes that the rays BD and BE must coincide.

In the formal reconstruction, the point E is incorporated directly into the synthetic representation of the hypothesis. We require

$$C - B - E$$

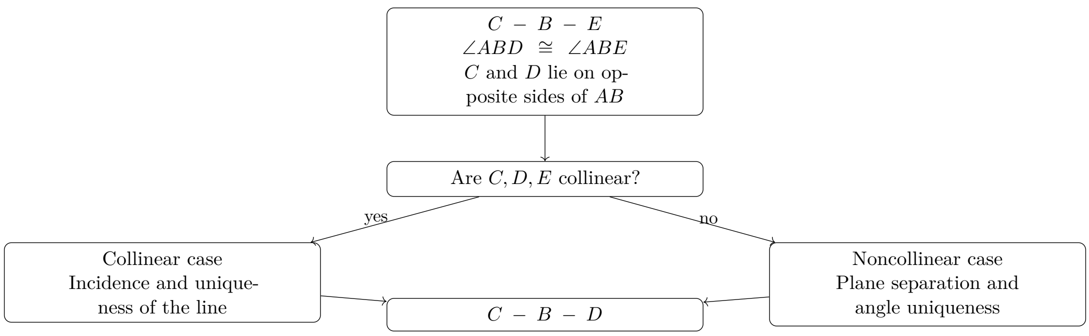
together with

$$\angle ABD \cong \angle ABE.$$

Thus the intermediate appeal to Proposition I.13 is no longer needed: the supplementary configuration required by Euclid's argument is already present in the formal hypothesis.

The remaining problem is to prove that BD and BE determine the same ray from B.

At this point the reconstruction divides into two cases according to whether C, D, E are collinear.



In the collinear case, incidence and line uniqueness are sufficient.

In the noncollinear case, plane separation first places D and E on the same side of the base line. The uniqueness clause of Hilbert's angle-construction axiom then identifies the rays BD and BE. Ray calculus and transport of betweenness finally yield

$$C - B - D.$$

Schematically, the two proof architectures are

$$\begin{aligned} \text{Euclid} \quad &: \quad \text{extension} \to \text{I.13} \to \text{angle comparison} \to \text{ray coincidence}, \\ \text{Hilbert reconstruction} \quad &: \quad \text{encoded supplement} \to \text{case split} \to \text{line/angle uniqueness} \to \text{betweenness}. \end{aligned}$$

14.6 Hilbert Reconstruction

The Hilbert reconstruction begins by replacing Euclid's phrase "together equal to two right angles" by a purely synthetic configuration.

No numerical measure of angles and no operation of angle addition are introduced. Instead, the supplementary relation is represented by an opposite ray together with angle congruence.

14.6.1 Encoding the Angle Hypothesis

Choose a point E on the continuation of CB beyond B, so that

$$C - B - E.$$

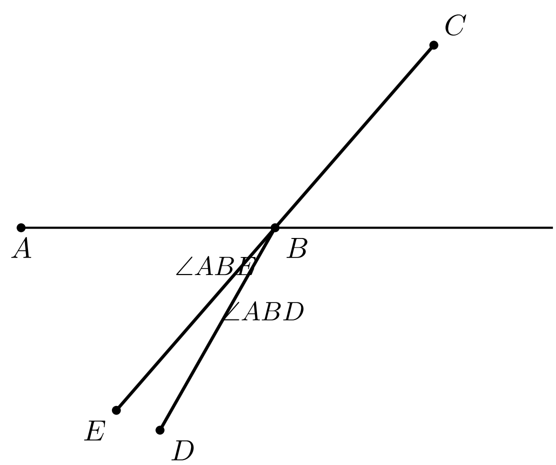
The rays BC and BE are therefore opposite. Consequently, $\angle ABE$ is the angle supplementary to $\angle ABC$.

The hypothesis that

$$\angle ABC \qquad \text{and} \qquad \angle ABD$$

are together equal to two right angles is represented by requiring

$$\angle ABD \cong \angle ABE.$$



In Lean this condition is packaged as

```
def HilbertAnglesEqualTwoRightAngles
[HilbertIncidence Geo]
(A B C D : Geo.Point) : Prop :=
exists E : Geo.Point,
Geo.Between C B E /\
Geo.AngleCongruent
A B D
A B E
```

Thus the witness E contains exactly the two pieces of geometric information needed by the reconstruction:

$$C - B - E$$

and

$$\angle ABD \cong \angle ABE.$$

This definition should not be interpreted as a numerical definition of the sum of two angles. It is a synthetic encoding of the particular supplementary configuration occurring in Proposition I.14.

14.6.2 The Collinear Case

Assume first that C, D, and E are collinear.

From the definition of `HilbertAnglesEqualTwoRightAngles` we already have

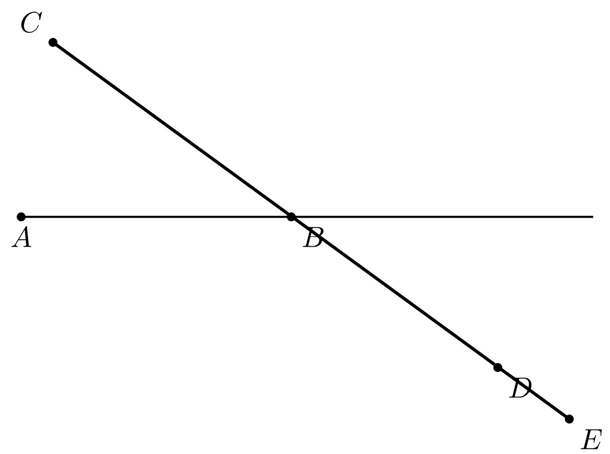
$$C - B - E.$$

Hence B lies on the line determined by C and E.

Since C and D lie on opposite sides of the base line AB, the segment CD meets that line. Let X be a point of intersection, so that

$$C - X - D$$

and X lies on the base line.



Both B and X now lie on two lines: the base line AB and the line containing C, D, E.

Suppose that

$$X \neq B.$$

By uniqueness of the line through two distinct points, the two lines would then have to coincide. In particular, C would lie on the base line.

This contradicts the hypothesis that C and D lie on opposite sides of the base line, since a point belonging to the line itself cannot be one of the two opposite-side points.

Therefore

$$X = B.$$

Substituting this equality into

$$C - X - D$$

gives

$$C - B - D.$$

Thus the collinear branch requires no angle argument at all. Once C, D, E are known to be collinear, the conclusion follows entirely from betweenness, plane separation, incidence, and uniqueness of the line through two distinct points.

14.6.3 The Noncollinear Case

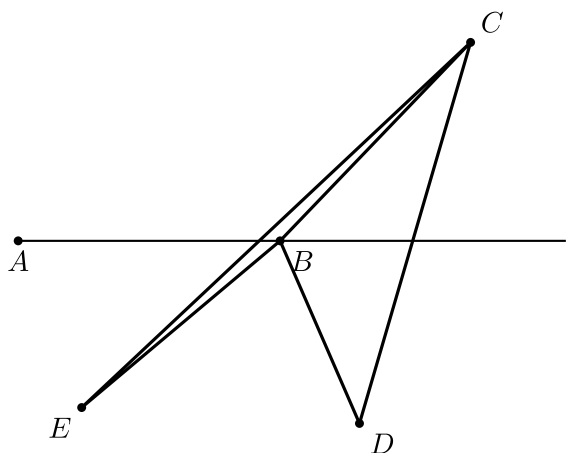
Assume now that C, D, and E are not collinear.

The segment CD meets the base line because C and D lie on opposite sides of it. The segment CE also meets the base line, namely at B, since

$$C - B - E.$$

Because C, D, E form a genuine triangle, the Hilbert plane-separation theory implies that the remaining endpoints D and E lie on the same side of the base line. Thus

$$\text{SameSide}(D, E; AB).$$



D and E on the same side of AB

We now use Hilbert's angle-construction axiom.

The angle $\angle ABE$ is nondegenerate. Construct, on the side of the base line containing D, a ray BZ satisfying

$$\angle ABZ \cong \angle ABE.$$

Hilbert III.4 includes a uniqueness clause: on the prescribed side of the base ray there is only one ray realizing the prescribed angle.

Since E lies on the same side of the base line as D, and

$$\angle ABE \cong \angle ABE,$$

the uniqueness clause gives

$$\mathrm{SameRay}(B, Z, E).$$

On the other hand, the encoded hypothesis of Proposition I.14 gives

$$\angle ABD \cong \angle ABE,$$

and hence, by symmetry,

$$\angle ABE \cong \angle ABD.$$

The point D lies on the side on which BZ was constructed. Applying the same uniqueness clause again gives

$$\mathrm{SameRay}(B, Z, D).$$

We therefore have two same-ray relations with the common reference ray BZ:

$$\mathrm{SameRay}(B, Z, E), \qquad \mathrm{SameRay}(B, Z, D).$$

The Book Zero ray calculus now yields

$$\mathrm{SameRay}(B, E, D).$$

Finally, from the encoded supplementary configuration we already know

$$C - B - E.$$

Replacing E by another point D on the same ray from B preserves the betweenness relation. Transport of betweenness along equal rays therefore gives

$$C - B - D.$$

This completes the noncollinear branch and hence the Hilbert reconstruction of Proposition I.14.

14.7 Lean Formalization

The reconstructed proposition is formalized by the theorem

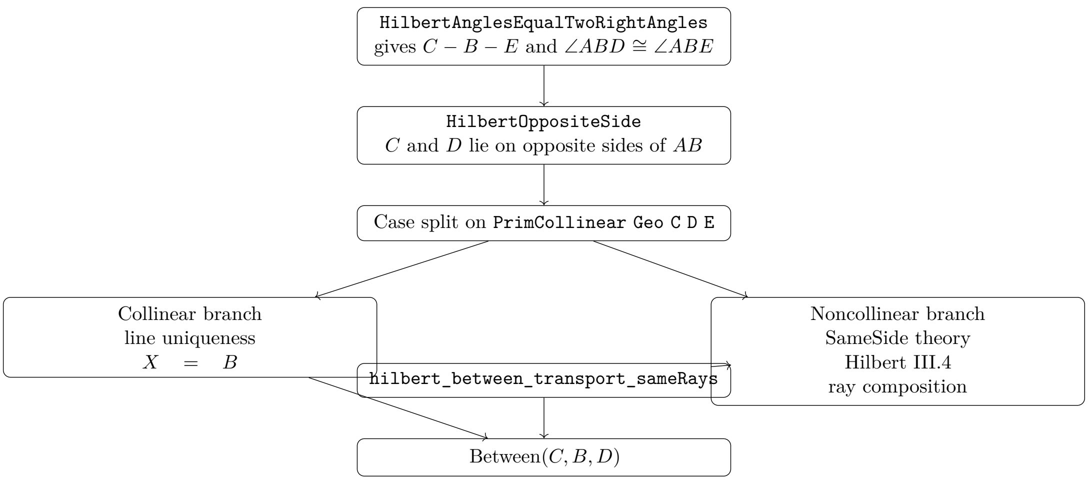
```
theorem euclid_proposition_14
[HilbertIncidence Geo]
[HilbertCongruence Geo]
(A B C D : Geo.Point)
(base : Geo.Line)
(hAB : A != B)
(hAbase :
HilbertIncidence.OnLine A base)
(hBbase :
HilbertIncidence.OnLine B base)
(hOppCD :
HilbertOppositeSide Geo C D base)
(hTwoRight :
HilbertAnglesEqualTwoRightAngles
Geo A B C D) :
Geo.Between C B D := by
```

The formal conclusion

$$\mathrm{Between}(C, B, D)$$

expresses directly that B lies between C and D, and hence that the rays BC and BD form one straight line.

The principal dependency structure of the proof is



The proof introduces no new geometric axiom. The new predicate

```
HilbertAnglesEqualTwoRightAngles
```

serves only as an interface representation of the angle hypothesis. The actual derivation uses the existing incidence, order, plane separation, angle-construction, and ray machinery.

A notable difference from Euclid's dependency graph is that the Lean theorem does not invoke Proposition I.13. The supplementary configuration needed in the classical proof has already been encoded in the hypothesis, and the identification of the relevant rays is obtained directly from Hilbert III.4.

14.8 Hilbert and Book Zero Components Used

The proof of Proposition I.14 uses several previously established components of the Hilbert and Book Zero layers. Their roles in the reconstruction are recorded here explicitly.

14.8.1 `hilbert_third_side_endpoints_sameSide`

In the noncollinear branch, the points C, D, E form a genuine triangle. The base line meets the interior of CD, because C and D lie on opposite sides of it, and it meets the interior of CE at B, since

$$C - B - E.$$

The theorem

```
hilbert_third_side_endpoints_sameSide
```

then shows that the remaining endpoints D and E lie on the same side of the base line.

Schematically,

$$\begin{array}{c} C - X - D, \\ C - B - E, \\ X, B \in AB \end{array} \qquad \Longrightarrow \qquad \mathrm{SameSide}(D, E; AB).$$

This is the plane-separation step that makes the uniqueness clause of Hilbert's angle-construction axiom applicable to both D and E.

14.8.2 Hilbert III.4: Uniqueness of Angle Construction

Hilbert's angle-construction axiom provides not only the existence of a copy of a given angle but also uniqueness on a prescribed side of the base ray.

In Proposition I.14 a ray BZ is constructed so that

$$\angle ABZ \cong \angle ABE.$$

Both E and D lie on the prescribed side of the base line. The uniqueness clause can therefore be applied twice, yielding

$$\mathrm{SameRay}(B, Z, E)$$

and

$$\mathrm{SameRay}(B, Z, D).$$

This is the formal replacement for Euclid's final inference that equal angles on the same side determine the same straight line.

14.8.3 `bookZero_36_ray3`

The Book Zero theorem

```
bookZero_36_ray3
```

is a ray-composition principle.

In the present proof,

$$\mathrm{SameRay}(B, Z, E)$$

and

$$\mathrm{SameRay}(B, Z, D)$$

imply

$$\mathrm{SameRay}(B, E, D).$$

Thus the auxiliary ray BZ introduced by Hilbert III.4 can be eliminated, leaving the geometrically relevant statement that BE and BD are the same ray.

14.8.4 `hilbert_between_transport_sameRays`

The final step uses

```
hilbert_between_transport_sameRays
```

to transport strict betweenness along corresponding rays.

We know

$$C - B - E$$

and have established

$$\mathrm{SameRay}(B, E, D).$$

The ray from B toward C is unchanged, while E is replaced by the point D on the same ray from B. Therefore the theorem gives

$$C - B - D.$$

This is precisely the formal conclusion of Proposition I.14.

The dependency path of the noncollinear branch can therefore be summarized as

$$\begin{array}{c} \text{plane separation} \\ \downarrow \\ \mathrm{SameSide}(D, E; AB) \\ \downarrow \\ \text{Hilbert III.4} \\ \downarrow \\ \mathrm{SameRay}(B, Z, E), \mathrm{SameRay}(B, Z, D) \\ \downarrow \\ \texttt{bookZero\_36\_ray3} \\ \downarrow \\ \mathrm{SameRay}(B, E, D) \\ \downarrow \\ \texttt{hilbert\_between\_transport\_sameRays} \\ \downarrow \\ C - B - D. \end{array}$$

Chapter 15

Proposition I.15

15.1 Introduction

Euclid's Proposition I.15 establishes the equality of vertical angles formed by two intersecting straight lines.

If two straight lines intersect at a point, each pair of opposite angles is equal. In modern terminology, these are the vertical-angle relations.

Euclid proves the proposition by applying Proposition I.13 to two adjacent linear pairs and subtracting a common angle from two sums that are each equal to two right angles.

The Hilbert reconstruction is considerably shorter. The corresponding vertical-angle theorem has already been derived in the Hilbert layer as

```
hilbert_vertical_angles
```

from Hilbert's theory of adjacent angles. Proposition I.15 therefore becomes an application of an existing synthetic theorem rather than a new geometric derivation.

This proposition provides another example in which a classical Euclidean proof is compressed by the intermediate Hilbert interface: the mathematical content required by Euclid has already been isolated as a reusable theorem.

15.2 The Classical Configuration

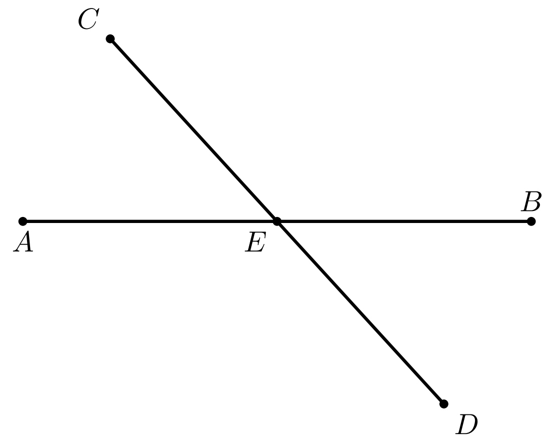
Let the straight lines AB and CD intersect at the point E, with

$$A - E - B$$

and

$$C - E - D.$$

The intersection determines four angles around E.



The opposite pairs are

$$\angle AEC \qquad \text{and} \qquad \angle BED,$$

and

$$\angle CEB \qquad \text{and} \qquad \angle DEA.$$

Proposition I.15 states that each pair is equal.

The betweenness relations

$$A - E - B, \qquad C - E - D$$

express synthetically that EA and EB are opposite rays and that EC and ED are opposite rays. Thus the classical configuration is already naturally adapted to Hilbert's order geometry.

15.3 Statement

Euclid's Proposition I.15 states:

> If two straight lines cut one another, they make the vertical angles equal to one another.

For the configuration

$$A - E - B, \qquad C - E - D,$$

the proposition gives the two vertical-angle congruences

$$\angle AEC \cong \angle BED$$

and

$$\angle CEB \cong \angle DEA.$$

In the Hilbert reconstruction we assume, in addition, that the two lines are genuinely distinct. This is expressed by

$$\neg\,\mathrm{Collinear}(A, E, C).$$

The formal statement is therefore

$$\begin{array}{l} A - E - B, \\ C - E - D, \\ \neg\,\mathrm{Collinear}(A, E, C) \end{array} \quad \Longrightarrow \quad \begin{cases} \angle AEC \cong \angle BED, \\ \angle CEB \cong \angle DEA. \end{cases}$$

The noncollinearity hypothesis is the synthetic expression of a proper intersection of two distinct straight lines. It also guarantees that the angles occurring in the proposition are nondegenerate.

Thus the Lean formulation makes explicit a condition that is implicit in Euclid's phrase "two straight lines cut one another." Strict betweenness specifies the two pairs of opposite rays, while noncollinearity excludes the case in which all five points lie on a single straight line.

15.4 Classical Proof

Euclid first proves the equality of one pair of vertical angles.

Since the straight line AE stands upon the straight line CD, Proposition I.13 gives

$$\angle CEA + \angle AED$$

equal to two right angles.

Again, since the straight line DE stands upon the straight line AB, Proposition I.13 gives

$$\angle AED + \angle DEB$$

equal to two right angles.

Therefore

$$\angle CEA + \angle AED = \angle AED + \angle DEB.$$

Subtracting the common angle $\angle AED$ from both sides, by Common Notion 3, gives

$$\angle CEA = \angle DEB.$$

Thus one pair of vertical angles is equal.

In the same way,

$$\angle CEB = \angle AED.$$

Therefore, when two straight lines cut one another, they make the vertical angles equal to one another.

15.5 Proof Architecture

The formal reconstruction and Euclid's original proof establish the same geometric fact at different levels of the theory.

Euclid derives the vertical-angle theorem from Proposition I.13. Each of two adjacent angle pairs is equal to two right angles. Since the two sums contain a common angle, Common Notion 3 allows that angle to be subtracted, leaving the two vertical angles equal.

Thus the classical dependency is

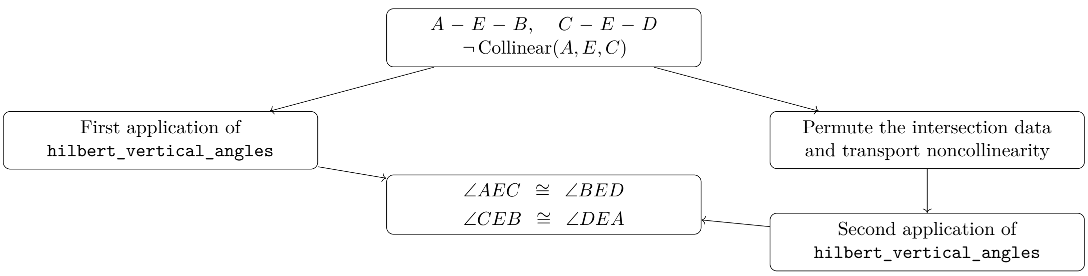
$$\text{I.13} \longrightarrow \text{two linear-pair equalities} \longrightarrow \text{Common Notion 3} \longrightarrow \text{I.15.}$$

The Hilbert development has already isolated the corresponding geometric argument in the theorem

```
hilbert_vertical_angles
```

Consequently, Proposition I.15 does not require a new reconstruction of angle addition or subtraction. The first pair of vertical angles follows by a direct application of this theorem.

The second pair is obtained by viewing the same intersection with the roles of the two straight lines exchanged. The only additional work is to transport the required noncollinearity hypothesis to the reordered triple of points.



Schematically, the two proof architectures are therefore

$$\begin{aligned} \text{Euclid} &: \text{I.13} \to \text{angle sums} \to \text{C.N.3} \to \text{vertical angles}, \\ \text{Hilbert reconstruction} &: \text{intersection data} \to \texttt{hilbert\_vertical\_angles} \to \text{vertical angles}. \end{aligned}$$

The difference is architectural rather than mathematical. The Hilbert layer has already packaged the vertical-angle argument as a reusable synthetic theorem, so the Euclidean proposition becomes a thin interface theorem over previously established geometry.

15.6 Hilbert Reconstruction

The Hilbert reconstruction starts directly from the synthetic data of the intersection:

$$A - E - B, \qquad C - E - D,$$

together with

$$\neg\,\mathrm{Collinear}(A, E, C).$$

Unlike Euclid's proof, no explicit notion of a sum of angles equal to two right angles is required. The relevant geometric content has already been established in the Hilbert layer as the vertical-angle theorem.

15.6.1 The First Pair of Vertical Angles

The theorem

```
hilbert_vertical_angles
```

has precisely the form needed for the first pair.

Applied to

$$A - E - B, \qquad C - E - D, \qquad \neg\,\mathrm{Collinear}(A, E, C),$$

it gives directly

$$\angle AEC \cong \angle BED.$$

Thus the first conclusion of Proposition I.15 requires no auxiliary construction.

15.6.2 The Second Pair of Vertical Angles

For the second pair we view the same intersection with the two straight lines exchanged.

From

$$A - E - B$$

the symmetry of strict betweenness gives

$$B - E - A.$$

We also require the noncollinearity condition

$$\neg \text{Collinear}(C, E, B).$$

Suppose, for contradiction, that

$$\text{Collinear}(C, E, B).$$

Since $A - E - B$, the points A, E, B are collinear. Permuting the first collinearity relation and using transitivity of collinearity through the distinct points E and B would then imply

$$\text{Collinear}(A, E, C),$$

contrary to the original hypothesis.

Hence

$$\neg \text{Collinear}(C, E, B).$$

We may therefore apply `hilbert_vertical_angles` a second time, now to

$$C - E - D, \qquad B - E - A.$$

This yields

$$\angle CEB \cong \angle DEA.$$

Combining the two applications gives the complete conclusion

$$\angle AEC \cong \angle BED, \qquad \angle CEB \cong \angle DEA.$$

The Hilbert reconstruction is therefore substantially shorter than Euclid's original argument. The classical use of Proposition I.13 and Common Notion 3 has already been absorbed into the previously proved vertical-angle theorem; the only additional formal work is the permutation of the intersection data required for its second application.

15.7 Lean Formalization

The reconstructed proposition is formalized by the theorem

```
theorem euclid_proposition_15
[HilbertIncidence Geo]
[HilbertCongruence Geo]
(A B C D E : Geo.Point)
(hAEB : Geo.Between A E B)
(hCED : Geo.Between C E D)
(hAEC : not PrimCollinear Geo A E C) :
Geo.AngleCongruent A E C B E D /\
Geo.AngleCongruent C E B D E A := by
```

The conclusion records both pairs of vertical angles:

$$\angle AEC \cong \angle BED$$

and

$$\angle CEB \cong \angle DEA.$$

The first pair is obtained directly:

```
have hFirst :
Geo.AngleCongruent A E C B E D :=
hilbert_vertical_angles
Geo A E C B D
hAEB
hCED
hAEC
```

For the second pair, the first line is read in the opposite direction:

```
have hBEC : Geo.Between B E A :=
(HilbertOrder.between_incidence
A E B hAEB).2.2.2.2
```

The required noncollinearity condition is then derived by contradiction. If C, E, B were collinear, cyclic permutation gives E, B, C collinear. Since A, E, B are already collinear and $E \neq B$, transitivity would imply that A, E, C are collinear, contradicting `hAEC`.

In Lean this is expressed by

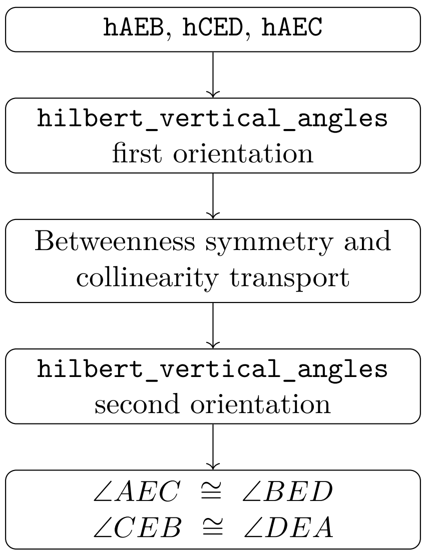
```
have hCEB :
not PrimCollinear Geo C E B := by
intro h

have hEBC :
PrimCollinear Geo E B C :=
PrimCollinearCycle Geo C E B h

have hAEBcol :
PrimCollinear Geo A E B :=
(HilbertOrder.between_incidence
A E B hAEB).2.2.2.1

have hEB : E != B :=
(HilbertOrder.between_incidence
A E B hAEB).2.1

have hAECcol :
PrimCollinear Geo A E C :=
```

```
hilbert_primCollinear_trans
Geo A E B C
hEB
hAEBcol
hEBC

exact hAEC hAECcol
```

The second pair of vertical angles is now another direct application:

```
have hSecond :
Geo.AngleCongruent C E B D E A :=
hilbert_vertical_angles
Geo C E B D A
hCED
hBEC
hCEB

exact <hFirst, hSecond>
```

Thus the formal dependency structure is



No new geometric axiom or auxiliary geometric construction is introduced in the formalization. Proposition I.15 is obtained entirely from the existing Hilbert incidence, order, collinearity, and vertical-angle theory.

15.8 Hilbert and Book Zero Components Used

Proposition I.15 is formally short because its main geometric content has already been isolated in the Hilbert layer.

The proof uses one principal theorem together with elementary order and collinearity machinery required to apply it in the two orientations of the intersection.

15.8.1 `hilbert_vertical_angles`

The central result is

```
hilbert_vertical_angles
```

which states that if

$$C - E - B, \qquad D - E - F,$$

and

$$\neg\,\mathrm{Collinear}(C, E, D),$$

then

$$\angle CED \cong \angle BEF.$$

This theorem is the immediate vertical-angle corollary developed in the Hilbert layer after the theorem on adjacent congruent angles.

In Proposition I.15 it is used twice.

The first application gives

$$\angle AEC \cong \angle BED.$$

The second application, after reversing the first straight line and transporting noncollinearity, gives

$$\angle CEB \cong \angle DEA.$$

Thus essentially all of the geometric content of Euclid's Proposition I.15 is already contained in `hilbert_vertical_angles`.

15.8.2 `HilbertOrder.between_incidence`

The theorem

```
HilbertOrder.between_incidence
```

extracts the elementary order and incidence consequences of strict betweenness.

From

$$A - E - B$$

the proof obtains both

$$B - E - A$$

and the collinearity relation

$$\text{Collinear}(A, E, B).$$

It also supplies the inequality

$$E \neq B,$$

which is needed for the transitivity of collinearity.

Thus a single betweenness hypothesis provides all of the auxiliary order data required for the second application of the vertical-angle theorem.

15.8.3 PrimCollinearCycle

The permutation theorem

```
PrimCollinearCycle
```

cyclically reorders a collinearity relation.

In the contradiction argument,

$$\text{Collinear}(C, E, B)$$

is transformed into

$$\text{Collinear}(E, B, C).$$

This places the points in exactly the orientation required by the transitivity theorem used in the next step.

15.8.4 hilbert_primCollinear_trans

The theorem

```
hilbert_primCollinear_trans
```

expresses the transitivity of collinearity when two distinct common points determine the same line.

In Proposition I.15 we combine

$$\text{Collinear}(A, E, B)$$

with

$$\text{Collinear}(E, B, C)$$

and the inequality

$$E \neq B.$$

The result is

$$\text{Collinear}(A, E, C),$$

contradicting the original noncollinearity hypothesis.

Hence

$$\neg\,\text{Collinear}(C, E, B),$$

which is precisely the hypothesis needed for the second invocation of `hilbert_vertical_angles`.

The complete formal dependency may therefore be summarized as

$$\begin{gathered}
A - E - B, \quad C - E - D, \quad \neg\,\text{Collinear}(A, E, C) \\
\downarrow \\
\texttt{hilbert_vertical_angles} \\
\downarrow \\
\angle AEC \cong \angle BED \\
\text{betweenness symmetry} + \text{collinearity transport} \\
\downarrow \\
\neg\,\text{Collinear}(C, E, B) \\
\downarrow \\
\texttt{hilbert_vertical_angles} \\
\downarrow \\
\angle CEB \cong \angle DEA.
\end{gathered}$$

No genuinely new Book Zero theorem is required for Proposition I.15. The proposition is therefore primarily an interface theorem over the already established Hilbert vertical-angle theory.

Chapter 16

Proposition I.16

16.1 Introduction

Euclid's Proposition I.16 is the exterior-angle inequality for a triangle. If a side of a triangle is extended beyond one of its vertices, the exterior angle formed at that vertex is greater than either of the two remote interior angles.

For a nondegenerate triangle ABC, extend BC through C to a point D, so that

$$B - C - D.$$

The proposition asserts

$$\angle BAC < \angle ACD \qquad \text{and} \qquad \angle ABC < \angle ACD.$$

Unlike Proposition I.15, this proposition is not merely a thin application of one previously packaged theorem. Its reconstruction requires a substantial piece of Hilbert order-and-congruence geometry: an interior ray must be transported between congruent angles, and the resulting angle comparison must be expressed without numerical angle measure.

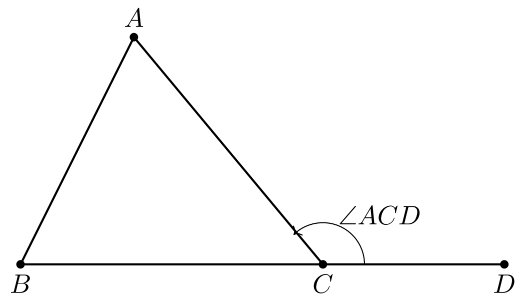
The formal development therefore isolates the geometric machinery in the Hilbert interface and leaves the Euclidean proposition itself comparatively short. In particular, the final Lean file contains three theorems:

- `euclid_proposition_16_first`,
- `euclid_proposition_16_second`,
- `euclid_proposition_16`.

16.2 The Classical Configuration

Let ABC be a nondegenerate triangle and let the side BC be produced through C to D:

$$B - C - D.$$



The angle $\angle ACD$ is exterior to the triangle. The two interior opposite angles are $\angle BAC$ and $\angle ABC$.

The formal hypotheses are

$$\neg\,\mathrm{Collinear}(A, B, C) \qquad \text{and} \qquad B - C - D.$$

The noncollinearity hypothesis makes the triangle nondegenerate, while strict betweenness records that D lies on the continuation of BC beyond C.

16.3 Statement

Euclid states Proposition I.16 as follows:

> If one side of a triangle is produced, the exterior angle is greater than either of the two interior opposite angles.

In the Hilbert reconstruction, strict comparison of angles is represented by `HilbertAngleLess`. Thus the formal conclusion is

$$\mathrm{HilbertAngleLess}(BAC, ACD) \quad \wedge \quad \mathrm{HilbertAngleLess}(ABC, ACD).$$

More explicitly,

$$\neg\,\mathrm{Collinear}(A, B, C), \quad B - C - D \Longrightarrow \begin{cases} \angle BAC < \angle ACD, \\ \angle ABC < \angle ACD. \end{cases}$$

The relation `HilbertAngleLess` is synthetic. It does not compare numerical measures. An angle is smaller than another when a congruent copy of the first can be realized as a proper interior subangle of the second.

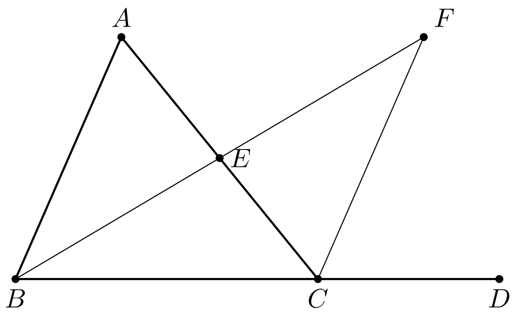
16.4 Euclid's Classical Proof

Euclid proves the first inequality by a midpoint construction.

Let E be the midpoint of AC. Join BE and extend it beyond E to a point F such that

$$BE = EF.$$

Join CF.



Since $AE = EC$, $BE = EF$, and the vertical angles $\angle AEB$ and $\angle CEF$ are equal by Proposition I.15, the triangles AEB and CEF are congruent by Proposition I.4. Hence

$$\angle BAE = \angle ECF.$$

Since E lies on AC, this gives

$$\angle BAC = \angle ACF.$$

But the ray CF lies inside the exterior angle ACD. Therefore

$$\angle BAC < \angle ACD.$$

For the second opposite angle Euclid repeats the same argument after extending another side of the triangle. The equality of the corresponding vertical exterior angles then identifies the resulting exterior angle with $\angle ACD$.

Thus the classical architecture is

$$\text{midpoint construction} \to \text{I.15} \to \text{SAS (I.4)} \to \text{interior subangle} \to \text{I.16}.$$

16.5 Proof Architecture of the Reconstruction

The Hilbert reconstruction retains the synthetic structure of Euclid's argument but packages the difficult order-theoretic steps into reusable interface theorems.

For the first opposite angle the auxiliary construction produces points M and E with

$$A - M - C, \qquad B - M - E,$$

and the required segment and angle congruences. From these data the interface proves that AB is parallel to CE. The theorem `hilbert_exterior_angle_less` then concludes directly that

$$\angle BAC < \angle ACD.$$

Schematically:

$$
\begin{gathered}
\neg\,\mathrm{Collinear}(A,B,C)\\
\downarrow\\
\texttt{hilbert\_exterior\_angle\_aux}\\
\downarrow\\
A-M-C,\ B-M-E,\ \text{congruence data}\\
\downarrow\\
\texttt{hilbert\_exterior\_angle\_aux\_parallel}\\
\downarrow\\
AB \parallel CE\\
\downarrow\\
\texttt{hilbert\_exterior\_angle\_less}\\
\downarrow\\
\angle BAC < \angle ACD.
\end{gathered}
$$

The second inequality is deliberately reduced to the first. Extend AC through C to F. Apply the first part to triangle BAC to obtain

$$\angle ABC < \angle BCF.$$

The lines ACF and BCD intersect at C, so Proposition I.15 gives the vertical-angle congruence

$$\angle BCF \cong \angle ACD.$$

Transporting the right-hand side of the strict angle comparison across this congruence yields

$$\angle ABC < \angle ACD.$$

Thus the second half has the compact architecture

$$\text{first half of I.16} \to \text{I.15 / vertical angles} \to \text{transport of } < \to \text{second half of I.16.}$$

16.6 Hilbert Reconstruction

16.6.1 The First Opposite Interior Angle

The theorem `euclid_proposition_16_first` begins from

$$\neg\,\mathrm{Collinear}(A,B,C), \qquad B-C-D.$$

It invokes

`hilbert_exterior_angle_aux`

to obtain points M, E and the auxiliary configuration required for the exterior-angle argument.

The same configuration is passed to

`hilbert_exterior_angle_aux_parallel`,

which proves

$$AB \parallel CE.$$

Finally,

`hilbert_exterior_angle_less`

combines the auxiliary construction, the extension $B - C - D$, and the parallelism to establish

$$\angle BAC < \angle ACD.$$

The Euclidean theorem itself therefore contains no unresolved construction. The substantial work has been moved into the Hilbert interface, where it can be reused independently of Proposition I.16.

16.6.2 The Second Opposite Interior Angle

For the second inequality, the proof first derives $A \neq C$ from the noncollinearity of A, B, C. Hilbert's order axiom then extends AC through C to a point F:

$$A - C - F.$$

The triangle is viewed in the order BAC. Applying `euclid_proposition_16_first` to this reordered triangle gives

$$\angle ABC < \angle BCF.$$

It remains to identify $\angle BCF$ with the desired exterior angle $\angle ACD$. From $B - C - D$ we obtain the reversed betweenness relation

$$D - C - B.$$

The two straight lines ACF and BCD therefore form a pair of vertical angles at C. The theorem `VerticalAngles` yields, after the required symmetry and reversal operations,

$$\angle BCF \cong \angle ACD.$$

The theorem

`hilbert_angleLess_transport_right`

then transports the comparison across the congruent right-hand angle:

$$\angle ABC < \angle BCF, \qquad \angle BCF \cong \angle ACD \quad \Longrightarrow \quad \angle ABC < \angle ACD.$$

16.7 Lean Formalization

The first half is formalized as follows:

```
theorem euclid_proposition_16_first
    [HilbertIncidence Geo]
    [HilbertCongruence Geo]
    (A B C D : Geo.Point)
    (hABC : not PrimCollinear Geo A B C)
    (hBCD : Geo.Between B C D) :
    HilbertAngleLess Geo B A C A C D := by

  rcases
```

```
    hilbert_exterior_angle_aux
      Geo A B C hABC with
   <M, E,
    hAMC,
    hAMMC,
    hBME,
    hBMME,
    hAngle>

  have hParallel :
    Geo.Parallel A B C E :=
   hilbert_exterior_angle_aux_parallel
    Geo A B C M E
    hABC
    hAMC
    hBME
    hAngle

  exact
   hilbert_exterior_angle_less
    Geo A B C D M E
    hABC
    hAMC
    hBME
    hBCD
    hAngle
    hParallel
```

The second half deliberately reuses the first:

```
have hLessBCF :
  HilbertAngleLess Geo A B C B C F :=
 euclid_proposition_16_first
  Geo
  B A C F
  hBACnc
  hACF
```

The vertical-angle relation is then constructed at C and normalized to the required orientation:

```
have hVerticalRaw :
  Geo.AngleCongruent A C D F C B :=
 VerticalAngles
  Geo
  A C D F B
  hACF
  hDCA
  hACDnc

have hVerticalSymm :
  Geo.AngleCongruent F C B A C D :=
 Geo.angle_congruent_symmetry
  A C D
  F C B
  hVerticalRaw

have hVertical :
  Geo.AngleCongruent B C F A C D :=
```

```
  (Geo.angle_congruent_reverse_first
    F C B A C D).mp
    hVerticalSymm
```

Finally the strict inequality is transported:

```
exact
  hilbert_angleLess_transport_right
    Geo
    A B C
    B C F
    A C D
    hLessBCF
    hACDnc
    hVertical
```

The complete proposition simply combines the two parts:

```
theorem euclid_proposition_16
    [HilbertIncidence Geo]
    [HilbertCongruence Geo]
    (A B C D : Geo.Point)
    (hABC : not PrimCollinear Geo A B C)
    (hBCD : Geo.Between B C D) :
    HilbertAngleLess Geo B A C A C D /\
    HilbertAngleLess Geo A B C A C D := by

  have hFirst :=
    euclid_proposition_16_first
      Geo A B C D hABC hBCD

  have hSecond :=
    euclid_proposition_16_second
      Geo A B C D hABC hBCD

  exact <hFirst, hSecond>
```

16.8 Hilbert and Book Zero Components Used

16.8.1 `HilbertAngleLess`

The proposition requires a genuinely synthetic notion of strict angle comparison. The relation `HilbertAngleLess` expresses that a congruent copy of one angle occurs as a proper interior subangle of another. This avoids introducing numerical angle measure and keeps Proposition I.16 inside pure incidence, order, and congruence geometry.

16.8.2 `hilbert_exterior_angle_aux`

This theorem supplies the auxiliary points used in the first exterior-angle construction. It packages the midpoint-style configuration that underlies Euclid's original proof and returns the betweenness and congruence data required by the subsequent argument.

16.8.3 `hilbert_exterior_angle_aux_parallel`

From the auxiliary configuration, this theorem proves

$$AB \parallel CE.$$

The parallel relation is not an additional Euclidean postulate here; it is a derived relation inside the neutral Hilbert development used to organize the exterior-angle argument.

16.8.4 `hilbert_exterior_angle_less`

This is the principal interface theorem used by the first half of I.16. It converts the auxiliary construction and its parallelism into the strict angle comparison

$$\angle BAC < \angle ACD.$$

Its proof depends on the interior-subangle machinery developed in the Hilbert layer.

16.8.5 Interior-subangle transport

A substantial part of the work required for Proposition I.16 lies below the final Euclidean file. The Hilbert interface establishes that an interior ray can be transported between congruent angles while preserving the corresponding subangle.

The completed infrastructure includes the segment and crossbar mechanisms needed for that transport, in particular the ability to transport strict segment comparison through congruence, transport an interior point between congruent segments, and preserve ray–segment intersection when the endpoints are moved along the same two rays.

This is the part of the reconstruction where the formal proof exposes geometric structure that is largely implicit in the paper proof.

16.8.6 `VerticalAngles`

The second half uses the vertical-angle theorem already exposed by Proposition I.15. After AC is extended through C to F, the angles

$$\angle BCF \quad \text{and} \quad \angle ACD$$

are vertical. Their congruence allows the second inequality to be reduced to the first.

16.8.7 `hilbert_angleLess_transport_right`

This theorem expresses invariance of strict angle comparison under congruence of the right-hand angle. In the final step,

$$\angle ABC < \angle BCF$$

and

$$\angle BCF \cong \angle ACD$$

imply

$$\angle ABC < \angle ACD.$$

16.9 Dependency Summary

The formal dependency structure can be summarized as

$$
\begin{array}{c}
\text{Hilbert incidence + order + congruence}\\
\downarrow\\
\text{segment construction and ray/order theory}\\
\downarrow\\
\text{interior-subangle transport / crossbar machinery}\\
\downarrow\\
\texttt{hilbert\_exterior\_angle\_less}\\
\downarrow\\
\texttt{euclid\_proposition\_16\_first}\\
\downarrow\\
\text{vertical angles + angle-less transport}\\
\downarrow\\
\texttt{euclid\_proposition\_16\_second}\\
\downarrow\\
\texttt{euclid\_proposition\_16}.
\end{array}
$$

The final Euclidean theorem is therefore compact, but its compactness is meaningful: the difficult geometry has been factored into reusable Hilbert-interface results rather than hidden in an axiom or a local `sorry`.

16.10 Conclusion

Proposition I.16 marks a significant step beyond the immediately preceding propositions. It introduces a strict comparison of angles and requires a robust synthetic notion of interiority. In Euclid's presentation this is handled by a midpoint construction, SAS, and the observation that one angle is a proper part of another. In the formal Hilbert reconstruction the same content is decomposed into explicit reusable mechanisms for rays, betweenness, congruence, crossbar behavior, and transport of strict angle comparison.

The final Lean theorem is short precisely because those mechanisms have been established independently. No additional axiom is introduced in `Proposition16.lean`, and the completed development compiles without `sorry`.

Chapter 17

Proposition I.17

17.1 Introduction

Euclid's Proposition I.17 states that in every triangle any two interior angles are together less than two right angles.

For a nondegenerate triangle ABC, the three conclusions are

$$\angle BAC + \angle ABC < 2R,$$

$$\angle BAC + \angle ACB < 2R,$$

and

$$\angle ABC + \angle ACB < 2R.$$

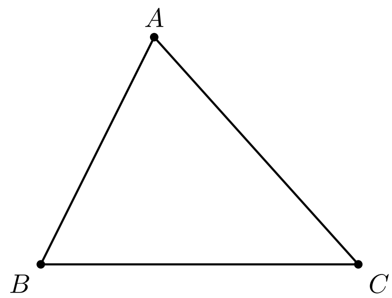
In Euclid's classical proof, Proposition I.16 supplies an exterior-angle inequality and Proposition I.13 identifies an adjacent pair of angles with two right angles. The informal argument then adds the same angle to both sides of an inequality.

The formal reconstruction takes a different route. No numerical angle measure and no binary operation of angle addition are introduced. Instead, the phrase "two angles are together less than two right angles" is represented synthetically: the first angle is required to be strictly smaller than the supplementary angle of the second. This follows the same design principle already used in the reconstruction of Proposition I.14, where "equal to two right angles" was encoded by comparison with a supplementary angle.

The final Lean development consists of a new interface predicate, three pairwise forms of Proposition I.17, and one wrapper theorem combining them.

17.2 The Classical Configuration

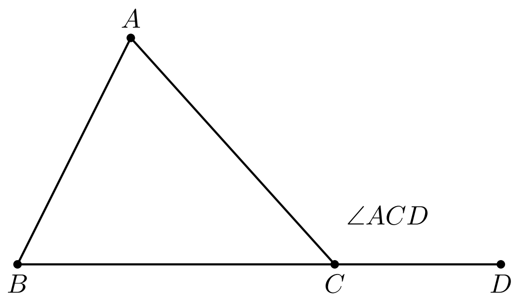
Let ABC be a nondegenerate triangle.



Euclid asserts that every pair among the three interior angles has sum strictly smaller than two right angles.

For example, extend BC beyond C to a point D:

$$B - C - D.$$



Proposition I.16 gives

$$\angle BAC < \angle ACD \qquad \text{and} \qquad \angle ABC < \angle ACD.$$

Since CB and CD are opposite rays, $\angle ACB$ and $\angle ACD$ form a supplementary pair. Thus both

$$\angle BAC + \angle ACB < 2R$$

and

$$\angle ABC + \angle ACB < 2R$$

follow from the same extension.

The remaining pair is obtained by extending CB beyond B.

17.3 Classical Proof

Euclid proves one representative case and then observes that the same argument applies to the other pairs.

Extend BC beyond C to D. By Proposition I.16,

$$\angle ABC < \angle ACD.$$

Add $\angle ACB$ to both sides:

$$\angle ABC + \angle ACB < \angle ACD + \angle ACB.$$

By Proposition I.13, the adjacent angles $\angle ACD$ and $\angle ACB$ are together equal to two right angles. Hence

$$\angle ABC + \angle ACB < 2R.$$

The other two pairs follow in the same way after choosing the appropriate side extension.

The classical dependency pattern is therefore

$$\text{I.16 exterior-angle inequality} \longrightarrow \text{add the remaining interior angle} \longrightarrow \text{I.13 supplementary pair} \longrightarrow < 2R.$$

The formal reconstruction deliberately removes the middle arithmetic step.

17.4 Synthetic Representation of "Less Than Two Right Angles"

The central design choice is the predicate

`HilbertAnglesLessThanTwoRightAngles`.

For two angles α and β, instead of introducing a sum $\alpha + \beta$, choose a supplementary angle β' to β. Then

$$\alpha + \beta < 2R$$

is represented by

$$\alpha < \beta'.$$

In the formal language, if the second angle is $\angle CPD$, a witness E is chosen with

$$D - P - E.$$

Then PE is the opposite ray to PD, so $\angle CPE$ is supplementary to $\angle CPD$. The definition requires

$$\angle AOB < \angle CPE.$$

The Lean definition is:

```
def HilbertAnglesLessThanTwoRightAngles
    [HilbertIncidence Geo]
    (A O B C P D : Geo.Point) : Prop :=
  exists E : Geo.Point,
    Geo.Between D P E /\
    HilbertAngleLess Geo A O B C P E
```

This representation is entirely synthetic. It records only strict betweenness and strict comparison of angles. No real-valued measure, no degree notation, and no algebraic operation on angles is required.

17.5 Proof Architecture of the Reconstruction

The reconstruction is especially compact because Proposition I.16 already gives exactly the inequalities needed against exterior angles.

For the two pairs containing $\angle ACB$, extend BC beyond C:

$$B - C - D.$$

Then Proposition I.16 gives

$$\angle BAC < \angle ACD$$

and

$$\angle ABC < \angle ACD.$$

Since $\angle ACD$ is the supplement of $\angle ACB$, these are precisely the synthetic forms of

$$\angle BAC + \angle ACB < 2R$$

and

$$\angle ABC + \angle ACB < 2R.$$

For the remaining pair, extend CB beyond B:

$$C - B - E.$$

Apply Proposition I.16 to the reordered triangle ACB. This gives

$$\angle CAB < \angle ABE.$$

Reversal of the first angle identifies $\angle CAB$ with $\angle BAC$, while $\angle ABE$ is supplementary to $\angle ABC$. Hence

$$\angle BAC + \angle ABC < 2R.$$

Schematically:

$$\begin{array}{c}
\neg\,\text{Collinear}(A, B, C) \\
\downarrow \\
\text{extend } BC \text{ beyond } C \\
\downarrow \\
B - C - D \\
\downarrow \\
\text{I.16}
\end{array}$$

$$\begin{array}{cc}
\swarrow & \searrow \\
\angle BAC < \angle ACD & \angle ABC < \angle ACD \\
\downarrow & \downarrow \\
\angle BAC + \angle ACB < 2R & \angle ABC + \angle ACB < 2R
\end{array}$$

and separately

$$\begin{array}{c}
\text{extend } CB \text{ beyond } B \\
\downarrow \\
C - B - E \\
\downarrow \\
\text{I.16 on triangle } ACB \\
\downarrow \\
\angle CAB < \angle ABE \\
\downarrow \\
\angle BAC + \angle ABC < 2R.
\end{array}$$

17.6 Lean Formalization

17.6.1 The Pair $\angle BAC$ and $\angle ACB$

The first theorem extends BC through C and uses the first half of Proposition I.16.

```
theorem euclid_proposition_17_BAC_ACB
    [HilbertIncidence Geo]
    [HilbertCongruence Geo]
    (A B C : Geo.Point)
    (hABC : not PrimCollinear Geo A B C) :
    HilbertAnglesLessThanTwoRightAngles
      Geo B A C A C B := by

  have hBCA :
      not PrimCollinear Geo B C A := by
    intro h

    have hACB :
        PrimCollinear Geo A C B :=
      PrimCollinearSymm Geo B C A h

    have hABC' :
        PrimCollinear Geo A B C :=
      PrimCollinearRotate Geo A C B hACB

    exact hABC hABC'

  have hBC : B != C :=
    hilbert_noncollinear_ne_first
      Geo B C A hBCA

  rcases
      HilbertOrder.between_extension
        B C hBC with
    <D, hBCD>

  refine <D, hBCD, ?_>

  exact
    euclid_proposition_16_first
      Geo A B C D hABC hBCD
```

The witness D simultaneously records the supplementary configuration and serves as the exterior-angle target in I.16.

17.6.2 The Pair $\angle ABC$ and $\angle ACB$

The second theorem uses the same extension but invokes the second half of I.16.

```
theorem euclid_proposition_17_ABC_ACB
    [HilbertIncidence Geo]
    [HilbertCongruence Geo]
    (A B C : Geo.Point)
    (hABC : not PrimCollinear Geo A B C) :
    HilbertAnglesLessThanTwoRightAngles
```

```
    Geo A B C A C B := by

  have hBCA :
      not PrimCollinear Geo B C A := by
    intro h

    have hACB :
        PrimCollinear Geo A C B :=
      PrimCollinearSymm Geo B C A h

    have hABC' :
        PrimCollinear Geo A B C :=
      PrimCollinearRotate Geo A C B hACB

    exact hABC hABC'

  have hBC : B != C :=
    hilbert_noncollinear_ne_first
      Geo B C A hBCA

  rcases
      HilbertOrder.between_extension
        B C hBC with
    <D, hBCD>

  refine <D, hBCD, ?_>

  exact
    euclid_proposition_16_second
      Geo A B C D hABC hBCD
```

Thus the two inequalities associated with the vertex C differ only in which conclusion of I.16 is selected.

17.6.3 The Pair $\angle BAC$ and $\angle ABC$

For the remaining pair, extend CB through B to E and apply I.16 to the triangle in the order ACB.

```
theorem euclid_proposition_17_BAC_ABC
    [HilbertIncidence Geo]
    [HilbertCongruence Geo]
    (A B C : Geo.Point)
    (hABC : not PrimCollinear Geo A B C) :
    HilbertAnglesLessThanTwoRightAngles
      Geo B A C A B C := by

  have hCBA :
      not PrimCollinear Geo C B A := by
    intro h

    have hABC' :
        PrimCollinear Geo A B C :=
      PrimCollinearSymm Geo C B A h

    exact hABC hABC'
```

```
have hCB : C != B :=
  hilbert_noncollinear_ne_first
    Geo C B A hCBA

rcases
    HilbertOrder.between_extension
      C B hCB with
  <E, hCBE>

refine <E, hCBE, ?_>

have hACB :
    not PrimCollinear Geo A C B := by
  intro h

  have hABC' :
      PrimCollinear Geo A B C :=
    PrimCollinearRotate Geo A C B h

  exact hABC hABC'

have hLessCAB :
    HilbertAngleLess Geo C A B A B E :=
  euclid_proposition_16_first
    Geo A C B E hACB hCBE

rcases hLessCAB with
  <hCABnc, hABEnc, X, hMeet, hAngle>

have hBACnc :
    not PrimCollinear Geo B A C := by
  intro h
  exact hABC
    (PrimCollinearSwap Geo B A C h)

have hAngle' :
    Geo.AngleCongruent B A C A B X :=
  (Geometry.Geo.angle_congruent_reverse_first
    Geo C A B A B X).mp hAngle

exact
  <hBACnc,
    hABEnc,
    X,
    hMeet,
    hAngle'>
```

The only additional work here is normalization of the orientation of the first angle: I.16 returns $\angle CAB < \angle ABE$, whereas the final interface uses $\angle BAC$. The unordered-angle representation makes this a reversal operation rather than a new geometric argument.

17.6.4 The Final Wrapper

The final proposition simply combines the three pairwise results.

```
theorem euclid_proposition_17
    [HilbertIncidence Geo]
```

```
  [HilbertCongruence Geo]
  (A B C : Geo.Point)
  (hABC : not PrimCollinear Geo A B C) :
  HilbertAnglesLessThanTwoRightAngles Geo B A C A B C /\
  HilbertAnglesLessThanTwoRightAngles Geo B A C A C B /\
  HilbertAnglesLessThanTwoRightAngles Geo A B C A C B := by

have h1 :=
  euclid_proposition_17_BAC_ABC
    Geo A B C hABC

have h2 :=
  euclid_proposition_17_BAC_ACB
    Geo A B C hABC

have h3 :=
  euclid_proposition_17_ABC_ACB
    Geo A B C hABC

exact <h1, h2, h3>
```

17.7 Relationship with Propositions I.13, I.14, and I.16

Proposition I.17 sits at an interesting point in the reconstruction because its classical statement appears to require angle arithmetic, while the existing synthetic interface allows that arithmetic to be avoided.

17.7.1 Proposition I.13

I.13 identifies two adjacent angles on a straight line as a supplementary pair. In the Book Zero reconstruction this is represented structurally by `BookZeroSupplement`, rather than by an equation involving angle sums.

For I.17 we use the same geometric idea but do not need to invoke the theorem I.13 explicitly. The witness produced by `between_extension` already records the opposite ray required to construct the supplementary angle.

17.7.2 Proposition I.14

I.14 introduced the useful design pattern of representing "equal to two right angles" by congruence with a supplementary angle. The I.17 predicate is the strict-inequality analogue of that pattern:

$$\alpha + \beta = 2R \quad \rightsquigarrow \quad \alpha \cong \mathrm{supp}(\beta),$$

while

$$\alpha + \beta < 2R \quad \rightsquigarrow \quad \alpha < \mathrm{supp}(\beta).$$

Thus I.17 extends an already established interface convention rather than introducing a new angle algebra.

17.7.3 Proposition I.16

I.16 is the decisive geometric input. Once a side is extended, it provides strict comparison of each remote interior angle with the exterior angle. Since that exterior angle is automatically supplementary to the adjacent interior angle, the conclusions of I.16 are already in the exact form required by `HilbertAnglesLessThanTwoRightAngles`.

This gives the particularly short dependency path

$$\text{betweenness extension} \longrightarrow \text{I.16} \longrightarrow \text{synthetic } < 2R.$$

17.8 Book Zero and Hilbert Components Used

17.8.1 `HilbertAngleLess`

The strict comparison relation introduced for I.16 remains the fundamental notion of angle inequality. It is synthetic: one angle is less than another when a congruent copy of the first occurs as a proper interior subangle of the second.

17.8.2 `HilbertOrder.between_extension`

This construction supplies the points D and E on the required opposite rays:

$$B - C - D \qquad \text{or} \qquad C - B - E.$$

The witness is simultaneously the side extension used by I.16 and the supplementary-ray witness used by the definition of "less than two right angles."

17.8.3 Collinearity Symmetry and Rotation

The three applications of I.16 require the triangle vertices in different orders. The small uses of `PrimCollinearSymm`, `PrimCollinearRotate`, and `PrimCollinearSwap` normalize noncollinearity hypotheses without adding any geometry.

17.8.4 Angle Reversal

In the third pair, Proposition I.16 naturally produces the angle $\angle CAB$, while the public form of I.17 uses $\angle BAC$. The theorem `angle_congruent_reverse_first` transfers the congruence across reversal of the first angle.

17.9 Dependency Summary

The formal dependency structure is

$$\begin{array}{c}
\text{Hilbert incidence + order + congruence} \\
\downarrow \\
\texttt{HilbertAngleLess} \\
\downarrow \\
\text{Euclid I.16} \\
\downarrow \\
\texttt{HilbertAnglesLessThanTwoRightAngles} \\
\downarrow \\
\texttt{I.17_BAC_ABC} \quad \texttt{I.17_BAC_ACB} \quad \texttt{I.17_ABC_ACB} \\
\downarrow \\
\texttt{euclid_proposition_17}.
\end{array}$$

No additional geometric axiom is introduced. More importantly, no general arithmetic of angles is introduced merely to reproduce Euclid's phrase "together less than two right angles." The proposition is expressed entirely in the existing synthetic language of rays, betweenness, supplementary configurations, and strict angle comparison.

17.10 Conclusion

Proposition I.17 is a useful example of the difference between reproducing a classical proof literally and reconstructing its mathematical content over a synthetic interface.

Euclid's paper proof uses an implicit monotonicity principle for angle addition: from $\alpha < \beta$ one passes to $\alpha + \gamma < \beta + \gamma$. A literal formalization would therefore require an angle-sum operation and an order theory for such sums.

The Hilbert/Book Zero reconstruction avoids that additional layer. The statement "two angles are together less than two right angles" is encoded directly by comparison of one angle with the supplement of the other. Once this representation is chosen, Proposition I.16 supplies exactly the needed inequalities, and Proposition I.17 becomes a short structural consequence of side extension and exterior-angle comparison.

The completed Lean development compiles cleanly and contains no `sorry`.

Chapter 18

Proposition I.18

18.1 Introduction

Euclid's Proposition I.18 states that in a triangle the greater side is opposite the greater angle.

For a nondegenerate triangle ABC, if

$$AB < AC,$$

then

$$\angle ACB < \angle ABC.$$

The classical proof is short but structurally rich. A copy of the shorter side AB is laid off on the longer side AC. This produces a point D with

$$A - D - C \qquad \text{and} \qquad AD \cong AB.$$

The auxiliary triangle ABD is therefore isosceles. Proposition I.5 identifies its base angles, while Proposition I.16 compares one of these angles with an angle of the original triangle. Finally, the fact that an interior subangle is smaller than the whole angle completes the comparison.

The formal reconstruction follows this synthetic argument closely. It does not introduce numerical angle measures. Instead, it uses the strict comparison relation `HilbertAngleLess` already developed for Proposition I.16.

The reconstruction also isolates two general facts about strict angle comparison:

`hilbert_interior_angle_less`

and

`hilbert_angleLess_trans`.

These belong naturally to the Hilbert interface rather than to Proposition I.18 itself.

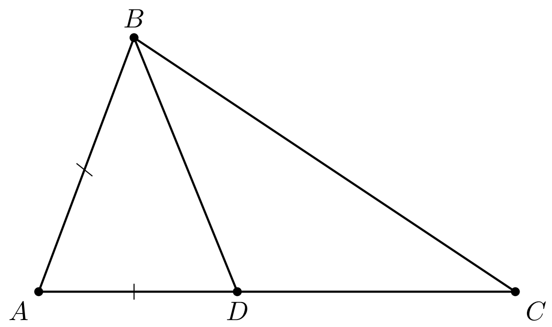
18.2 The Classical Configuration

Let ABC be a nondegenerate triangle and suppose

$$AB < AC.$$

Choose D on AC so that

$$A - D - C \qquad \text{and} \qquad AD \cong AB.$$



The construction creates two auxiliary triangles.

First, triangle ABD is isosceles because

$$AB \cong AD.$$

Hence Proposition I.5 gives

$$\angle ABD \cong \angle ADB.$$

Second, consider triangle BCD. Since $A - D - C$, the ray DA is the continuation of DC through D. Thus $\angle BDA$ is an exterior angle of triangle BCD.

Proposition I.16 therefore gives

$$\angle BCD < \angle BDA.$$

Because A, D, C are collinear with D between A and C, the rays CD and CA coincide. Hence

$$\angle BCD = \angle BCA.$$

Combining these observations gives

$$\angle BCA < \angle BDA \cong \angle ABD.$$

Finally, the ray BD lies inside $\angle ABC$, since it meets the segment AC at the interior point D. Therefore

$$\angle ABD < \angle ABC.$$

By transitivity,

$$\angle BCA < \angle ABC,$$

which is the desired conclusion.

18.3 Classical Proof

Assume that in triangle ABC,

$$AC > AB.$$

Choose a point D on AC such that

$$AD = AB.$$

Since $AC > AB$, the point D lies strictly between A and C.

The triangle ABD is isosceles, so by Proposition I.5,

$$\angle ABD = \angle ADB.$$

Now $\angle ADB$ is an exterior angle of triangle BCD. Proposition I.16 gives

$$\angle ADB > \angle BCD.$$

Since D lies on AC,

$$\angle BCD = \angle BCA.$$

Consequently,

$$\angle ABD > \angle BCA.$$

But BD lies inside $\angle ABC$, so

$$\angle ABC > \angle ABD.$$

Therefore

$$\angle ABC > \angle BCA.$$

The classical dependency pattern is thus

$$\begin{gathered} AB < AC \\ \downarrow \\ A - D - C, \quad AB \cong AD \\ \downarrow \\ \text{I.5} \\ \downarrow \\ \angle ABD \cong \angle ADB \\ \downarrow \\ \text{I.16 on } BCD \\ \downarrow \\ \angle BCA < \angle ADB \\ \downarrow \\ \angle BCA < \angle ABD \\ \downarrow \\ \angle ABD < \angle ABC \\ \downarrow \\ \angle BCA < \angle ABC. \end{gathered}$$

18.4 Strict Angle Comparison

The formal statement uses the synthetic relation

`HilbertAngleLess`.

No numerical measure of angles is introduced. Informally,

$$\alpha < \beta$$

means that a congruent copy of α occurs as a proper interior subangle of β.

This relation was already required for Proposition I.16. Proposition I.18 reveals that two further structural properties are needed if strict angle comparison is to behave as a genuine order relation.

The first is that an interior subangle is smaller than the whole angle.

```
hilbert_interior_angle_less
```

Geometrically, if a ray from the vertex of $\angle AOB$ meets the opposite side AB internally, then the angle cut off by that ray is strictly smaller than $\angle AOB$.

The second property is transitivity.

```
hilbert_angleLess_trans
```

It expresses the expected implication

$$\alpha < \beta \quad \text{and} \quad \beta < \gamma \quad \Longrightarrow \quad \alpha < \gamma.$$

Both results are independent of Proposition I.18 and were therefore added to the general Hilbert interface.

18.5 Proof Architecture of the Reconstruction

The formal proof separates naturally into five geometric stages.

18.5.1 Recovering the Interior Point

The hypothesis

$$AB < AC$$

is represented by `HilbertSegmentLess`.

Unpacking it produces a witness D satisfying

$$A - D - C$$

and

$$AB \cong AD.$$

Thus the first step of Euclid's construction is already contained in the definition of strict segment comparison.

18.5.2 The Isosceles Triangle

From

$$AB \cong AD$$

the existing theorem

`hilbert_isosceles_base_angles`

gives

$$\angle ABD \cong \angle ADB.$$

This is precisely Proposition I.5 in the form required by the reconstruction.

18.5.3 The Exterior-Angle Comparison

Since

$$A - D - C,$$

we also have

$$C - D - A.$$

Apply Proposition I.16 to triangle BCD with DA as the extension of DC. This yields

$$\angle BCD < \angle BDA.$$

The ray from C through D is the same ray as the ray from C through A. Hence the first angle can be normalized:

$$\angle BCD = \angle BCA.$$

Thus

$$\angle BCA < \angle BDA.$$

Using the isosceles-angle congruence and the symmetry of the unordered angle representation gives

$$\angle BCA < \angle ABD.$$

18.5.4 The Interior Subangle

The point D lies on the segment AC. Consequently the ray BD meets the interior of that segment.

Therefore BD lies inside $\angle ABC$, and `hilbert_interior_angle_less` gives

$$\angle ABD < \angle ABC.$$

18.5.5 Transitivity

We now have

$$\angle BCA < \angle ABD$$

and

$$\angle ABD < \angle ABC.$$

Applying `hilbert_angleLess_trans` gives

$$\angle BCA < \angle ABC.$$

The final public statement writes the first angle as $\angle ACB$. Since the formal angle representation is insensitive to reversal of its two rays, this is a final orientation normalization rather than an additional geometric argument.

18.6 Lean Formalization

The public theorem has the form

```
theorem euclid_proposition_18
[HilbertIncidence Geo]
[HilbertCongruence Geo]
(A B C : Geo.Point)
(hABC : not PrimCollinear Geo A B C)
(hABltAC : HilbertSegmentLess Geo A B A C) :
HilbertAngleLess Geo A C B A B C := by
```

The hypothesis `hABltAC` is immediately unpacked:

```
rcases hABltAC with
<D, hADC, hAB_AD>
```

Thus `hADC` records

$$A - D - C,$$

while `hAB_AD` records

$$AB \cong AD.$$

The isosceles part of the argument is then:

```
have hIso :
Geo.AngleCongruent A B D A D B :=
hilbert_isosceles_base_angles
Geo
A B D
hABD
hAB_AD
```

The exterior-angle step uses Proposition I.16:

```
have hBCD_BDA :
HilbertAngleLess Geo B C D B D A :=
euclid_proposition_16_second
Geo B C D A hBCD hCDA
```

Here

$$C - D - A$$

is obtained from the original betweenness relation

$$A - D - C.$$

The ray identity $CD = CA$ is expressed by `HilbertSameRay`:

```
have hRayCDA :
HilbertSameRay Geo C D A :=
hilbert_sameRay_of_between
Geo C D A hCDA
```

This permits normalization of

$$\angle BCD$$

to

$$\angle BCA.$$

After transporting the right-hand angle across the isosceles congruence, the proof obtains

```
have hBCA_ABD :
HilbertAngleLess Geo B C A A B D := ...
```

which represents

$$\angle BCA < \angle ABD.$$

For the second comparison, the ray BD meets the segment AC at D:

```
have hInsideBD :
HilbertRayMeetsSegment Geo B D A C :=
<D, hADC, hRayBDD>
```

Hence

```
have hABD_ABC :
HilbertAngleLess Geo A B D A B C :=
hilbert_interior_angle_less
Geo
B D A C
hABC
hInsideBD
```

which is exactly

$$\angle ABD < \angle ABC.$$

Transitivity now gives

```
have hFinal :
HilbertAngleLess Geo B C A A B C :=
hilbert_angleLess_trans
Geo
B C A
A B D
A B C
hBCA_ABD
hABD_ABC
```

The final lines merely reverse the first angle from

$$\angle BCA$$

to

$$\angle ACB,$$

matching the public orientation of the proposition.

18.7 Relationship with Earlier Propositions

18.7.1 Proposition I.5

Proposition I.5 supplies the equality of the base angles in the auxiliary isosceles triangle ABD:

$$AB \cong AD \implies \angle ABD \cong \angle ADB.$$

This is the bridge between the comparison of sides and the comparison of angles.

18.7.2 Proposition I.16

Proposition I.16 supplies the strict inequality. Applied to triangle BCD, it compares the remote interior angle at C with the exterior angle at D:

$$\angle BCD < \angle BDA.$$

Thus I.18 depends essentially on the exterior-angle theorem proved two propositions earlier.

18.7.3 Proposition I.17

Proposition I.17 is not required by the proof of I.18. The formal dependency follows Euclid's direct argument through I.5 and I.16.

This is useful structurally: I.18 does not rely on an angle-sum theory developed for I.17. Both propositions instead reuse the synthetic strict-angle relation introduced for I.16.

18.8 Hilbert and Book Zero Components Used

18.8.1 `HilbertSegmentLess`

The hypothesis that one side is shorter than another already contains the point required by Euclid's construction. Unpacking

$$AB < AC$$

produces

$$A - D - C \qquad \text{and} \qquad AB \cong AD.$$

Thus no separate segment-layoff construction is required inside Proposition I.18.

18.8.2 `hilbert_isosceles_base_angles`

This theorem supplies the I.5 step for the auxiliary triangle ABD.

18.8.3 `HilbertSameRay`

Same-ray machinery is used to recognize that, because $A - D - C$, the rays CD and CA determine the same side of the angle at C.

This turns the I.16 result about $\angle BCD$ into the required comparison involving the original angle $\angle BCA$.

18.8.4 `hilbert_interior_angle_less`

This new interface theorem expresses the elementary principle that a proper interior part of an angle is smaller than the whole angle.

In I.18 it gives

$$\angle ABD < \angle ABC$$

from the fact that BD meets AC internally.

18.8.5 `hilbert_angleLess_trans`

This new interface theorem provides transitivity of strict angle comparison.

It combines

$$\angle BCA < \angle ABD$$

and

$$\angle ABD < \angle ABC$$

into the final inequality.

18.9 Dependency Summary

The formal dependency structure is

$$\begin{array}{c}
\text{Hilbert incidence + order + congruence} \\
\downarrow \\
\texttt{HilbertSegmentLess} \quad \texttt{HilbertAngleLess} \\
\downarrow \\
A - D - C, \quad AB \cong AD \\
\downarrow \\
\texttt{hilbert_isosceles_base_angles} \quad \text{(I.5)} \\
\downarrow \\
\angle ABD \cong \angle ADB \\
\downarrow \\
\text{Euclid I.16} \\
\downarrow \\
\angle BCA < \angle ABD \\
\downarrow \\
\texttt{hilbert_interior_angle_less} \\
\downarrow \\
\angle ABD < \angle ABC \\
\downarrow \\
\texttt{hilbert_angleLess_trans} \\
\downarrow \\
\texttt{euclid_proposition_18}.
\end{array}$$

No new geometric axiom is introduced.

The two results added to the Hilbert interface are structural properties of the already existing strict-angle relation rather than assumptions specific to Proposition I.18.

18.10 Conclusion

Proposition I.18 is a particularly clear example of how Euclid's construction survives formal reconstruction.

The classical proof introduces a point D on the longer side, creates an isosceles triangle, applies the exterior-angle theorem, and finally observes that a part of an angle is smaller than the whole. The Lean proof follows exactly this geometry.

What changes is not the mathematical argument but the amount of structure that must be made explicit. The informal steps

$$\angle BCA < \angle ABD < \angle ABC$$

require a formal notion of strict angle comparison, a theorem asserting that an interior subangle is smaller than its containing angle, and a proof of transitivity.

Consequently, Proposition I.18 does more than add another theorem of Book I. It strengthens the reusable language of synthetic angle order. The new results `hilbert_interior_angle_less` and `hilbert_angleLess_trans` are available for subsequent propositions independently of I.18.

The completed Lean development compiles cleanly and contains no `sorry`.

Chapter 19

Proposition I.19

19.1 Introduction

Euclid's Proposition I.19 is the converse of Proposition I.18. In a nondegenerate triangle, the greater angle is opposite the greater side.

For a triangle ABC, if

$$\angle ACB < \angle ABC,$$

then

$$AB < AC.$$

The classical proof is indirect. One compares the two opposite sides AB and AC. Exactly one of the following three possibilities must occur:

$$AB \cong AC, \qquad AB < AC, \qquad AC < AB.$$

The first case contradicts Proposition I.5, because equal sides would imply congruent base angles. The third case contradicts Proposition I.18, because the greater side AB would force the opposite angle at C to be greater than the angle at B. Hence only

$$AB < AC$$

remains.

The formal reconstruction makes this trichotomy explicit through the new general theorem

`hilbert_segment_trichotomy`.

It also isolates a general irreflexivity theorem for strict angle comparison:

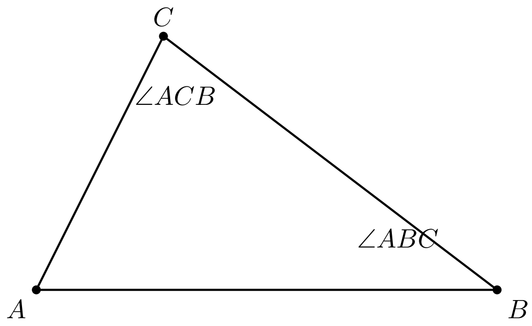
`hilbert_angleLess_irrefl`.

These two results are reusable structural components of the Hilbert layer rather than local devices specific to Proposition I.19.

19.2 The Classical Configuration

Let ABC be a nondegenerate triangle and suppose

$$\angle ACB < \angle ABC.$$



The theorem asserts that the side opposite the larger angle must be larger:

$$AB < AC.$$

No new auxiliary point is required. The proof is entirely comparative: it uses the trichotomy of the two sides and eliminates the two alternatives incompatible with the assumed angle inequality.

19.3 Classical Proof

Assume

$$\angle ACB < \angle ABC.$$

Compare the two sides AB and AC.

If

$$AB = AC,$$

then triangle ABC is isosceles. Proposition I.5 gives

$$\angle ABC = \angle ACB,$$

contradicting the hypothesis that

$$\angle ACB < \angle ABC.$$

If instead

$$AC < AB,$$

then Proposition I.18, applied to triangle ACB, yields

$$\angle ABC < \angle ACB,$$

which again contradicts the original hypothesis.

The only remaining possibility is

$$AB < AC.$$

Thus the greater angle is subtended by the greater side.

The classical dependency pattern is therefore

$$\begin{array}{ccc} & \angle ACB < \angle ABC & \\ & \downarrow & \\ & AB \cong AC \ \vee \ AB < AC \ \vee \ AC < AB & \\ & \downarrow & \\ AB \cong AC & AB < AC & AC < AB \\ \downarrow & & \downarrow \\ \text{I.5 contradiction} & \text{desired case} & \text{I.18 contradiction} \end{array}$$

19.4 Segment Trichotomy

The key structural theorem introduced for the formal proof is

`hilbert_segment_trichotomy`.

Its conclusion has the form

$$AB \cong CD \ \vee \ AB < CD \ \vee \ CD < AB.$$

The proof does not introduce a new axiom. It is derived from segment construction and the order theory of collinear points.

A copy of AB is laid off on the ray CD, producing a point X such that

$$CX \cong AB.$$

If $X = D$, then

$$AB \cong CD.$$

Otherwise, C, D, X are distinct collinear points. The Hilbert betweenness trichotomy gives three possibilities. The case

$$D - C - X$$

is incompatible with the fact that X lies on the ray CD. The remaining two orders give respectively

$$CD < CX$$

and

$$CX < CD.$$

Transporting these inequalities through

$$CX \cong AB$$

yields the required segment trichotomy.

Thus the theorem packages a general completeness property of strict segment comparison.

19.5 Irreflexivity of Strict Angle Comparison

The second structural theorem used in I.19 is

`hilbert_angleLess_irrefl`.

It states

$$\neg(\alpha < \alpha).$$

In the synthetic definition of `HilbertAngleLess`, the relation

$$\alpha < \beta$$

means that a congruent copy of α occurs as a proper interior subangle of β.

Therefore

$$\alpha < \alpha$$

would imply that an angle is congruent to one of its own proper interior subangles. This is excluded by the previously proved theorem

`hilbert_interior_subangle_not_congruent_whole`.

The irreflexivity result provides exactly the contradiction mechanism needed in both non-target branches of Proposition I.19.

19.6 Proof Architecture of the Reconstruction

The Lean proof divides naturally into three stages.

19.6.1 Nondegeneracy of the Compared Side

From the noncollinearity hypothesis

$$\neg\,\mathrm{Collinear}(A, B, C)$$

the proof first derives

$$A \neq C.$$

This is the nondegeneracy condition required to apply segment construction in `hilbert_segment_trichotomy`.

19.6.2 Apply Segment Trichotomy

The theorem

`hilbert_segment_trichotomy`

is applied to the segments AB and AC.

This produces three branches:

$$AB \cong AC, \qquad AB < AC, \qquad AC < AB.$$

The middle branch is exactly the desired conclusion.

19.6.3 Eliminate the Two Impossible Branches

In the congruent-side branch, Proposition I.5 gives

$$\angle ABC \cong \angle ACB.$$

The assumed strict inequality

$$\angle ACB < \angle ABC$$

is transported across that congruence, producing

$$\angle ACB < \angle ACB.$$

This contradicts `hilbert_angleLess_irrefl`.

In the branch

$$AC < AB,$$

Proposition I.18 applied to triangle ACB gives

$$\angle ABC < \angle ACB.$$

Combining this with the original inequality by transitivity yields

$$\angle ACB < \angle ACB,$$

again contradicting irreflexivity.

Only

$$AB < AC$$

remains.

19.7 Lean Formalization

The public theorem is:

```
theorem euclid_proposition_19
    [HilbertIncidence Geo]
    [HilbertCongruence Geo]
    (A B C : Geo.Point)
    (hABC : not PrimCollinear Geo A B C)
    (hAngle :
      HilbertAngleLess Geo A C B A B C) :
    HilbertSegmentLess Geo A B A C := by
```

The proof first derives the nondegeneracy of AC:

```
  have hAC : A != C :=
    hilbert_noncollinear_ne_first
      Geo A C B
      (by
        intro h
        exact hABC
          (PrimCollinearRotate Geo A C B h))
```

Then segment trichotomy is applied:

```
rcases
   hilbert_segment_trichotomy
     Geo
     A B
     A C
     hAC with
  hEq | hABltAC | hACltAB
```

The middle branch closes immediately:

```
$\cdot$ exact hABltAC
```

For the congruent-side branch, Proposition I.5 gives:

```
have hIso :
   Geo.AngleCongruent A B C A C B :=
  hilbert_isosceles_base_angles
   Geo
   A B C
   hABC
   hEq
```

The strict inequality is transported to the congruent target angle:

```
have hSame :
   HilbertAngleLess Geo A C B A C B :=
  hilbert_angleLess_transport_right
   Geo
   A C B
   A B C
   A C B
   hAngle
   hACB
   hIso
```

This contradicts irreflexivity:

```
exact
  False.elim
   ((hilbert_angleLess_irrefl
      Geo A C B)
     hSame)
```

In the final branch, Proposition I.18 gives the opposite strict inequality:

```
have hOpp :
   HilbertAngleLess Geo A B C A C B :=
  euclid_proposition_18
   Geo
   A C B
   hACB
   hACltAB
```

Transitivity then creates an impossible strict self-comparison:

```
have hCycle :
```

```
      HilbertAngleLess Geo A C B A C B :=
    hilbert_angleLess_trans
      Geo
      A C B
      A B C
      A C B
      hAngle
      hOpp
```

and finally:

```
  exact
    False.elim
      ((hilbert_angleLess_irrefl
          Geo A C B)
        hCycle)
```

19.8 Relationship with Earlier Propositions

19.8.1 Proposition I.5

I.5 is used to eliminate the equality case. If

$$AB \cong AC,$$

then the base angles are congruent:

$$\angle ABC \cong \angle ACB.$$

This is incompatible with the assumed strict angle inequality.

19.8.2 Proposition I.18

I.18 is used to eliminate the reversed side inequality. If

$$AC < AB,$$

then applying I.18 to triangle ACB gives

$$\angle ABC < \angle ACB.$$

Together with the hypothesis

$$\angle ACB < \angle ABC,$$

this produces a strict cycle and therefore a contradiction.

19.8.3 Logical Relation between I.18 and I.19

The two propositions form a converse pair:

$$AB < AC \quad \Longrightarrow \quad \angle ACB < \angle ABC$$

for I.18, and

$$\angle ACB < \angle ABC \quad \Longrightarrow \quad AB < AC$$

for I.19.

Together they establish the equivalence between ordering of opposite sides and ordering of opposite angles in a nondegenerate triangle.

19.9 Hilbert Components Used

19.9.1 `hilbert_segment_trichotomy`

This theorem provides the complete three-way comparison of two segments and is the main new structural component required by I.19.

19.9.2 `hilbert_angleLess_irrefl`

This theorem excludes strict self-comparison of an angle. It is used in both contradictory branches of the proof.

19.9.3 `hilbert_isosceles_base_angles`

This is the formal version of Proposition I.5 needed to eliminate the equality case.

19.9.4 `hilbert_angleLess_transport_right`

This theorem transports a strict angle comparison across congruence of the containing angle. It converts the equality branch into an impossible self-comparison.

19.9.5 `hilbert_angleLess_trans`

Transitivity combines the original strict inequality with the opposite inequality supplied by I.18.

19.10 Dependency Summary

The formal dependency structure is

$$\begin{array}{ccc}
 & \text{Hilbert incidence + order + congruence} & \\
 & \downarrow & \\
 & \texttt{hilbert_segment_trichotomy} \quad \texttt{hilbert_angleLess_irrefl} & \\
 & \downarrow & \\
 & AB \cong AC \ \vee \ AB < AC \ \vee \ AC < AB & \\
 & \downarrow & \\
\text{I.5} & & \text{I.18} \\
\downarrow & & \downarrow \\
\text{contradiction} & AB < AC & \text{contradiction} \\
 & \downarrow & \\
 & \texttt{euclid_proposition_19}. &
\end{array}$$

No new geometric axiom is introduced.

19.11 Conclusion

Proposition I.19 is formally shorter than I.18, but it exposes an important structural issue: the proof requires a complete comparison principle for segments.

Once the theorem `hilbert_segment_trichotomy` is available, the Euclidean argument becomes exactly the expected three-case proof. Equal sides are ruled out by I.5, the reversed inequality is ruled out by I.18, and the remaining case is the desired conclusion.

The proof also clarifies the role of strict angle order. Instead of introducing an explicit asymmetry theorem, the development uses the more fundamental statement `hilbert_angleLess_irrefl` together with transitivity. This keeps the order theory modular and avoids unnecessary duplication.

Thus I.19 strengthens the reusable Hilbert layer in two directions: completeness of segment comparison and irreflexivity of strict angle comparison. The Euclidean theorem itself then becomes a concise application of those general principles.

The completed Lean development compiles cleanly and contains no `sorry`.

Chapter 20

Proposition I.20

20.1 Introduction

Euclid's Proposition I.20 is the triangle inequality in purely synthetic form. For a nondegenerate triangle ABC, Euclid proves that any two sides taken together are greater than the third. For one of the three cyclic cases the statement is

$$BA + AC > BC.$$

The Hilbert reconstruction does not introduce a numerical length function and does not define an arithmetic sum of real numbers. Instead, the sum $BA + AC$ is represented geometrically. Extend BA beyond A to a point D such that

$$B - A - D \qquad \text{and} \qquad AD \cong AC.$$

Then the whole segment BD represents the concatenation of BA with a copy of AC. The required inequality becomes

$$BC < BD.$$

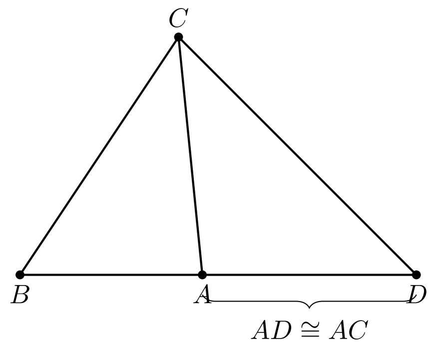
Thus the formal theorem asserts the existence of such a point D together with the strict segment comparison $BC < BD$.

The completed proof remains entirely inside neutral Hilbert geometry. It uses incidence, order, congruence, Proposition I.5, the synthetic angle-order relation, and Proposition I.19. No parallel postulate, numerical angle measure, or numerical segment length is used.

20.2 The Classical Configuration

Let ABC be a nondegenerate triangle. Extend BA beyond A to a point D and choose D so that

$$AD \cong AC.$$



The construction records

$$B - A - D, \qquad AD \cong AC.$$

The target is

$$BC < BD.$$

Geometrically, this is exactly the inequality

$$BC < BA + AC.$$

20.3 Euclid's Classical Proof

After extending BA to D so that $AD = AC$, join CD.

The auxiliary triangle ADC is isosceles. Proposition I.5 therefore gives

$$\angle ADC = \angle ACD.$$

Since A lies between B and D, the ray CA lies inside the angle BCD. Hence

$$\angle ACD < \angle BCD.$$

Combining this with the isosceles-triangle equality gives

$$\angle BDC < \angle BCD.$$

Now apply Proposition I.19 to triangle BCD. The greater angle is opposite the greater side, so

$$BC < BD.$$

Finally, because $B - A - D$ and $AD = AC$, the segment BD is the geometric concatenation of BA and AC. Therefore

$$BA + AC > BC.$$

The classical dependency pattern is

$$\begin{gathered} B - A - D,\ AD \cong AC \\ \downarrow \\ \angle ADC \cong \angle ACD \quad \text{(I.5)} \\ \downarrow \\ \angle BDC < \angle BCD \\ \downarrow \quad \text{(I.19)} \\ BC < BD. \end{gathered}$$

20.4 Synthetic Form of the Triangle Inequality

The formal reconstruction deliberately avoids introducing an arithmetic addition operation on segments. The public result is expressed by an extension point D:

$$\exists D\ (B - A - D\ \wedge\ AD \cong AC\ \wedge\ BC < BD).$$

This is the natural Hilbert form of Euclid's phrase "two sides taken together are greater than the remaining one."

The representation has two advantages. First, it remains purely geometric. Second, it makes explicit which part of the statement belongs to order and congruence geometry: the "sum" is a concatenated segment, not an externally imposed numerical quantity.

20.5 Construction of the Extension Point

The point D is constructed in two stages.

First, Hilbert's order axiom extends the line beyond A:

$$B - A - R.$$

In Lean this is obtained from

`HilbertOrder.between_extension`.

Second, a copy of AC is laid off from A on the ray AR using

`HilbertCongruence.segment_construction`.

This produces a point D satisfying

$$AD \cong AC$$

and placing D on the same ray from A as R.

The theorem

`hilbert_between_transport_sameRays`

then transports the original betweenness relation $B - A - R$ to

$$B - A - D.$$

Thus the geometric "sum" segment BD is obtained entirely from existing order and congruence infrastructure.

20.6 The Interior-Angle Step

The key order-theoretic observation is that the ray CA meets the open segment DB at the point A itself. Since

$$D - A - B,$$

the witness for

```
HilbertRayMeetsSegment Geo C A D B
```

is simply A.

The strict angle comparison is then introduced directly with

`hilbert_angleLess_intro`.

The first angle is $\angle ACD$. The corresponding proper subangle of $\angle DCB$ is represented by $\angle DCA$, which is the same unoriented angle as $\angle ACD$. The representation-level symmetry of angles is supplied by

`angle_congruent_reverse_second`

together with reflexivity of angle congruence.

This yields

$$\angle ACD < \angle DCB.$$

No auxiliary half-plane or perpendicular construction is required. The entire strict comparison follows from the ray meeting the opposite segment.

20.7 Transport Through the Isosceles Triangle

From

$$AD \cong AC$$

and the noncollinearity of A, D, C, Proposition I.5 is available in the form

`hilbert_isosceles_base_angles`.

It gives

$$\angle ADC \cong \angle ACD.$$

For Proposition I.20 we also introduced the reusable theorem

`hilbert_angleLess_transport_left`.

It expresses invariance of strict angle comparison under replacing the smaller angle by a congruent angle.

Applied here, it transports

$$\angle ACD < \angle DCB$$

across

$$\angle ADC \cong \angle ACD,$$

producing

$$\angle ADC < \angle DCB.$$

Since $B - A - D$, the rays DA and DB coincide. The theorem `hilbert_angle_eq_of_sameRay_first`

therefore identifies the angle $\angle ADC$ with $\angle BDC$. Hence

$$\angle BDC < \angle DCB.$$

Finally, reversal of the two boundary rays at C identifies

$$\angle DCB \cong \angle BCD.$$

The existing theorem

`hilbert_angleLess_transport_right`

then gives

$$\angle BDC < \angle BCD.$$

20.8 Application of Proposition I.19

At this point the proof has produced exactly the hypothesis needed for Proposition I.19 in triangle BCD:

$$\angle BDC < \angle BCD.$$

Applying

`euclid_proposition_19`

yields

$$BC < BD.$$

This is the synthetic triangle inequality for the chosen pair of sides. The other two cyclic cases follow by renaming the vertices.

20.9 Lean Formalization

The public theorem is:

```
theorem euclid_proposition_20
    [HilbertIncidence Geo]
    [HilbertCongruence Geo]
    (A B C : Geo.Point)
    (hABC : not PrimCollinear Geo A B C) :
    exists D : Geo.Point,
      Geo.Between B A D /\
      Geo.Congruent A D A C /\
      HilbertSegmentLess Geo B C B D := by
```

The first stage derives $A \neq B$ and extends BA beyond A:

```
have hAB : A != B :=
  hilbert_noncollinear_ne_first
    Geo A B C hABC

have hBA : B != A :=
  hAB.symm

rcases
    HilbertOrder.between_extension
      B A hBA with
  <R, hBAR>
```

A copy of AC is then laid off on the ray AR:

```
have hAR : A != R :=
  (HilbertOrder.between_incidence
    B A R hBAR).2.1

rcases
    HilbertCongruence.segment_construction
      (Geo := Geo)
      A C
      A R
      hAR with
  <D, hRayARD, hAD_AC>
```

Betweenness is transported to the actual constructed point D:

```
have hRayABA :
    HilbertSameRay Geo A B B :=
  hilbert_sameRay_refl
    Geo A B hBA

have hBAD :
    Geo.Between B A D :=
  hilbert_between_transport_sameRays
    Geo
    B A R
    B D
    hBAR
    hRayABA
    hRayARD
```

The theorem then packages the construction and continues with the strict segment comparison:

```
refine <D, hBAD, hAD_AC, ?_>
```

The interior ray is represented directly by the point A:

```
have hDAB :
    Geo.Between D A B :=
  (HilbertOrder.between_incidence
    B A D hBAD).2.2.2.2

have hRayCAA :
    HilbertSameRay Geo C A A :=
  hilbert_sameRay_refl
```

```
    Geo C A hAC

have hInside :
    HilbertRayMeetsSegment Geo C A D B :=
  <A, hDAB, hRayCAA>
```

Angle reversal is obtained through the representation-level congruence laws:

```
have hAngleRefl :
    Geo.AngleCongruent A C D A C D :=
  Geometry.Geo.angle_congruent_reflexive
    Geo A C D

have hAngleSwap :
    Geo.AngleCongruent A C D D C A :=
  (Geo.angle_congruent_reverse_second
    A C D
    A C D).mp
    hAngleRefl
```

This gives the strict subangle relation:

```
have hLessACD_DCB :
    HilbertAngleLess Geo A C D D C B :=
  hilbert_angleLess_intro
    Geo
    A C D
    D C B
    A
    hACD
    hDCB
    hInside
    hAngleSwap
```

Proposition I.5 and left transport give the corresponding comparison with the base angle at D:

```
have hIso :
    Geo.AngleCongruent A D C A C D :=
  hilbert_isosceles_base_angles
    Geo
    A D C
    hADC
    hAD_AC

have hLessADC_DCB :
    HilbertAngleLess Geo A D C D C B :=
  hilbert_angleLess_transport_left
    Geo
    A C D
    A D C
    D C B
    hLessACD_DCB
    hADC
    hIso
```

The ray DA is then replaced by the same ray DB:

```
have hRayDAB :
```

```
    HilbertSameRay Geo D A B :=
  hilbert_sameRay_of_between
    Geo D A B hDAB

have hRayDBA :
    HilbertSameRay Geo D B A :=
  hilbert_sameRay_symm
    Geo D A B hRayDAB

have hBDC_ACD :
    Geo.AngleCongruent B D C A C D := by
  unfold Geometry.Geo.AngleCongruent at hIso |-
  rw [hilbert_angle_eq_of_sameRay_first
    Geo D B A C hRayDBA]
  exact hIso
```

Using left transport again gives

```
have hLessBDC_DCB :
    HilbertAngleLess Geo B D C D C B :=
  hilbert_angleLess_transport_left
    Geo
    A C D
    B D C
    D C B
    hLessACD_DCB
    hBDC
    hBDC_ACD
```

Finally, the right-hand angle is reversed and Proposition I.19 closes the proof:

```
have hAngleReflDCB :
    Geo.AngleCongruent D C B D C B :=
  Geometry.Geo.angle_congruent_reflexive
    Geo D C B

have hDCB_BCD :
    Geo.AngleCongruent D C B B C D :=
  (Geo.angle_congruent_reverse_second
    D C B
    D C B).mp
    hAngleReflDCB

have hLessBDC_BCD :
    HilbertAngleLess Geo B D C B C D :=
  hilbert_angleLess_transport_right
    Geo
    B D C
    D C B
    B C D
    hLessBDC_DCB
    hBCD
    hDCB_BCD

exact
  euclid_proposition_19
    Geo
    B C D
```

```
    hBCD
    hLessBDC_BCD
```

20.10 New Reusable Infrastructure

Proposition I.20 required one new general theorem in the Hilbert interface:

`hilbert_angleLess_transport_left`.

Its role is symmetric to the previously available

`hilbert_angleLess_transport_right`.

Together they express that strict synthetic angle comparison is invariant under congruent replacement of either compared angle. This is not specific to the triangle inequality and is expected to be reusable in later propositions.

The proof also reuses the existing construction pattern

`between_extension` $\longrightarrow$ `segment_construction` $\longrightarrow$ `between_transport_sameRays`,

which already occurs elsewhere in the Hilbert development.

20.11 Dependency Summary

The formal dependency structure of Proposition I.20 can be summarized as

$$\begin{array}{c}
\text{Hilbert incidence + order + congruence} \\
\downarrow \\
\text{between extension + segment construction} \\
\downarrow \\
B - A - D, \quad AD \cong AC \\
\downarrow \\
\text{I.5 + HilbertAngleLess} \\
\downarrow \\
\angle BDC < \angle BCD \\
\downarrow \quad \text{I.19} \\
BC < BD \\
\downarrow \\
BA + AC > BC \quad \text{(synthetic interpretation).}
\end{array}$$

No Euclidean parallel axiom appears in this chain. Proposition I.20 is therefore a theorem of neutral geometry in the present Hilbert reconstruction.

20.12 Conclusion

Proposition I.20 is the first place in this sequence where Euclid's language of adding lengths is naturally reconstructed without introducing numerical length. The formal statement replaces the expression $BA + AC$ by an explicit extension segment BD satisfying

$$B - A - D, \qquad AD \cong AC.$$

The rest of the argument is purely synthetic. Proposition I.5 converts the copied side into an angle congruence, the interior-ray definition produces a strict angle inequality, congruence transport normalizes the two angles, and Proposition I.19 converts the angle inequality back into the desired side inequality.

The resulting Lean proof therefore reproduces the classical geometry while exposing its exact logical structure. No new axiom is introduced, and the only new general infrastructure is the left-transport theorem for strict angle comparison.

Chapter 21

Proposition I.21

21.1 Introduction

Euclid's Proposition I.21 concerns two straight lines drawn from the ends of one side of a triangle to a point inside the triangle.

Let ABC be a nondegenerate triangle. Let D be an interior point, and let the line BD meet the side AC at E, so that

$$A - E - C \qquad \text{and} \qquad B - D - E.$$

Euclid asserts simultaneously that

$$BD + DC < BA + AC$$

and

$$\angle BAC < \angle BDC.$$

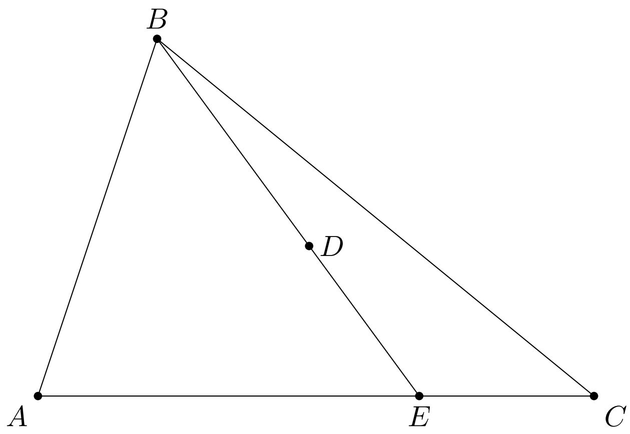
The proposition is therefore the first result in this part of Book I whose reconstruction naturally separates into two substantial comparison arguments: one for segments and one for angles.

The formal reconstruction preserves this separation. The segment inequality is obtained by combining Proposition I.20 with a synthetic "addition of unequal segments" argument. The angle inequality is proved independently, using Proposition I.16 and the strict angle comparison machinery developed in the preceding propositions.

A particularly important feature of the formalization is that no numerical length function is introduced. Expressions such as $BA+AC$ and $BD+DC$ are represented by ordinary constructed segments.

21.2 The Classical Configuration

Let ABC be a nondegenerate triangle. Choose a point D inside the triangle and let the line BD meet AC at E.



Thus

$$A - E - C, \qquad B - D - E.$$

For the segment inequality, Euclid first compares the broken line $BA + AE$ with BE by applying Proposition I.20 to triangle BAE. The remaining pieces EC and the corresponding copied segment are then added to obtain the comparison with $BA + AC$.

For the angle inequality, the point E allows the exterior-angle theorem to be applied to appropriate auxiliary triangles. The resulting strict angle comparisons are transported along the collinear rays determined by A, E, C and B, D, E.

The formal proof makes these two uses of the same configuration explicit.

21.3 Classical Proof

Since D lies inside triangle ABC, the line BD meets AC at an interior point E.

Apply Proposition I.20 to triangle ABE. It gives

$$BE < BA + AE.$$

Add EC to the two sides in the geometric sense. Since

$$AE + EC = AC,$$

we obtain

$$BE + EC < BA + AC.$$

Now apply Proposition I.20 to triangle CDE. Since $B - D - E$,

$$BD + DE = BE.$$

Proposition I.20 gives

$$DC + DE > CE,$$

or equivalently

$$CE < DC + DE.$$

Combining the appropriate segment comparisons in the classical configuration yields

$$BD + DC < BA + AC.$$

For the angle part, the exterior-angle theorem I.16 is applied to the triangles determined by the interior point and the two transversals. It yields strict comparisons which, after identifying the collinear rays, give

$$\angle BAC < \angle BDC.$$

Thus both assertions of I.21 follow:

$$BD + DC < BA + AC \qquad \text{and} \qquad \angle BAC < \angle BDC.$$

The classical proof is extremely compact because segment addition, subtraction, and replacement of collinear rays are read directly from the diagram. These are precisely the operations that must be made explicit in the formal reconstruction.

21.4 Synthetic Segment Arithmetic

The segment part of I.21 exposes an important feature of the Hilbert reconstruction.

The formal development does not introduce a numerical length function, nor does it define an algebraic addition operation on segments. Instead, a sum is represented by a constructed segment.

For example, a point X satisfying

$$B - A - X \qquad \text{and} \qquad AX \cong AC$$

makes BX a representative of

$$BA + AC.$$

Similarly, a point Y satisfying

$$B - D - Y \qquad \text{and} \qquad DY \cong DC$$

makes BY a representative of

$$BD + DC.$$

Consequently the desired inequality is represented formally by

$$BY < BX.$$

In Lean this is

```
HilbertSegmentLess Geo B Y B X
```

This observation made an explicit predicate for segment sums unnecessary. The existing language of betweenness, congruence, and strict segment comparison is sufficient.

This is not merely a simplification of I.21. It is evidence that a useful fragment of segment arithmetic can be expressed internally in the synthetic language of Book Zero.

21.5 Proof Architecture of the Reconstruction

The Lean proof divides naturally into three major components.

21.5.1 The Segment Helper

The first reusable component is

`euclid_proposition_21_segment_helper`

It begins with

$$A - E - C, \qquad B - A - H,$$

together with

$$AH \cong AE$$

and

$$BE < BH.$$

A point F is constructed on the ray AH with

$$AF \cong AC.$$

Since $AE < AC$, congruence transport gives

$$AH < AF.$$

The Book Zero order machinery then yields

$$A - H - F.$$

Together with $B - A - H$, this produces

$$B - A - F \qquad \text{and} \qquad B - H - F.$$

Thus BF represents $BA + AC$.

Next a point G is constructed beyond E with

$$EG \cong EC.$$

Hence

$$B - E - G,$$

and BG represents $BE + EC$.

From

$$A - E - C, \qquad A - H - F,$$

and

$$AE \cong AH, \qquad AC \cong AF,$$

the theorem

`bookZero_differenceOfParts`

gives

$$EC \cong HF.$$

The inequality $BE < BH$ can therefore be extended by congruent terminal parts. The theorem `bookZero_53_lessThanAdditive`

yields

$$BG < BF.$$

This is the synthetic form of

$$BE + EC < BA + AC.$$

21.5.2 The Angle Branch

The angle comparison is isolated as

`euclid_proposition_21_angle`

with conclusion

`HilbertAngleLess Geo B A C B D C`

representing

$$\angle BAC < \angle BDC.$$

The proof first reconstructs the noncollinearity information required for the auxiliary triangles. This is a substantial formal task: Euclid simply reads these facts from the diagram.

Proposition I.16 then supplies the strict exterior-angle comparisons. The results are transported through collinear and same-ray configurations, and strict angle comparison is propagated to the required pair of angles.

The essential point is that no numerical angle measure is used. The proof remains entirely inside the synthetic relation `HilbertAngleLess`.

21.5.3 Final Segment Comparison

The main theorem first applies Proposition I.20 to triangle BAE. It obtains a point H with

$$B - A - H, \qquad AH \cong AE,$$

and

$$BE < BH.$$

The segment helper extends this inequality to a comparison of constructed representatives.

The remaining part of the proof constructs a point Y satisfying

$$B - D - Y, \qquad DY \cong DC.$$

The point X on the other side satisfies

$$B - A - X, \qquad AX \cong AC.$$

After a sequence of congruence transports and applications of the Book Zero comparison lemmas, the proof reaches

```
hBY_BX :
    HilbertSegmentLess Geo B Y B X
```

which is exactly

$$BD + DC < BA + AC.$$

21.6 Lean Formalization

The public theorem has the form

```
theorem euclid_proposition_21
    [HilbertIncidence Geo]
    [HilbertCongruence Geo]
    (A B C D E : Geo.Point)
    (hABC : Not (PrimCollinear Geo A B C))
    (hAEC : Geo.Between A E C)
    (hBDE : Geo.Between B D E) :
    (exists X Y : Geo.Point,
      Geo.Between B A X /\
      Geo.Congruent A X A C /\
      Geo.Between B D Y /\
      Geo.Congruent D Y D C /\
      HilbertSegmentLess Geo B Y B X) /\
    HilbertAngleLess Geo B A C B D C := by
```

The existential part is the exact synthetic encoding of the segment inequality.

The conditions

```
Geo.Between B A X
Geo.Congruent A X A C
```

say that BX represents $BA + AC$.

Likewise

```
Geo.Between B D Y
Geo.Congruent D Y D C
```

say that BY represents $BD + DC$.

The comparison

```
HilbertSegmentLess Geo B Y B X
```

therefore expresses the first assertion of I.21.

The second assertion appears directly as

```
HilbertAngleLess Geo B A C B D C
```

The two branches are deliberately proved separately and assembled only at the end.

21.7 Relationship with Earlier Propositions

21.7.1 Proposition I.16

Proposition I.16 provides the exterior-angle inequalities needed for the angle branch.

This is the principal earlier proposition behind the proof of

$$\angle BAC < \angle BDC.$$

The formal reconstruction also reuses the same strict-angle language that was introduced and strengthened in the development surrounding I.16–I.18.

21.7.2 Proposition I.20

Proposition I.20 is fundamental to the segment branch.

Applied to triangle BAE, it supplies the initial inequality

$$BE < BA + AE.$$

In the formal proof the sum $BA + AE$ is represented by a constructed point H:

$$B - A - H, \qquad AH \cong AE,$$

so that

$$BE < BH.$$

This inequality is then enlarged synthetically by adding congruent parts.

7.3 The cumulative character of I.21

Proposition I.21 is markedly more cumulative than the immediately preceding propositions.

Its proof is no longer dominated by one construction. Instead it requires a coordinated use of order, congruence, segment comparison, ray transport, angle comparison, I.16, I.20, and numerous Book Zero lemmas.

For this reason I.21 is a useful test of whether the preceding formal development has produced a genuine working language rather than a collection of isolated theorem statements.

21.8 Hilbert and Book Zero Components Used

21.8.1 HilbertSegmentLess

Strict segment comparison supplies a synthetic replacement for numerical inequalities between lengths.

Its witness-based definition also makes it possible to recover interior points and to turn inequalities into betweenness configurations.

21.8.2 Congruence transport

The proof repeatedly changes representatives of the same segment. This requires strict inequalities to be transported through segment congruence.

Among the relevant Book Zero results are

```
bookZero_30_lessThanCongruence
bookZero_32_lessThanCongruence2
```

These operations play the role that substitution of equal numerical lengths would play in an algebraic proof.

21.8.3 Difference of parts

The theorem

```
bookZero_differenceOfParts
```

expresses the synthetic subtraction principle.

From two collinear decompositions with congruent wholes and congruent initial parts, it identifies the remaining parts.

In I.21 it produces

$$EC \cong HF.$$

21.8.4 Addition of inequalities

The theorem

```
bookZero_53_lessThanAdditive
```

is especially significant here.

It permits an existing strict segment inequality to be extended by congruent parts. In I.21 it turns

$$BE < BH$$

into the corresponding comparison after EC and HF have been added.

This is one of the clearest examples so far of Book Zero functioning as a synthetic arithmetic of segments.

8.5 Segment-inequality transitivity

The theorem

```
bookZero_52_lessThanTransitive
```

allows the several intermediate comparisons to be composed into the final inequality.

21.8.5 Sum of parts

The existing theorem

```
bookZero_sumOfParts
```

is used when two adjacent congruent pieces must be recognized as giving congruent whole segments.

Together with difference-of-parts and inequality transport, this provides much of the elementary segment calculus required by I.21.

21.9 What I.21 Reveals About Book Zero

The reconstruction of I.21 gives a useful answer to a structural question about the project.

It might appear natural to introduce explicit objects representing formal sums of segments. Indeed, an intermediate version of the development contained such a definition.

The completed proof shows that this is not presently necessary.

The combination of

$$\text{betweenness} + \text{congruence} + \text{strict segment comparison}$$

already permits sums to be constructed, compared, enlarged, reduced, and transported.

Thus the expressions

$$AB + CD, \qquad AB < CD,$$

need not force the development prematurely into an algebra of lengths.

The geometric witnesses themselves carry the arithmetic.

This is important for the intended role of Book Zero. Its purpose is not merely to shorten proofs. It supplies the intermediate operations by which ordinary synthetic reasoning can be expressed without abandoning the primitive Hilbert geometry.

I.21 is the strongest example of this role encountered so far.

21.10 Dependency Summary

The segment branch can be summarized schematically as

$$
\begin{gathered}
A - E - C, \quad B - D - E, \quad ABC \text{ noncollinear} \\
\downarrow \\
\text{I.20 on } BAE \\
\downarrow \\
BE < BA + AE \\
\downarrow \\
\text{construct congruent remainders} \\
\downarrow \\
\texttt{bookZero\_differenceOfParts} \\
\downarrow \\
\texttt{bookZero\_53\_lessThanAdditive} \\
\downarrow \\
\text{congruence transport and transitivity} \\
\downarrow \\
BD + DC < BA + AC.
\end{gathered}
$$

The angle branch is

$$
\begin{gathered}
A - E - C, \quad B - D - E, \quad ABC \text{ noncollinear} \\
\downarrow \\
\text{noncollinearity and ray reconstruction} \\
\downarrow \\
\text{Euclid I.16} \\
\downarrow \\
\text{strict angle transport} \\
\downarrow \\
\angle BAC < \angle BDC.
\end{gathered}
$$

Finally,

$$
\begin{array}{ccc}
BD + DC < BA + AC & & \angle BAC < \angle BDC \\
& \searrow \;\; \swarrow & \\
& \texttt{euclid\_proposition\_21}. &
\end{array}
$$

No new geometric axiom is introduced.

21.11 Formalization Notes

Several points that are nearly invisible in the classical proof become substantial in Lean.

First, every assertion that a point lies on the same line, on the same ray, or on the opposite continuation of a ray must be derived explicitly.

Second, the notation $AB + AC$ cannot silently denote a number. A concrete point must be constructed whose position and congruence properties make an ordinary segment represent that sum.

Third, inequalities cannot simply be rewritten by replacing equal lengths. Congruence transport requires explicit theorems.

Fourth, the two assertions of I.21 genuinely have different proof architectures. Keeping them separate in the formal development makes their dependencies visible and produces reusable intermediate results.

The resulting Lean proof is much longer than Euclid's proof, but the additional length is not caused by a different mathematical idea. It records the incidence, order, congruence, and transport operations which the classical diagram suppresses.

21.12 Conclusion

Proposition I.21 is an important transition point in the formalization of Book I.

The classical proposition combines a comparison of broken lines with a comparison of angles. Its proof assumes a mature ability to manipulate segment sums, strict inequalities, collinear configurations, and exterior angles.

The Lean reconstruction preserves this synthetic structure. It does not replace Euclid's argument by coordinates, numerical lengths, or angle measures.

More importantly, the reconstruction demonstrates that the Book Zero layer is beginning to perform its intended function. Results such as difference of parts, addition of inequalities, congruence transport, ray conversion, and strict-comparison transitivity combine into a usable language in which a substantially more complicated Euclidean argument can be expressed.

The fact that no separate `HilbertSegmentSum` object is needed is part of this conclusion. Synthetic segment arithmetic can, at least at this stage, remain geometry rather than become an external algebra of lengths.

The Hilbert reconstruction of Euclid I.21 is complete. It introduces no new geometric axiom, and the complete development through Proposition I.21 compiles successfully.

Appendix A

Euclid I.1–I.20 in Greenberg's Hilbert-Style Development

A.1 Purpose and scope

This appendix records a source audit of Euclid Book I, Propositions I.1–I.20, against Marvin Jay Greenberg's *Euclidean and Non-Euclidean Geometries: Development and History*, fourth edition (2008) [4].

The purpose of the audit is deliberately stricter than identifying mathematically related results. For each proposition we ask:

- whether Greenberg explicitly identifies the Euclidean proposition;
- whether he supplies a complete proof, a proof skeleton, a hint, an exercise, an axiom, or only related infrastructure;
- whether the argument is genuinely synthetic or is presented using numerical measurement;
- what additional foundational issue Greenberg explicitly records, such as continuity, betweenness, separation, or the status of SAS;
- how the result relates to the present Lean reconstruction.

No observation from the source audit is intentionally discarded merely because it is not needed for the present proof. Such remarks may become important later when Book I is extended beyond Proposition I.20 or when the Hilbert and Book Zero interfaces are reorganized.

A.2 Why the fourth edition matters

The fourth edition is not a bibliographical reprint of the third. Greenberg explicitly reports substantial changes to Chapters 3 and 4. In particular, the proof of the crossbar theorem was moved from the exercises into the text; the terminology of Hilbert planes was aligned with Hartshorne; Dedekind's axiom was no longer assumed as part of the basic Euclidean setup; the existence proof for midpoints was moved from the exercises into the text; and the discussion of measurement was rewritten to stress that real-number measurement is convenient rather than mathematically necessary.

These changes directly affect the status of several propositions in I.1–I.20, especially I.9, I.10, I.16, I.17, and I.20.

A.3 Comparison table

Table A.1: Euclid I.1–I.20 in Greenberg's fourth edition

Euclid	**Greenberg counterpart**	**Status in Greenberg**	**Source-audit remarks and relation to the Lean reconstruction**
I.1	Construction of an equilateral triangle on a given segment.	Greenberg explicitly treats Euclid's first proposition as containing a continuity gap. The gap is filled by the circle-circle continuity principle.	This is a foundational warning, not merely a proof detail. Greenberg emphasizes that a weak quadratic continuity principle is needed here. Thus I.1 has a different status from the later neutral results derived from incidence, betweenness, and congruence alone. This should be retained when the formal dependency graph of Book I is documented.
I.2	Euclid's second postulate; extension of a segment and laying off a congruent copy.	Greenberg asks the reader explicitly to prove Euclid's second postulate and later states that it follows easily from Axioms C-1 and B-2.	This is the primitive mechanism behind geometric segment addition. Greenberg later uses exactly this observation in his discussion of I.20. The point is important for Book Zero: "addition" of segments need not be numerical; it is implemented synthetically by extension plus segment construction.
I.3	Cutting off from a greater segment a segment congruent to a given smaller segment.	No separate Greenberg proposition labelled as Euclid I.3 was located in the fourth-edition text. Its content is absorbed into segment construction, segment ordering, and betweenness.	The absence of a named counterpart is itself informative. Modern Hilbert-style organization decomposes I.3 into lower-level operations rather than treating it as an independent theorem. The Lean proof should therefore be compared with C-1, Proposition 3.12, and Proposition 3.13 rather than with a single Greenberg theorem.
I.4	SAS triangle congruence.	SAS is Congruence Axiom C-6. Greenberg explicitly rejects Euclid's superposition argument as a proof in Euclid's stated system.	This proposition becomes foundational in the Hilbert-style system. Greenberg stresses that SAS cannot be proved from the preceding incidence, betweenness, and congruence axioms without adding an appropriate congruence axiom. This remains an important distinction between Euclid's proposition order and the dependency order of the formal Hilbert layer.
I.5	Proposition 3.10: base angles of an isosceles triangle.	A complete synthetic proof is printed, using SAS.	This is a direct Hilbert-style reconstruction of the mathematical content of I.5. It is one of the cleanest reference proofs for the formal development.

Continued on next page

Table A.1 – continued from previous page

Euclid	**Greenberg counterpart**	**Status in Greenberg**	**Source-audit remarks and relation to the Lean reconstruction**
I.6	Proposition 3.18: converse of Proposition 3.10.	The proposition is stated, but the proof is assigned as Exercise 27.	Greenberg confirms the theorem and its dependency location but does not supply the completed derivation in the main text. The formal Lean proof therefore fills a genuine proof-level omission in the textbook presentation, without claiming a new mathematical theorem.
I.7	Uniqueness configuration for triangles on the same base and on the same side.	No explicit fourth-edition counterpart labelled as Euclid I.7 was located in the source audit. Related uniqueness is distributed through triangle laying-off, congruence axioms, side-of-line machinery, and subsequent congruence criteria.	This row should not be simplified to "absent." The relevant mathematics is present as infrastructure, but Greenberg does not appear to preserve I.7 as a named theorem. For future work it is worth recording exactly which formal uniqueness lemma replaces I.7 in the Hilbert interface.
I.8	Proposition 3.22: SSS criterion for congruence.	SSS is stated as a proposition. Its proof is left as Exercise 32, with a substantial hint reducing the argument by the corollary to SAS and a same/opposite-side configuration.	Greenberg supplies the mathematical route but not a completed printed proof. This is especially relevant to the formal project because SSS was previously a foundational debt in the Hilbert interface. The exercise shows that Greenberg regards SSS as derivable from the Hilbert congruence machinery rather than as an additional axiom.
I.9	Proposition 4.4(a): every angle has a unique bisector.	The proposition is stated in the text; Exercise 6(b) asks for the proof of existence and uniqueness. Greenberg says the angle-bisector construction follows easily from the midpoint construction.	Greenberg explicitly comments on Euclid's own I.9: Euclid uses I.1 and also relies tacitly on a betweenness fact visible in the diagram, namely that a constructed point lies on the required opposite side. This note should be retained because it identifies the hidden order/separation content that a Lean proof must expose.
I.10	Proposition 4.3: every segment has a unique midpoint.	A major change from the third edition: the existence proof is now in the text as an explicit 18-step proof skeleton. The reader is asked to justify the steps; uniqueness remains Exercise 6(a).	Greenberg explicitly says that Euclid's proof of I.10 tacitly requires the crossbar theorem. His replacement construction avoids using an equilateral triangle and therefore avoids importing the continuity issue from I.1. This distinction is valuable for future optimization of the formal dependency graph.
I.11	Perpendicular erected at a point on a line; part of Proposition 3.16.	Proposition 3.16 gives a complete proof of existence of a perpendicular through an arbitrary point, with a separate case when the point lies on the line.	The modern theorem merges the content of I.11 and I.12. When the point lies on the line, Greenberg uses the existence of a right angle and angle laying-off. The result is obtained without invoking Euclid's circle construction.

Continued on next page

Table A.1 – continued from previous page

Euclid	Greenberg counterpart	Status in Greenberg	Source-audit remarks and relation to the Lean reconstruction
I.12	Perpendicular dropped from a point to a line; part of Proposition 3.16.	A complete proof is printed for the case in which the point is not on the line.	Greenberg explicitly warns that Euclid's original construction of I.12 tacitly assumes line-circle continuity. His proof of Proposition 3.16 avoids that route; indeed, he notes that using line-circle continuity there would be circular because his later derivation of that continuity principle uses Proposition 3.16. This is a particularly important dependency note for any future foundational cleanup.
I.13	Supplementary-angle configuration when a line stands on another line.	No direct proposition labelled I.13 was located. The relevant theory is distributed among the definition of supplementary angles, betweenness, Proposition 3.14 on supplements of congruent angles, and angle addition (Proposition 3.19).	The source audit therefore treats I.13 as an example of reorganization: Greenberg proves and uses more general structural facts rather than retaining the Euclidean proposition as a named unit. In Lean, the corresponding proof obligations naturally live in the angle and ray interface.
I.14	Converse straight-line criterion from adjacent supplementary angles.	No direct proposition labelled I.14 was located in the fourth-edition text. Its ingredients occur in angle laying-off uniqueness, supplementary-angle theory, ray order, and angle addition/subtraction.	As with I.13, the mathematical content is absorbed into a stronger modern infrastructure. This should be preserved as a source-audit observation rather than recorded simply as "not found."
I.15	Proposition 3.15(a): vertical angles are congruent.	The theorem is stated in the text. Exercise 25 asks the reader to deduce it from Proposition 3.14.	Thus the result is present but the proof is not fully written out in the main text. The dependency on supplements of congruent angles matches the synthetic route used in formal reconstruction.
I.16	Exterior Angle Theorem 4.2.	A complete proof is printed. Greenberg then separately analyzes Euclid's original proof and fills its diagrammatic gap with betweenness and plane separation.	This is one of the richest entries in the audit. Greenberg stresses that the missing assumption is not that "lines are infinite in extent." What is needed is the betweenness theory, especially plane separation B-4 and its consequences. He also notes that the theorem fails in elliptic geometry because the relevant separation/parallel structure fails. These remarks are potentially important well beyond I.16.

Continued on next page

Table A.1 – continued from previous page

Euclid	Greenberg counterpart	Status in Greenberg	Source-audit remarks and relation to the Lean reconstruction
I.17	Corollary 2 to the Exterior Angle Theorem: the sum of the degree measures of any two angles of a triangle is less than 180 degrees.	A complete short proof is printed, but it is stated in the language of numerical angle measure.	This is not the purely synthetic presentation used in the Lean reconstruction. Greenberg is explicit, however, that the introduction of numbers is a convenience: Hartshorne's treatment by congruence classes is mathematically preferable and valid in all Hilbert planes. Thus the numerical wording should not be mistaken for a foundational need for real numbers.
I.18–I.19	Proposition 4.5: in a triangle, the greater side lies opposite the greater angle and conversely.	Greenberg combines I.18 and I.19 into one proposition. The proof is Exercise 8, with a specific synthetic hint: choose a point on a side, use an isosceles triangle, apply the exterior angle theorem, and use trichotomy for the converse.	This is strong confirmation of the dependency structure reconstructed in Lean. The textbook does not provide the full formal derivation, so the Lean development must supply all order, noncollinearity, ray, and angle-transport details suppressed by the hint.
I.20	Triangle inequality.	A complete seven-step proof is printed. It is written using segment length notation and addition, but Greenberg immediately explains that the numerical language is only a convenience.	This row changes significantly relative to a superficial reading. Greenberg says that Euclid's proof is the same proof, with "two sides taken together" interpreted as geometric addition of segment congruence classes. He explains that Euclid II is obtained from C-1 and B-2 and that C-3 makes segment-class addition well-defined. Hence Archimedes' axiom and the full measurement theorem are not required. This is very close in mathematical structure to the present synthetic Lean proof. Greenberg also records Heron's alternative proof as an exercise: bisect the angle, use the crossbar theorem, the exterior angle theorem, and Proposition 4.5.

A.4 Structural observations from the audit

The fourth edition makes several general points that are directly relevant to the formal reconstruction.

First, Greenberg states that the main purpose of Chapter 3 is to fill gaps in Euclid's presentation, while also acknowledging that not every gap is filled there. He emphasizes that almost all elementary synthetic Euclidean geometry can nevertheless be developed from his axiomatic system.

Second, Greenberg separates three kinds of foundational phenomena that are easy to blur in an informal reading of Euclid:

1. **incidence, betweenness, and congruence**, which define a Hilbert plane and support a large neutral fragment;
2. **quadratic continuity**, needed for constructions such as Euclid I.1 and, later, the converse to the triangle inequality;
3. **measurement by real numbers**, which is convenient but not required for the elementary synthetic content under study.

Third, the fourth edition repeatedly points out that statements which look obvious in diagrams encode nontrivial order information. The crossbar theorem is the most visible example. Its proof was promoted from an exercise to the main text in this edition, which makes the book especially useful as a reference for the formal project.

Fourth, Greenberg's organization confirms that proposition numbering is not a dependency structure. Several Euclidean propositions disappear as named theorems because their content is absorbed into more general machinery; conversely, some single Greenberg propositions combine two Euclidean propositions.

A.5 Notes that should be retained for future work

The following source observations are not needed merely to certify I.1–I.20, but should be retained because they may matter later.

- Greenberg defines a Hilbert plane as a model of the thirteen incidence, betweenness, and congruence axioms.
- His Euclidean setup adds Hilbert's Euclidean parallel axiom and circle-circle continuity; Dedekind's axiom is deliberately not assumed in the fourth edition.
- He explicitly distinguishes circle-circle continuity from Archimedes' axiom and from Dedekind completeness.
- The converse to the triangle inequality, Euclid I.22, is tied directly to circle-circle continuity. This is likely to become important almost immediately after the present I.1–I.20 checkpoint.
- Greenberg repeatedly refers to Hartshorne for the technically complete treatment of elementary geometry without real numbers, especially for congruence classes of angles and their addition.
- The fourth edition states that numerical measurement is introduced in Chapter 4 to shorten the exposition, not because the geometry requires real numbers.
- Greenberg explicitly rejects the widespread claim that Euclid I.16 needs an additional assumption that lines are "infinite in extent." His diagnosis is that the missing machinery is betweenness and plane separation.
- The existence proof for a midpoint in Proposition 4.3 is not Euclid's own construction. Greenberg deliberately gives a route which avoids dependence on equilateral triangles and therefore avoids importing the continuity requirement of I.1.
- Greenberg gives Heron's alternative proof of the triangle inequality as an exercise. This provides a second synthetic route to I.20 that may later be useful for comparing proof graphs or designing tactics.

A.6 Result of the fourth-edition query

The fourth edition still does not provide a continuous sequence of twenty fully written synthetic proofs corresponding one-for-one to Euclid I.1–I.20. It provides something more useful for foundational analysis: a reorganized Hilbert-style theory in which the Euclidean results have different statuses.

Some are axiomatic in the modern system (most notably SAS/I.4); some have complete synthetic proofs (for example I.5, I.11–I.12, and I.16); some are propositions with proofs assigned as exercises (for example I.6, I.8, I.9, I.15, and I.18–I.19); some are absorbed into general order or congruence infrastructure (notably I.3, I.7, I.13, and I.14); and some are printed using numerical language even though Greenberg explicitly explains how the numerical layer can be eliminated (I.17 and especially I.20).

For the present project this means that Greenberg should be used as a foundational and dependency reference, not as a source from which the Lean proofs can simply be transcribed. The formal development exposes the intermediate operations that textbook prose suppresses: permutations of collinearity, orientation of betweenness, same-ray transport, side-of-line arguments, normalization of unoriented angles, transport of segment and angle order under congruence, and repeated noncollinearity obligations.

A.7 Current checkpoint

The present source audit concerns the completed initial block Euclid Book I, Propositions I.1–I.20. It should be treated as a durable research record for that checkpoint. When later propositions are formalized, new Greenberg observations should be appended rather than replacing or silently deleting the present notes.

Appendix B

Euclid I.1–I.20 in Hartshorne's Hilbert-Plane Development

B.1 Purpose and scope

This appendix records a source audit of Euclid Book I, Propositions I.1–I.20, against Robin Hartshorne's *Geometry: Euclid and Beyond* [5].

The audit has a specific purpose. Hartshorne begins with Euclid's own proofs and constructions, analyzes where their logical structure is incomplete, and then reconstructs the relevant part of Book I inside a Hilbert-plane framework. For each proposition I.1–I.20, we record:

- whether Hartshorne regards Euclid's proof as valid as it stands;
- whether the proposition is replaced by an axiom or by a more general Hilbert-plane theorem;
- whether Hartshorne supplies a complete proof, a proof sketch, or an exercise;
- which hidden order, betweenness, separation, congruence, or existence issue he explicitly identifies;
- which points are especially relevant to a fully formal synthetic reconstruction.

The purpose is not to claim novelty for the classical mathematics. Rather, the audit identifies the exact level at which a conventional synthetic proof still leaves obligations that must be made explicit in a formal proof assistant.

B.2 Hartshorne's organization of the problem

Hartshorne separates the discussion into two stages. Chapter 1 studies Euclid's own geometry and repeatedly asks whether a proof depends on a diagram, an unstated positional assumption, or a construction whose existence has not yet been justified. Chapter 2 then introduces Hilbert's axioms of incidence, betweenness, and congruence and reconstructs the early propositions inside that framework.

The key structural statement appears in the section on Hilbert planes: apart from I.1 and I.22, the results of Book I up to that point can be recovered in an arbitrary Hilbert plane. Proposition

I.1 requires an additional intersection or continuity assumption, and I.22 is another exception. This makes I.1–I.20 a particularly natural block for testing the Hilbert reconstruction.

B.3 Comparison table

Table B.1: Euclid I.1–I.20 in Hartshorne's Hilbert-plane development

Euclid	**Hartshorne counterpart**	**Status in Hartshorne**	**Source-audit remarks and relevance to formalization**
I.1	Construction of an equilateral triangle on a given segment.	Hartshorne treats I.1 as the first genuine exception to the Hilbert-plane recovery of Book I. The intersection of the two circles used by Euclid cannot be derived from the basic Hilbert-plane axioms alone.	Chapter 1 already isolates the hidden existence problem: the diagram shows two circles meeting, but Euclid's stated postulates do not guarantee that they do. Hartshorne therefore requires an additional intersection or continuity principle. This is a foundational dependency, not a minor gap in presentation.
I.2	Transporting a given segment to a specified point.	Hartshorne states that I.2 is effectively replaced by the segment congruence construction axiom C1.	The Euclidean proposition becomes primitive infrastructure in the Hilbert system. This is important for later segment arithmetic and for the synthetic interpretation of "adding" segments by extending a ray and laying off a congruent copy.
I.3	Cutting off from a greater segment a segment equal to a given smaller segment.	Hartshorne groups I.3 with I.2 as effectively replaced by C1 together with the order theory of segments.	The proposition is no longer structurally fundamental as a named theorem. Its mathematical content is decomposed into segment construction, uniqueness, betweenness, and segment ordering.
I.4	SAS triangle congruence.	SAS is taken as axiom C6. Hartshorne analyzes Euclid's superposition argument and concludes that it relies on an unstated theory of rigid motions.	This is one of the sharpest differences between Euclid's proposition order and a Hilbert-style dependency order. The proof by superposition is not accepted as a derivation from Euclid's explicit postulates and common notions.
I.5	Base angles of an isosceles triangle.	Hartshorne explicitly reinterprets Euclid's proof in the Hilbert system and gives the proof in detail using segment construction and SAS.	This is a direct example of the program of recovering Euclid inside a Hilbert plane. Hartshorne also points out the hidden betweenness facts needed to justify some of Euclid's positional claims.
I.6	Converse of the isosceles base-angle theorem.	Hartshorne says that Euclid's argument is essentially valid, but one point must be rewritten using the formally defined order relation on angles.	The phrase "less than" cannot be left intuitive. The proof depends on having an explicit angle-order relation and on using its asymmetry and trichotomy correctly. This is precisely the kind of obligation that becomes visible in Lean.

Continued on next page

Table B.1 – continued from previous page

Euclid	**Hartshorne counterpart**	**Status in Hartshorne**	**Source-audit remarks and relevance to formalization**
I.7	Uniqueness configuration for triangles on the same base and on the same side.	Hartshorne identifies a hidden dependence on the relative position of the lines and points. The missing betweenness justification is assigned for completion in the Hilbert-plane exercises.	This proposition is a canonical example of a proof whose diagram suppresses case distinctions. Hartshorne stresses that one must know where the lines and intersection points lie before the angle inequalities used in the proof are legitimate.
I.8	SSS triangle congruence.	Hartshorne replaces Euclid's superposition proof with a new theorem, Proposition 10.1 (SSS), proved from the Hilbert congruence axioms.	A complete Hilbert-style proof is supplied. The construction creates a copy of one triangle on the opposite side of a side of the other and uses angle addition and SAS. The positional cases are explicitly noted.
I.9	Construction of an angle bisector.	Hartshorne reconstructs I.9 using Hilbert tools. Since I.1 is not available in an arbitrary Hilbert plane, he replaces Euclid's equilateral-triangle step by the construction of an isosceles triangle.	He explicitly notes that it is not enough to perform the construction: one must prove that the constructed ray really lies in the interior of the original angle. That requires the order/separation machinery.
I.10	Construction of the midpoint of a segment.	Hartshorne reconstructs the proposition by the same strategy: replace Euclid's use of I.1 by an isosceles triangle. The corresponding Hilbert-tools construction is left as Exercise 10.2.	The proposition is therefore recoverable in a Hilbert plane without using the circle-intersection principle required by I.1. This is an important alternative dependency route.
I.11	Erecting a perpendicular at a point of a line.	Hartshorne explains that the construction can be recovered using Hilbert tools and assigns the explicit construction as Exercise 10.3.	The reconstruction also establishes the existence of right angles, which is not taken for granted beforehand. The proof is therefore part of the basic congruence-and-angle infrastructure rather than a compass construction in Euclid's original sense.
I.12	Dropping a perpendicular from a point to a line.	Hartshorne explicitly says that Euclid's original compass construction does not work in an arbitrary Hilbert plane. A new existence proof is required; the explicit construction is Exercise 10.4.	This is a significant foundational note. The failure is not that the theorem is false, but that Euclid's chosen construction imports an intersection principle unavailable in the bare Hilbert system.
I.13	Angles formed when a straight line stands on another straight line.	Hartshorne replaces the Euclidean proposition by Proposition 9.2 on supplements of congruent angles.	He remarks explicitly that Proposition 9.2 may be viewed as the proper Hilbert-style replacement for I.13. The modern statement is more general and fits directly into the angle-congruence infrastructure.

Continued on next page

Table B.1 – continued from previous page

Euclid	**Hartshorne counterpart**	**Status in Hartshorne**	**Source-audit remarks and relevance to formalization**
I.14	Converse straight-line criterion from adjacent supplementary angles.	Hartshorne treats I.14 as an easy consequence of the preceding supplement theory and asks for a careful Hilbert-plane formulation in Exercise 10.7.	The theorem is mathematically routine only after the modern notions of rays, supplementary angles, and angle congruence have been made precise. The formal version therefore belongs naturally to the low-level angle interface.
I.15	Vertical angles are equal.	Corollary 9.3: vertical angles are congruent. A complete short proof is given from the theory of supplementary angles.	This is a clean example in which Hartshorne replaces Euclid's common notion argument by an explicitly stated congruence theorem.
I.16	Exterior Angle Theorem.	Proposition 10.3 gives a full Hilbert-plane proof.	Hartshorne returns to the exact point he had criticized in Chapter 1: the constructed ray must really lie inside the relevant angle. He closes this gap using betweenness and plane separation. Thus the proof makes explicit the hidden positional content of Euclid's diagram.
I.17	Any two angles of a triangle are together less than two right angles.	Hartshorne says that I.17–I.21 are valid in a Hilbert plane after reinterpretation. For I.17 he avoids treating "the sum of two angles" as a primitive object and reformulates the statement using supplements and angle order.	This is especially relevant to formalization: the familiar informal statement must be replaced by a proposition expressible in the available angle language.
I.18	The greater side of a triangle subtends the greater opposite angle.	Hartshorne regards the Euclidean argument as recoverable but assigns a careful Hilbert-plane proof as Exercise 10.6.	The exercise explicitly calls attention to betweenness and inequalities. These are exactly the two classes of hidden obligations that dominate a formal reconstruction of I.18.
I.19	The greater angle of a triangle subtends the greater opposite side.	Included in Hartshorne's statement that I.17–I.21 are valid in a Hilbert plane after the preceding reconstruction of order and congruence.	Hartshorne does not single out a new main-text proof comparable to Proposition 10.3. Its validity rests on the now-formalized order relations for sides and angles and on the preceding comparison theorem.
I.20	Triangle inequality.	Hartshorne says that I.17–I.21 are valid in a Hilbert plane and assigns a careful proof of I.20 as Exercise 10.8.	The theorem therefore belongs to neutral Hilbert geometry. Hartshorne's exercise status is important: the mathematics is declared recoverable, but the complete chain of synthetic justifications is left to the reader. A formal proof must expose all those omitted steps explicitly.

B.4 Foundational observations retained from Hartshorne

Several remarks in Hartshorne are important beyond the immediate I.1–I.20 audit and should be preserved for later work.

- Hartshorne repeatedly uses a proof from a diagram as a warning sign. When a step depends on the relative position of points, lines, or rays, that position must be derived from the axioms rather than read from the figure.
- Betweenness and plane separation are not secondary bookkeeping devices. They are the mechanism that repairs several of the early Euclidean proofs, especially I.7, I.9, and I.16.
- Hartshorne's Hilbert plane is built from incidence, betweenness, and congruence. The parallel postulate is kept separate so that one can identify which propositions belong to neutral geometry.
- The first major exception is I.1. The basic Hilbert-plane axioms do not guarantee the intersection of the two circles used by Euclid. Hence an additional intersection/continuity principle is required.
- Euclid's constructions and Hilbert's existence axioms should not be conflated. A Euclidean compass construction may fail as a proof in an arbitrary Hilbert plane even when the corresponding geometric object can still be proved to exist by another route.
- Hartshorne replaces Euclid's superposition method by axiomatic SAS. He interprets superposition as requiring a theory of rigid motions, which is not supplied by Euclid's stated postulates.
- Segment addition is formalized synthetically. Hartshorne defines the sum of two segments by laying off a congruent copy on a ray and proves that this construction is well-defined up to congruence.
- Segment order is then defined using betweenness and congruence, and Hartshorne proves its compatibility with congruence and its trichotomy properties.
- The angle theory is developed in parallel: supplementary angles, vertical angles, angle addition, and angle order are all made explicit before the later Euclidean propositions are recovered.
- I.17 illustrates an important language issue. An informal sum of angles need not correspond directly to a primitive formal object, so the theorem is reformulated using inequalities and supplements.
- Hartshorne explicitly identifies I.22 as another exception beyond the current checkpoint. Thus the continuity/intersection issue encountered at I.1 returns immediately after I.20 and is likely to matter in the next phase of Book I.

B.5 Result of the Hartshorne audit

Hartshorne does not present Euclid I.1–I.20 as a sequence of twenty independent modern theorems. Instead, he reorganizes the material according to the logical structure of a Hilbert plane.

Some Euclidean propositions become axioms or primitive constructions (I.2–I.4); some receive complete Hilbert-style proofs (notably I.5, I.8, I.15, and I.16); some are reconstructed by replacing

Euclid's original construction with one valid in a Hilbert plane (I.9–I.12); and several are declared recoverable but are left to the reader as careful exercises (including I.18 and I.20).

The most important feature of the source for formal work is not merely which theorem is available. Hartshorne repeatedly identifies the exact kind of hidden obligation that must be supplied: betweenness, side-of-line information, angle order, segment order, interior-of-angle claims, and existence of intersections.

For a proof assistant, these are not editorial details. They are explicit proof terms and intermediate lemmas. Hartshorne therefore provides an unusually close mathematical guide to the level of granularity required by a fully formal synthetic reconstruction, even where the final derivation is still left as an exercise.

B.6 Current checkpoint

This appendix concerns only Euclid Book I, Propositions I.1–I.20. Observations relevant to later propositions have been retained when Hartshorne explicitly connects them to the present material, but no claim is made here about the complete status of the remainder of Book I.

Bibliography

[1] Euclid, *The Thirteen Books of Euclid's Elements*, edited and translated by Sir Thomas L. Heath, 3 vols., Dover Publications, New York, 1956. (Reprint of the second edition, Cambridge University Press, 1926.)

[2] David Hilbert, *Grundlagen der Geometrie*, B. G. Teubner, Leipzig, 1899.

[3] Karol Borsuk and Wanda Szmielew, *Foundations of Geometry: Euclidean and Bolyai–Lobachevskian Geometry, Projective Geometry*, North-Holland Publishing Company, Amsterdam, 1960.

[4] Marvin Jay Greenberg, *Euclidean and Non-Euclidean Geometries: Development and History*, 4th ed., W. H. Freeman and Company, New York, 2008.

[5] Robin Hartshorne, *Geometry: Euclid and Beyond*, Springer, New York, 2000.

[6] Michael Beeson, *A Constructive Version of Tarski's Geometry*, Annals of Pure and Applied Logic, Vol. 166, No. 11, pp. 1199–1273, 2015.

[7] Michael Beeson, Julien Narboux, and Freek Wiedijk, "Proof-checking Euclid," *Annals of Mathematics and Artificial Intelligence*, Vol. 85, pp. 213–257, 2019.

[8] Michael Beeson and Larry Wos, "Axiomatizing Geometry," *Annals of Mathematics and Artificial Intelligence*, 2019.